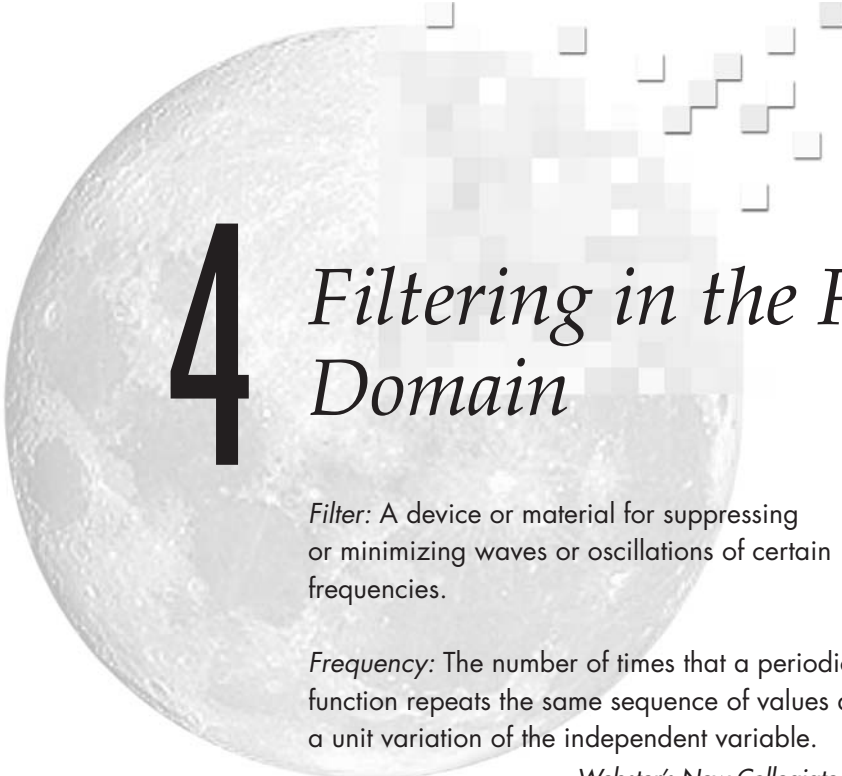




4

Filtering in the Frequency Domain

Filter: A device or material for suppressing or minimizing waves or oscillations of certain frequencies.

Frequency: The number of times that a periodic function repeats the same sequence of values during a unit variation of the independent variable.

Webster's New Collegiate Dictionary

Preview

Although significant effort was devoted in the previous chapter to spatial filtering, a thorough understanding of this area is impossible without having at least a working knowledge of how the Fourier transform and the frequency domain can be used for image filtering. You can develop a solid understanding of this topic without having to become a signal processing expert. The key lies in focusing on the fundamentals and their relevance to digital image processing. The notation, usually a source of trouble for beginners, is clarified significantly in this chapter by emphasizing the connection between image characteristics and the mathematical tools used to represent them. This chapter is concerned primarily with establishing a foundation for the Fourier transform and how it is used in basic image filtering. Later, in Chapters 5, 8, 10, and 11, we discuss other applications of the Fourier transform. We begin the discussion with a brief outline of the origins of the Fourier transform and its impact on countless branches of mathematics, science, and engineering. Next, we start from basic principles of function sampling and proceed step-by-step to derive the one- and two-dimensional discrete Fourier transforms, the basic staples of frequency domain processing. During this development, we also touch upon several important aspects of sampling, such as aliasing, whose treatment requires an understanding of the frequency domain and thus are best covered in this chapter. This material is followed by a formulation of filtering in the frequency domain and the development of sections that parallel the spatial

smoothing and sharpening filtering techniques discussed in Chapter 3. We conclude the chapter with a discussion of issues related to implementing the Fourier transform in the context of image processing. Because the material in Sections 4.2 through 4.4 is basic background, readers familiar with the concepts of 1-D signal processing, including the Fourier transform, sampling, aliasing, and the convolution theorem, can proceed to Section 4.5, where we begin a discussion of the 2-D Fourier transform and its application to digital image processing.

4.1 Background

4.1.1 A Brief History of the Fourier Series and Transform

The French mathematician Jean Baptiste Joseph Fourier was born in 1768 in the town of Auxerre, about midway between Paris and Dijon. The contribution for which he is most remembered was outlined in a memoir in 1807 and published in 1822 in his book, *La Théorie Analytique de la Chaleur* (*The Analytic Theory of Heat*). This book was translated into English 55 years later by Freeman (see Freeman [1878]). Basically, Fourier's contribution in this field states that any periodic function can be expressed as the sum of sines and/or cosines of different frequencies, each multiplied by a different coefficient (we now call this sum a *Fourier series*). It does not matter how complicated the function is; if it is periodic and satisfies some mild mathematical conditions, it can be represented by such a sum. This is now taken for granted but, at the time it first appeared, the concept that complicated functions could be represented as a sum of simple sines and cosines was not at all intuitive (Fig. 4.1), so it is not surprising that Fourier's ideas were met initially with skepticism.

Even functions that are not periodic (but whose area under the curve is finite) can be expressed as the integral of sines and/or cosines multiplied by a weighing function. The formulation in this case is the *Fourier transform*, and its utility is even greater than the Fourier series in many theoretical and applied disciplines. Both representations share the important characteristic that a function, expressed in either a Fourier series or transform, can be reconstructed (recovered) completely via an inverse process, with no loss of information. This is one of the most important characteristics of these representations because it allows us to work in the "Fourier domain" and then return to the original domain of the function without losing any information. Ultimately, it was the utility of the Fourier series and transform in solving practical problems that made them widely studied and used as fundamental tools.

The initial application of Fourier's ideas was in the field of heat diffusion, where they allowed the formulation of differential equations representing heat flow in such a way that solutions could be obtained for the first time. During the past century, and especially in the past 50 years, entire industries and academic disciplines have flourished as a result of Fourier's ideas. The advent of digital computers and the "discovery" of a fast Fourier transform (FFT) algorithm in the early 1960s (more about this later) revolutionized the field of signal processing. These two core technologies allowed for the first time practical processing of

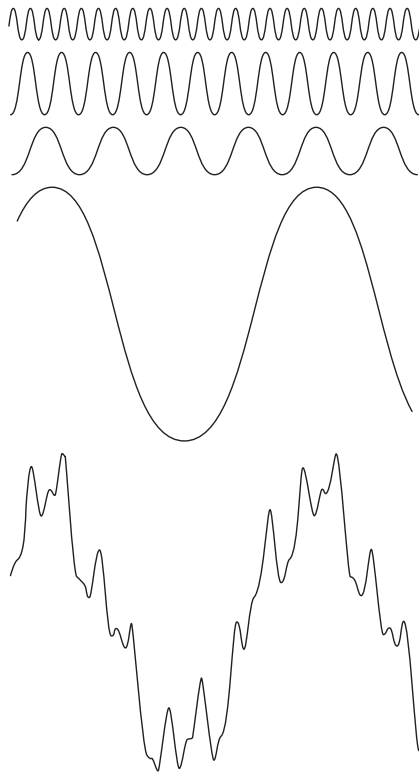


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

a host of signals of exceptional importance, ranging from medical monitors and scanners to modern electronic communications.

We will be dealing only with functions (images) of finite duration, so the Fourier transform is the tool in which we are interested. The material in the following section introduces the Fourier transform and the frequency domain. It is shown that Fourier techniques provide a meaningful and practical way to study and implement a host of image processing approaches. In some cases, these approaches are similar to the ones we developed in Chapter 3.

4.1.2 About the Examples in this Chapter

As in Chapter 3, most of the image filtering examples in this chapter deal with image enhancement. For example, smoothing and sharpening are traditionally associated with image enhancement, as are techniques for contrast manipulation. By its very nature, beginners in digital image processing find enhancement to be interesting and relatively simple to understand. Therefore, using

examples from image enhancement in this chapter not only saves having an extra chapter in the book but, more importantly, is an effective tool for introducing newcomers to filtering techniques in the frequency domain. We use frequency domain processing methods for other applications in Chapters 5, 8, 10, and 11.

4.2 Preliminary Concepts

In order to simplify the progression of ideas presented in this chapter, we pause briefly to introduce several of the basic concepts that underlie the material that follows in later sections.

4.2.1 Complex Numbers

A complex number, C , is defined as

$$C = R + jI \quad (4.2-1)$$

where R and I are real numbers, and j is an imaginary number equal to the square of -1 ; that is, $j = \sqrt{-1}$. Here, R denotes the *real part* of the complex number and I its *imaginary part*. Real numbers are a subset of complex numbers in which $I = 0$. The *conjugate* of a complex number C , denoted C^* , is defined as

$$C^* = R - jI \quad (4.2-2)$$

Complex numbers can be viewed geometrically as points in a plane (called the *complex plane*) whose abscissa is the *real axis* (values of R) and whose ordinate is the *imaginary axis* (values of I). That is, the complex number $R + jI$ is point (R, I) in the rectangular coordinate system of the complex plane.

Sometimes, it is useful to represent complex numbers in polar coordinates,

$$C = |C|(\cos \theta + j \sin \theta) \quad (4.2-3)$$

where $|C| = \sqrt{R^2 + I^2}$ is the length of the vector extending from the origin of the complex plane to point (R, I) , and θ is the angle between the vector and the real axis. Drawing a simple diagram of the real and complex axes with the vector in the first quadrant will reveal that $\tan \theta = (I/R)$ or $\theta = \arctan(I/R)$. The arctan function returns angles in the range $[-\pi/2, \pi/2]$. However, because I and R can be positive and negative independently, we need to be able to obtain angles in the full range $[-\pi, \pi]$. This is accomplished simply by keeping track of the sign of I and R when computing θ . Many programming languages do this automatically via so called *four-quadrant arctangent* functions. For example, MATLAB provides the function `atan2(Imag, Real)` for this purpose.

Using Euler's formula,

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (4.2-4)$$

where $e = 2.71828\dots$, gives the following familiar representation of complex numbers in polar coordinates,

$$C = |C|e^{j\theta} \quad (4.2-5)$$

where $|C|$ and θ are as defined above. For example, the polar representation of the complex number $1 + j2$ is $\sqrt{3}e^{j\theta}$, where $\theta = 64.4^\circ$ or 1.1 radians. The preceding equations are applicable also to complex functions. For example, a complex function, $F(u)$, of a variable u , can be expressed as the sum $F(u) = R(u) + jI(u)$, where $R(u)$ and $I(u)$ are the real and imaginary component functions. As previously noted, the complex conjugate is $F^*(u) = R(u) - jI(u)$, the magnitude is $|F(u)| = \sqrt{R(u)^2 + I(u)^2}$, and the angle is $\theta(u) = \arctan[I(u)/R(u)]$. We return to complex functions several times in the course of this and the next chapter.

4.2.2 Fourier Series

As indicated in Section 4.1.1, a function $f(t)$ of a continuous variable t that is periodic with period, T , can be expressed as the sum of sines and cosines multiplied by appropriate coefficients. This sum, known as a *Fourier series*, has the form

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t} \quad (4.2-6)$$

where

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T}t} dt \quad \text{for } n = 0, \pm 1, \pm 2, \dots \quad (4.2-7)$$

are the coefficients. The fact that Eq. (4.2-6) is an expansion of sines and cosines follows from Euler's formula, Eq. (4.2-4). We will return to the Fourier series later in this section.

4.2.3 Impulses and Their Sifting Property

Central to the study of linear systems and the Fourier transform is the concept of an impulse and its sifting property. A *unit impulse* of a continuous variable t located at $t = 0$, denoted $\delta(t)$, is *defined* as

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases} \quad (4.2-8a)$$

and is constrained also to satisfy the identity

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (4.2-8b)$$

Physically, if we interpret t as time, an impulse may be viewed as a spike of infinity amplitude and zero duration, having unit area. An impulse has the so-called *sifting property* with respect to integration,

$$\int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0) \quad (4.2-9)$$

provided that $f(t)$ is continuous at $t = 0$, a condition typically satisfied in practice. Sifting simply yields the *value* of the function $f(t)$ at the *location* of the impulse (i.e., the origin, $t = 0$, in the previous equation). A more general statement

An impulse is not a function in the usual sense. A more accurate name is a *distribution* or *generalized function*. However, one often finds in the literature the names *impulse function*, *delta function*, and *Dirac delta function*, despite the misnomer.

To *sift* means literally to separate, or to separate out by putting through a sieve.

of the sifting property involves an impulse located at an arbitrary point t_0 , denoted by $\delta(t - t_0)$. In this case, the sifting property becomes

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0) \quad (4.2-10)$$

which yields the value of the function at the impulse location, t_0 . For instance, if $f(t) = \cos(t)$, using the impulse $\delta(t - \pi)$ in Eq. (4.2-10) yields the result $f(\pi) = \cos(\pi) = -1$. The power of the sifting concept will become quite evident shortly.

Let x represent a *discrete* variable. The *unit discrete impulse*, $\delta(x)$, serves the same purposes in the context of discrete systems as the impulse $\delta(t)$ does when working with continuous variables. It is defined as

$$\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases} \quad (4.2-11a)$$

Clearly, this definition also satisfies the discrete equivalent of Eq. (4.2-8b):

$$\sum_{x=-\infty}^{\infty} \delta(x) = 1 \quad (4.2-11b)$$

The sifting property for discrete variables has the form

$$\sum_{x=-\infty}^{\infty} f(x) \delta(x) = f(0) \quad (4.2-12)$$

or, more generally using a discrete impulse located at $x = x_0$,

$$\sum_{x=-\infty}^{\infty} f(x) \delta(x - x_0) = f(x_0) \quad (4.2-13)$$

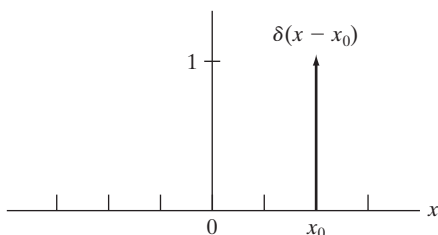
As before, we see that the sifting property simply yields the value of the function at the location of the impulse. Figure 4.2 shows the unit discrete impulse diagrammatically. Unlike its continuous counterpart, the discrete impulse is an ordinary function.

Of particular interest later in this section is an *impulse train*, $s_{\Delta T}(t)$, defined as the sum of infinitely many *periodic* impulses ΔT units apart:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T) \quad (4.2-14)$$

FIGURE 4.2

A unit discrete impulse located at $x = x_0$. Variable x is discrete, and δ is 0 everywhere except at $x = x_0$.



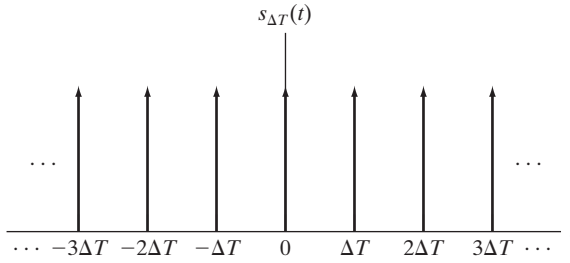


FIGURE 4.3 An impulse train.

Figure 4.3 shows an impulse train. The impulses can be continuous or discrete.

4.2.4 The Fourier Transform of Functions of One Continuous Variable

The *Fourier transform* of a continuous function $f(t)$ of a continuous variable, t , denoted $\mathfrak{F}\{f(t)\}$, is defined by the equation[†]

$$\mathfrak{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt \quad (4.2-15)$$

where μ is also a continuous variable. Because t is integrated out, $\mathfrak{F}\{f(t)\}$ is a function only of μ . We denote this fact explicitly by writing the Fourier transform as $\mathfrak{F}\{f(t)\} = F(\mu)$; that is, the Fourier transform of $f(t)$ may be written for convenience as

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt \quad (4.2-16)$$

Conversely, given $F(\mu)$, we can obtain $f(t)$ back using the *inverse Fourier transform*, $f(t) = \mathfrak{F}^{-1}\{F(\mu)\}$, written as

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu \quad (4.2-17)$$

where we made use of the fact that variable μ is integrated out in the inverse transform and wrote simple $f(t)$, rather than the more cumbersome notation $f(t) = \mathfrak{F}^{-1}\{F(\mu)\}$. Equations (4.2-16) and (4.2-17) comprise the so-called *Fourier transform pair*. They indicate the important fact mentioned in Section 4.1 that a function can be recovered from its transform.

Using Euler's formula we can express Eq. (4.2-16) as

$$F(\mu) = \int_{-\infty}^{\infty} f(t) [\cos(2\pi\mu t) - j\sin(2\pi\mu t)] dt \quad (4.2-18)$$

[†]Conditions for the existence of the Fourier transform are complicated to state in general (Champeney [1987]), but a sufficient condition for its existence is that the integral of the absolute value of $f(t)$, or the integral of the square of $f(t)$, be finite. Existence is seldom an issue in practice, except for idealized signals, such as sinusoids that extend forever. These are handled using generalized impulse functions. Our primary interest is in the discrete Fourier transform pair which, as you will see shortly, is guaranteed to exist for all finite functions.

For consistency in terminology used in the previous two chapters, and to be used later in this chapter in connection with images, we refer to the domain of variable t in general as the *spatial domain*.

If $f(t)$ is real, we see that its transform in general is complex. Note that the Fourier transform is an expansion of $f(t)$ multiplied by sinusoidal terms whose frequencies are determined by the values of μ (variable t is integrated out, as mentioned earlier). Because the only variable left after integration is frequency, we say that the domain of the Fourier transform is the *frequency domain*. We discuss the frequency domain and its properties in more detail later in this chapter. In our discussion, t can represent any continuous variable, and the units of the frequency variable μ depend on the units of t . For example, if t represents time in seconds, the units of μ are cycles/sec or Hertz (Hz). If t represents distance in meters, then the units of μ are cycles/meter, and so on. In other words, the units of the frequency domain are cycles per unit of the independent variable of the input function.

EXAMPLE 4.1:
Obtaining the
Fourier transform
of a simple
function.

■ The Fourier transform of the function in Fig. 4.4(a) follows from Eq. (4.2-16):

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt = \int_{-W/2}^{W/2} A e^{-j2\pi\mu t} dt \\ &= \frac{-A}{j2\pi\mu} \left[e^{-j2\pi\mu t} \right]_{-W/2}^{W/2} = \frac{-A}{j2\pi\mu} \left[e^{-j\pi\mu W} - e^{j\pi\mu W} \right] \\ &= \frac{A}{j2\pi\mu} \left[e^{j\pi\mu W} - e^{-j\pi\mu W} \right] \\ &= AW \frac{\sin(\pi\mu W)}{(\pi\mu W)} \end{aligned}$$

where we used the trigonometric identity $\sin \theta = (e^{j\theta} - e^{-j\theta})/2j$. In this case the complex terms of the Fourier transform combined nicely into a real sine

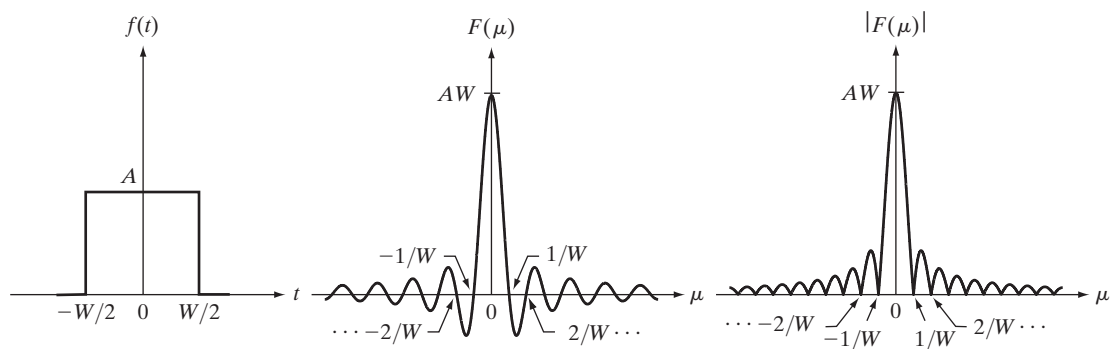


FIGURE 4.4 (a) A simple function; (b) its Fourier transform; and (c) the spectrum. All functions extend to infinity in both directions.

function. The result in the last step of the preceding expression is known as the *sinc* function:

$$\text{sinc}(m) = \frac{\sin(\pi m)}{(\pi m)} \quad (4.2-19)$$

where $\text{sinc}(0) = 1$, and $\text{sinc}(m) = 0$ for all other *integer* values of m . Figure 4.4(b) shows a plot of $F(\mu)$.

In general, the Fourier transform contains complex terms, and it is customary for display purposes to work with the magnitude of the transform (a real quantity), which is called the *Fourier spectrum* or the *frequency spectrum*:

$$|F(\mu)| = AT \left| \frac{\sin(\pi\mu W)}{(\pi\mu W)} \right|$$

Figure 4.4(c) shows a plot of $|F(\mu)|$ as a function of frequency. The key properties to note are that the locations of the zeros of both $F(\mu)$ and $|F(\mu)|$ are *inversely* proportional to the width, W , of the “box” function, that the height of the lobes decreases as a function of distance from the origin, and that the function extends to infinity for both positive and negative values of μ . As you will see later, these properties are quite helpful in interpreting the spectra of two-dimensional Fourier transforms of images. ■

■ The Fourier transform of a unit impulse located at the origin follows from Eq. (4.2-16):

EXAMPLE 4.2:
Fourier transform of an impulse and of an impulse train.

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} e^{-j2\pi\mu t} \delta(t) dt \\ &= e^{-j2\pi\mu 0} = e^0 \\ &= 1 \end{aligned}$$

where the third step follows from the sifting property in Eq. (4.2-9). Thus, we see that the Fourier transform of an impulse located at the origin of the spatial domain is a constant in the frequency domain. Similarly, the Fourier transform of an impulse located at $t = t_0$ is

$$\begin{aligned} F(\mu) &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} e^{-j2\pi\mu t} \delta(t - t_0) dt \\ &= e^{-j2\pi\mu t_0} \\ &= \cos(2\pi\mu t_0) - j \sin(2\pi\mu t_0) \end{aligned}$$

where the third line follows from the sifting property in Eq. (4.2-10) and the last line follows from Euler's formula. These last two lines are equivalent representations of a unit circle centered on the origin of the complex plane.

In Section 4.3, we make use of the Fourier transform of a periodic impulse train. Obtaining this transform is not as straightforward as we just showed for individual impulses. However, understanding how to derive the transform of an impulse train is quite important, so we take the time to derive it in detail here. We start by noting that the only difference in the *form* of Eqs. (4.2-16) and (4.2-17) is the sign of the exponential. Thus, if a function $f(t)$ has the Fourier transform $F(\mu)$, then the latter function evaluated at t , that is, $F(t)$, must have the transform $f(-\mu)$. Using this *symmetry* property and given, as we showed above, that the Fourier transform of an impulse $\delta(t - t_0)$ is $e^{-j2\pi\mu t_0}$, it follows that the function $e^{-j2\pi\mu t}$ has the transform $\delta(-\mu - t_0)$. By letting $-t_0 = a$, it follows that the transform of $e^{j2\pi at}$ is $\delta(-\mu + a) = \delta(\mu - a)$, where the last step is true because δ is not zero only when $\mu = a$, which is the same result for either $\delta(-\mu + a)$ or $\delta(\mu - a)$, so the two forms are equivalent.

The impulse train $s_{\Delta T}(t)$ in Eq. (4.2-14) is periodic with period ΔT , so we know from Section 4.2.2 that it can be expressed as a Fourier series:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t}$$

where

$$c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} s_{\Delta T}(t) e^{-j\frac{2\pi n}{\Delta T}t} dt$$

With reference to Fig. 4.3, we see that the integral in the interval $[-\Delta T/2, \Delta T/2]$ encompasses only the impulse of $s_{\Delta T}(t)$ that is located at the origin. Therefore, the preceding equation becomes

$$\begin{aligned} c_n &= \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-j\frac{2\pi n}{\Delta T}t} dt \\ &= \frac{1}{\Delta T} e^0 \\ &= \frac{1}{\Delta T} \end{aligned}$$

The Fourier series expansion then becomes

$$s_{\Delta T}(t) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}$$

Our objective is to obtain the Fourier transform of this expression. Because summation is a linear process, obtaining the Fourier transform of a sum is

the same as obtaining the sum of the transforms of the individual components. These components are exponentials, and we established earlier in this example that

$$\mathfrak{F}\left\{e^{j\frac{2\pi n}{\Delta T}t}\right\} = \delta\left(\mu - \frac{n}{\Delta T}\right)$$

So, $S(\mu)$, the Fourier transform of the periodic impulse train $s_{\Delta T}(t)$, is

$$\begin{aligned} S(\mu) &= \mathfrak{F}\{s_{\Delta T}(t)\} \\ &= \mathfrak{F}\left\{\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} \\ &= \frac{1}{\Delta T} \mathfrak{F}\left\{\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right\} \\ &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right) \end{aligned}$$

This fundamental result tells us that the Fourier transform of an impulse train with period ΔT is also an impulse train, whose period is $1/\Delta T$. This inverse proportionality between the periods of $s_{\Delta T}(t)$ and $S(\mu)$ is analogous to what we found in Fig. 4.4 in connection with a box function and its transform. This property plays a fundamental role in the remainder of this chapter. ■

4.2.5 Convolution

We need one more building block before proceeding. We introduced the idea of convolution in Section 3.4.2. You learned in that section that convolution of two functions involves flipping (rotating by 180°) one function about its origin and sliding it past the other. At each displacement in the sliding process, we perform a computation, which in the case of Chapter 3 was a sum of products. In the present discussion, we are interested in the convolution of two continuous functions, $f(t)$ and $h(t)$, of one *continuous* variable, t , so we have to use integration instead of a summation. The convolution of these two functions, denoted as before by the operator $\star$, is *defined* as

$$f(t) \star h(t) = \int_{-\infty}^{\infty} f(\tau) h(t - \tau) d\tau \quad (4.2-20)$$

where the minus sign accounts for the flipping just mentioned, t is the *displacement* needed to slide one function past the other, and τ is a dummy variable that is integrated out. We assume for now that the functions extend from $-\infty$ to ∞ .

We illustrated the basic mechanics of convolution in Section 3.4.2, and we will do so again later in this chapter and in Chapter 5. At the moment, we are

interested in finding the Fourier transform of Eq. (4.2-20). We start with Eq. (4.2-15):

$$\begin{aligned}\mathfrak{F}\{f(t) \star h(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau) h(t - \tau) d\tau \right] e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau) e^{-j2\pi\mu t} dt \right] d\tau\end{aligned}$$

The term inside the brackets is the Fourier transform of $h(t - \tau)$. We show later in this chapter that $\mathfrak{F}\{h(t - \tau)\} = H(\mu)e^{-j2\pi\mu\tau}$, where $H(\mu)$ is the Fourier transform of $h(t)$. Using this fact in the preceding equation gives us

$$\begin{aligned}\mathfrak{F}\{f(t) \star h(t)\} &= \int_{-\infty}^{\infty} f(\tau) [H(\mu) e^{-j2\pi\mu\tau}] d\tau \\ &= H(\mu) \int_{-\infty}^{\infty} f(\tau) e^{-j2\pi\mu\tau} d\tau \\ &= H(\mu) F(\mu)\end{aligned}$$

The same result would be obtained if the order of $f(t)$ and $h(t)$ were reversed, so convolution is commutative.

Recalling from Section 4.2.4 that we refer to the domain of t as the spatial domain, and the domain of μ as the frequency domain, the preceding equation tells us that the Fourier transform of the convolution of two functions in the spatial domain is equal to the product in the frequency domain of the Fourier transforms of the two functions. Conversely, if we have the product of the two transforms, we can obtain the convolution in the spatial domain by computing the inverse Fourier transform. In other words, $f(t) \star h(t)$ and $H(\mu)F(\mu)$ are a Fourier transform pair. This result is one-half of the *convolution theorem* and is written as

$$f(t) \star h(t) \Leftrightarrow H(\mu)F(\mu) \quad (4.2-21)$$

The double arrow is used to indicate that the expression on the right is obtained by taking the Fourier transform of the expression on the left, while the expression on the left is obtained by taking the *inverse* Fourier transform of the expression on the right.

Following a similar development would result in the other half of the convolution theorem:

$$f(t)h(t) \Leftrightarrow H(\mu) \star F(\mu) \quad (4.2-22)$$

which states that convolution in the frequency domain is analogous to multiplication in the spatial domain, the two being related by the forward and inverse Fourier transforms, respectively. As you will see later in this chapter, the convolution theorem is the foundation for filtering in the frequency domain.

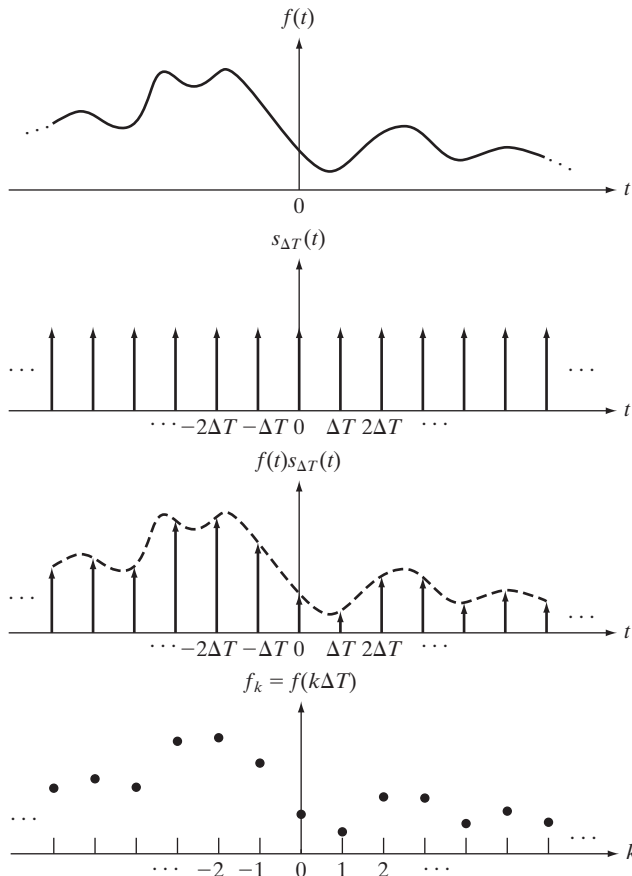
4.3 Sampling and the Fourier Transform of Sampled Functions

In this section, we use the concepts from Section 4.2 to formulate a basis for expressing sampling mathematically. This will lead us, starting from basic principles, to the Fourier transform of sampled functions.

4.3.1 Sampling

Continuous functions have to be converted into a sequence of discrete values before they can be processed in a computer. This is accomplished by using sampling and quantization, as introduced in Section 2.4. In the following discussion, we examine sampling in more detail.

With reference to Fig. 4.5, consider a continuous function, $f(t)$, that we wish to sample at uniform intervals (ΔT) of the independent variable t . We



a
b
c
d

FIGURE 4.5

(a) A continuous function. (b) Train of impulses used to model the sampling process. (c) Sampled function formed as the product of (a) and (b). (d) Sample values obtained by integration and using the sifting property of the impulse. (The dashed line in (c) is shown for reference. It is not part of the data.)

Taking samples ΔT units apart implies a *sampling rate* equal to $1/\Delta T$. If the units of ΔT are seconds, then the sampling rate is in samples/s. If the units of ΔT are meters, then the sampling rate is in samples/m, and so on.

assume that the function extends from $-\infty$ to ∞ with respect to t . One way to model sampling is to multiply $f(t)$ by a *sampling function* equal to a train of impulses ΔT units apart, as discussed in Section 4.2.3. That is,

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T) \quad (4.3-1)$$

where $\tilde{f}(t)$ denotes the sampled function. Each component of this summation is an impulse weighted by the value of $f(t)$ at the location of the impulse, as Fig. 4.5(c) shows. The *value* of each sample is then given by the “strength” of the weighted impulse, which we obtain by integration. That is, the value, f_k , of an arbitrary sample in the sequence is given by

$$\begin{aligned} f_k &= \int_{-\infty}^{\infty} f(t)\delta(t - k\Delta T) dt \\ &= f(k\Delta T) \end{aligned} \quad (4.3-2)$$

where we used the sifting property of δ in Eq. (4.2-10). Equation (4.3-2) holds for any integer value $k = \dots, -2, -1, 0, 1, 2, \dots$. Figure 4.5(d) shows the result, which consists of equally-spaced samples of the original function.

4.3.2 The Fourier Transform of Sampled Functions

Let $F(\mu)$ denote the Fourier transform of a continuous function $f(t)$. As discussed in the previous section, the corresponding sampled function, $\tilde{f}(t)$, is the product of $f(t)$ and an impulse train. We know from the convolution theorem in Section 4.2.5 that the Fourier transform of the product of two functions in the spatial domain is the convolution of the transforms of the two functions in the frequency domain. Thus, the Fourier transform, $\tilde{F}(\mu)$, of the sampled function $\tilde{f}(t)$ is:

$$\begin{aligned} \tilde{F}(\mu) &= \mathfrak{F}\{\tilde{f}(t)\} \\ &= \mathfrak{F}\{f(t)s_{\Delta T}(t)\} \\ &= F(\mu) \star S(\mu) \end{aligned} \quad (4.3-3)$$

where, from Example 4.2,

$$S(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right) \quad (4.3-4)$$

is the Fourier transform of the impulse train $s_{\Delta T}(t)$. We obtain the convolution of $F(\mu)$ and $S(\mu)$ directly from the definition in Eq. (4.2-20):

$$\begin{aligned}
 \tilde{F}(\mu) &= F(\mu) \star S(\mu) \\
 &= \int_{-\infty}^{\infty} F(\tau) S(\mu - \tau) d\tau \\
 &= \frac{1}{\Delta T} \int_{-\infty}^{\infty} F(\tau) \sum_{n=-\infty}^{\infty} \delta\left(\mu - \tau - \frac{n}{\Delta T}\right) d\tau \quad (4.3-5) \\
 &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} F(\tau) \delta\left(\mu - \tau - \frac{n}{\Delta T}\right) d\tau \\
 &= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F\left(\mu - \frac{n}{\Delta T}\right)
 \end{aligned}$$

where the final step follows from the sifting property of the impulse, as given in Eq. (4.2-10).

The summation in the last line of Eq. (4.3-5) shows that the Fourier transform $\tilde{F}(\mu)$ of the sampled function $\tilde{f}(t)$ is an *infinite, periodic* sequence of *copies* of $F(\mu)$, the transform of the original, continuous function. The separation between copies is determined by the value of $1/\Delta T$. Observe that although $\tilde{f}(t)$ is a sampled function, its transform $\tilde{F}(\mu)$ is *continuous* because it consists of copies of $F(\mu)$ which is a continuous function.

Figure 4.6 is a graphical summary of the preceding results.[†] Figure 4.6(a) is a sketch of the Fourier transform, $F(\mu)$, of a function $f(t)$, and Fig. 4.6(b) shows the transform, $\tilde{F}(\mu)$, of the sampled function. As mentioned in the previous section, the quantity $1/\Delta T$ is the sampling rate used to generate the sampled function. So, in Fig. 4.6(b) the sampling rate was high enough to provide sufficient separation between the periods and thus preserve the integrity of $F(\mu)$. In Fig. 4.6(c), the sampling rate was just enough to preserve $F(\mu)$, but in Fig. 4.6(d), the sampling rate was below the minimum required to maintain distinct copies of $F(\mu)$ and thus failed to preserve the original transform. Figure 4.6(b) is the result of an *over-sampled* signal, while Figs. 4.6(c) and (d) are the results of *critically-sampling* and *under-sampling* the signal, respectively. These concepts are the basis for the material in the following section.

4.3.3 The Sampling Theorem

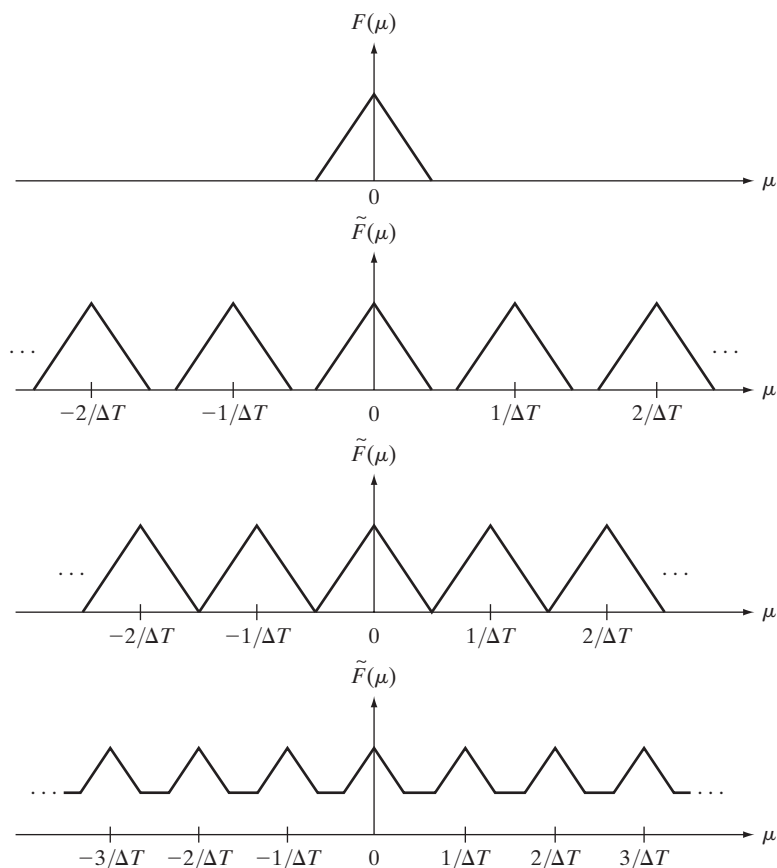
We introduced the idea of sampling intuitively in Section 2.4. Now we consider the sampling process formally and establish the conditions under which a continuous function can be *recovered uniquely* from a set of its samples.

[†]For the sake of clarity in illustrations, sketches of Fourier transforms in Fig. 4.6, and other similar figures in this chapter, ignore the fact that transforms typically are complex functions.

a
b
c
d

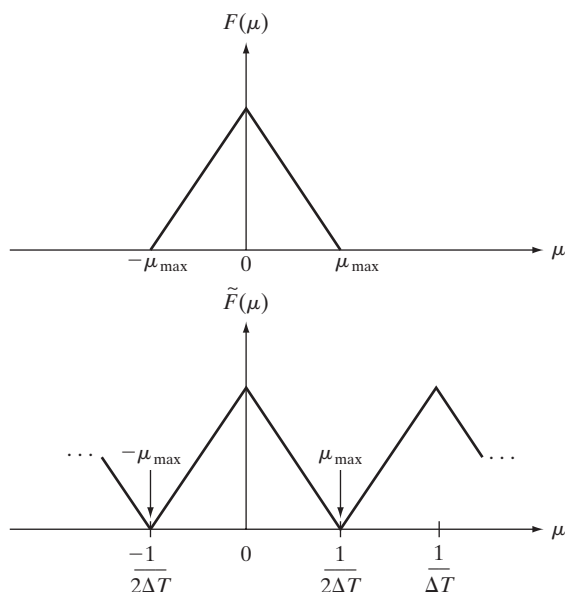
FIGURE 4.6

(a) Fourier transform of a band-limited function.
(b)–(d) Transforms of the corresponding sampled function under the conditions of over-sampling, critically-sampling, and under-sampling, respectively.



A function $f(t)$ whose Fourier transform is zero for values of frequencies outside a finite interval (band) $[-\mu_{\max}, \mu_{\max}]$ about the origin is called a *band-limited* function. Figure 4.7(a), which is a magnified section of Fig. 4.6(a), is such a function. Similarly, Fig. 4.7(b) is a more detailed view of the transform of a critically-sampled function shown in Fig. 4.6(c). A lower value of $1/\Delta T$ would cause the periods in $\tilde{F}(\mu)$ to merge; a higher value would provide a clean separation between the periods.

We can recover $f(t)$ from its sampled version if we can isolate a copy of $F(\mu)$ from the periodic sequence of copies of this function contained in $\tilde{F}(\mu)$, the transform of the sampled function $\tilde{f}(t)$. Recall from the discussion in the previous section that $\tilde{F}(\mu)$ is a *continuous, periodic* function with period $1/\Delta T$. Therefore, all we need is one complete period to characterize the entire transform. This implies that we can recover $f(t)$ from that single period by using the inverse Fourier transform.

a
b**FIGURE 4.7**

(a) Transform of a band-limited function.

(b) Transform resulting from critically sampling the same function.

Extracting from $\tilde{F}(\mu)$ a single period that is equal to $F(\mu)$ is possible if the separation between copies is sufficient (see Fig. 4.6). In terms of Fig. 4.7(b), sufficient separation is guaranteed if $1/2\Delta T > \mu_{\max}$ or

$$\frac{1}{\Delta T} > 2\mu_{\max} \quad (4.3-6)$$

This equation indicates that a continuous, band-limited function can be recovered completely from a set of its samples if the samples are acquired at a rate exceeding twice the highest frequency content of the function. This result is known as the *sampling theorem*.[†] We can say based on this result that no information is lost if a continuous, band-limited function is represented by samples acquired at a rate greater than twice the highest frequency content of the function. Conversely, we can say that the *maximum* frequency that can be “captured” by sampling a signal at a rate $1/\Delta T$ is $\mu_{\max} = 1/2\Delta T$. Sampling at the Nyquist rate sometimes is sufficient for perfect function recovery, but there are cases in which this leads to difficulties, as we illustrate later in Example 4.3. Thus, the sampling theorem specifies that sampling must exceed the Nyquist rate.

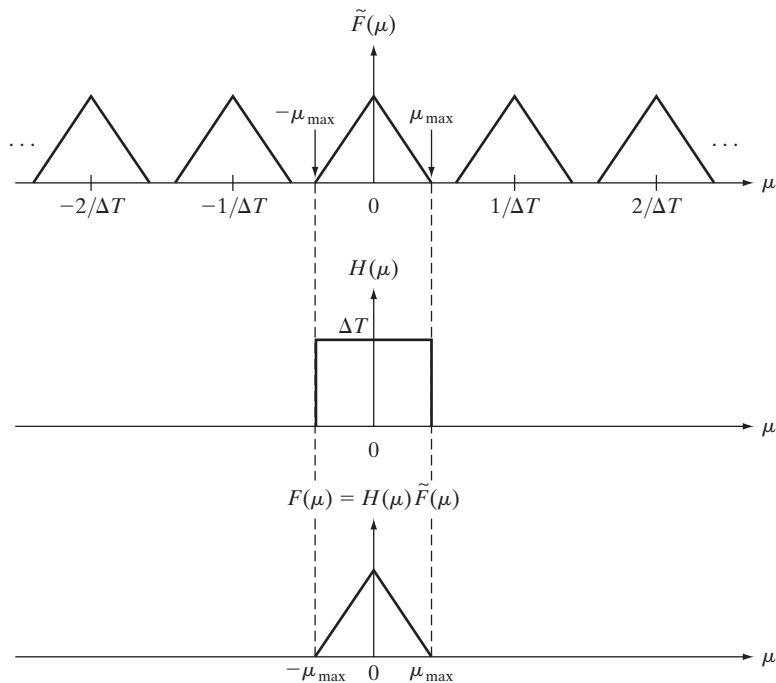
A sampling rate equal to exactly twice the highest frequency is called the *Nyquist rate*.

[†]The sampling theorem is a cornerstone of digital signal processing theory. It was first formulated in 1928 by Harry Nyquist, a Bell Laboratories scientist and engineer. Claude E. Shannon, also from Bell Labs, proved the theorem formally in 1949. The renewed interest in the sampling theorem in the late 1940s was motivated by the emergence of early digital computing systems and modern communications, which created a need for methods dealing with digital (sampled) data.

a
b
c

FIGURE 4.8

Extracting one period of the transform of a band-limited function using an ideal lowpass filter.



To see how the recovery of $F(\mu)$ from $\tilde{F}(\mu)$ is possible in principle, consider Fig. 4.8, which shows the Fourier transform of a function sampled at a rate slightly higher than the Nyquist rate. The function in Fig. 4.8(b) is defined by the equation

$$H(\mu) = \begin{cases} \Delta T & -\mu_{\max} \leq \mu \leq \mu_{\max} \\ 0 & \text{otherwise} \end{cases} \quad (4.3-7)$$

When multiplied by the periodic sequence in Fig. 4.8(a), this function isolates the period centered on the origin. Then, as Fig. 4.8(c) shows, we obtain $F(\mu)$ by multiplying $\tilde{F}(\mu)$ by $H(\mu)$:

$$F(\mu) = H(\mu)\tilde{F}(\mu) \quad (4.3-8)$$

Once we have $F(\mu)$ we can recover $f(t)$ by using the inverse Fourier transform:

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu \quad (4.3-9)$$

Equations (4.3-7) through (4.3-9) prove that, theoretically, it is possible to recover a band-limited function from samples of the function obtained at a rate exceeding twice the highest frequency content of the function. As we discuss in the following section, the requirement that $f(t)$ must be band-limited implies in general that $f(t)$ must extend from $-\infty$ to ∞ , a condition

The ΔT in Eq. (4.3-7) cancels out the $1/\Delta T$ in Eq. (4.3-5).

that cannot be met in practice. As you will see shortly, having to limit the duration of a function prevents perfect recovery of the function, except in some special cases.

Function $H(\mu)$ is called a *lowpass filter* because it passes frequencies at the low end of the frequency range but it eliminates (filters out) all higher frequencies. It is called also an *ideal* lowpass filter because of its infinitely rapid transitions in amplitude (between 0 and ΔT at location $-\mu_{\max}$ and the reverse at $\mu_{\max}$), a characteristic that cannot be achieved with physical electronic components. We can simulate ideal filters in software, but even then there are limitations, as we explain in Section 4.7.2. We will have much more to say about filtering later in this chapter. Because they are instrumental in recovering (reconstructing) the original function from its samples, filters used for the purpose just discussed are called *reconstruction filters*.

4.3.4 Aliasing

A logical question at this point is: What happens if a band-limited function is sampled at a rate that is less than twice its highest frequency? This corresponds to the under-sampled case discussed in the previous section. Figure 4.9(a) is the same as Fig. 4.6(d), which illustrates this condition. The net effect of lowering the sampling rate below the Nyquist rate is that the periods now overlap, and it becomes impossible to isolate a single period of the transform, regardless of the filter used. For instance, using the ideal lowpass filter in Fig. 4.9(b) would result in a transform that is corrupted by frequencies from adjacent periods, as Fig. 4.9(c) shows. The inverse transform would then yield a corrupted function of t . This effect, caused by under-sampling a function, is known as *frequency aliasing* or simply as *aliasing*. In words, aliasing is a process in which high frequency components of a continuous function “masquerade” as lower frequencies in the sampled function. This is consistent with the common use of the term *alias*, which means “a false identity.”

Unfortunately, except for some special cases mentioned below, aliasing is always present in sampled signals because, even if the original sampled function is band-limited, infinite frequency components are introduced the moment we limit the duration of the function, which we always have to do in practice. For example, suppose that we want to limit the duration of a band-limited function $f(t)$ to an interval, say $[0, T]$. We can do this by multiplying $f(t)$ by the function

$$h(t) = \begin{cases} 1 & 0 \leq t \leq T \\ 0 & \text{otherwise} \end{cases} \quad (4.3-10)$$

This function has the same basic shape as Fig. 4.4(a) whose transform, $H(\mu)$, has frequency components extending to infinity, as Fig. 4.4(b) shows. From the convolution theorem we know that the transform of the product of $h(t)f(t)$ is the convolution of the transforms of the functions. Even if the transform of $f(t)$ is band-limited, convolving it with $H(\mu)$, which involves sliding one function across the other, will yield a result with frequency

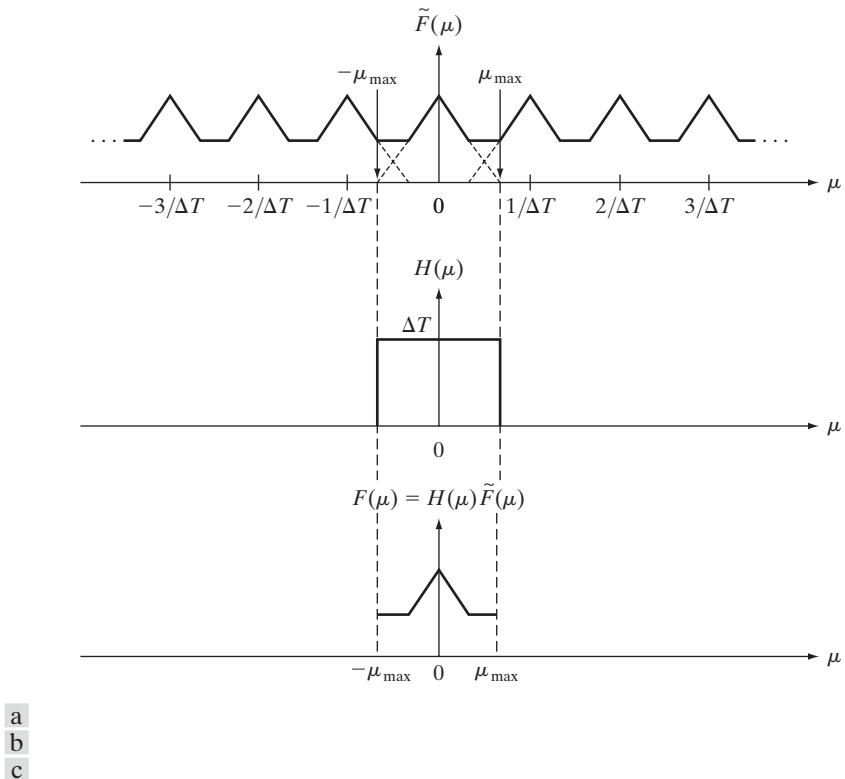


FIGURE 4.9 (a) Fourier transform of an under-sampled, band-limited function. (Interference from adjacent periods is shown dashed in this figure). (b) The same ideal lowpass filter used in Fig. 4.8(b). (c) The product of (a) and (b). The interference from adjacent periods results in aliasing that prevents perfect recovery of $F(\mu)$ and, therefore, of the original, band-limited continuous function. Compare with Fig. 4.8.

components extending to infinity. Therefore, no function of finite duration can be band-limited. Conversely, a function that is band-limited must extend from $-\infty$ to ∞ .[†]

We conclude that aliasing is an inevitable fact of working with sampled records of finite length for the reasons stated in the previous paragraph. In practice, the effects of aliasing can be *reduced* by smoothing the input function to attenuate its higher frequencies (e.g., by defocusing in the case of an image). This process, called *anti-aliasing*, has to be done *before* the function is sampled because aliasing is a sampling issue that cannot be “undone after the fact” using computational techniques.

[†]An important special case is when a function that extends from $-\infty$ to ∞ is band-limited *and* periodic. In this case, the function can be truncated and still be band-limited, *provided* that the truncation encompasses *exactly* an integral number of periods. A single truncated period (and thus the function) can be represented by a set of discrete samples satisfying the sampling theorem, taken over the truncated interval.

■ Figure 4.10 shows a classic illustration of aliasing. A pure sine wave extending infinitely in both directions has a single frequency so, obviously, it is band-limited. Suppose that the sine wave in the figure (ignore the large dots for now) has the equation $\sin(\pi t)$, and that the horizontal axis corresponds to time, t , in seconds. The function crosses the axis at $t = \dots -1, 0, 1, 2, 3 \dots$

The period, P , of $\sin(\pi t)$ is 2 s, and its frequency is $1/P$, or $1/2$ cycles/s. According to the sampling theorem, we can recover this signal from a set of its samples if the sampling rate, $1/\Delta T$, exceeds twice the highest frequency of the signal. This means that a sampling rate greater than 1 sample/s [$2 \times (1/2) = 1$], or $\Delta T < 1$ s, is required to recover the signal. Observe that sampling this signal at *exactly* twice the frequency (1 sample/s), with samples taken at $t = \dots -1, 0, 1, 2, 3 \dots$, results in $\dots \sin(-\pi), \sin(0), \sin(\pi), \sin(2\pi), \dots$, which are all 0. This illustrates the reason why the sampling theorem requires a sampling rate that exceeds twice the highest frequency, as mentioned earlier.

The large dots in Fig. 4.10 are samples taken uniformly at a rate of less than 1 sample/s (in fact, the separation between samples exceeds 2 s, which gives a sampling rate lower than $1/2$ samples/s). The sampled signal *looks* like a sine wave, but its frequency is about *one-tenth* the frequency of the original. This sampled signal, having a frequency well below anything present in the original continuous function is an example of aliasing. Given just the samples in Fig. 4.10, the seriousness of aliasing in a case such as this is that we would have no way of knowing that these samples are not a true representation of the original function. As you will see in later in this chapter, aliasing in images can produce similarly misleading results. ■

EXAMPLE 4.3:
Aliasing.

Recall that 1 cycle/s is defined as 1 Hz.

4.3.5 Function Reconstruction (Recovery) from Sampled Data

In this section, we show that reconstruction of a function from a set of its samples reduces in practice to interpolating between the samples. Even the simple act of displaying an image requires reconstruction of the image from its samples

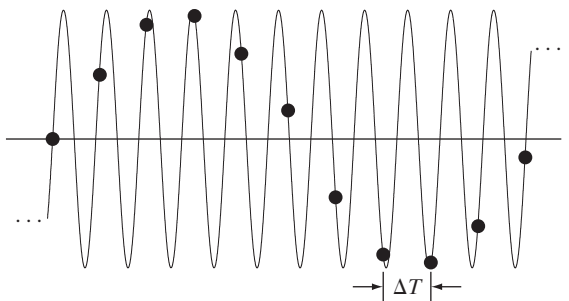


FIGURE 4.10 Illustration of aliasing. The under-sampled function (black dots) looks like a sine wave having a frequency much lower than the frequency of the continuous signal. The period of the sine wave is 2 s, so the zero crossings of the horizontal axis occur every second. ΔT is the separation between samples.

by the display medium. Therefore, it is important to understand the fundamentals of sampled data reconstruction. Convolution is central to developing this understanding, showing again the importance of this concept.

The discussion of Fig. 4.8 and Eq. (4.3-8) outlines the procedure for perfect recovery of a band-limited function from its samples using frequency domain methods. Using the convolution theorem, we can obtain the equivalent result in the spatial domain. From Eq. (4.3-8), $F(\mu) = H(\mu)\tilde{F}(\mu)$, so it follows that

$$\begin{aligned} f(t) &= \mathfrak{S}^{-1}\{F(\mu)\} \\ &= \mathfrak{S}^{-1}\{H(\mu)\tilde{F}(\mu)\} \\ &= h(t) \star \tilde{f}(t) \end{aligned} \quad (4.3-11)$$

where the last step follows from the convolution theorem, Eq. (4.2-21). It can be shown (Problem 4.6) that substituting Eq. (4.3-1) for $\tilde{f}(t)$ into Eq. (4.3-11) and then using Eq. (4.2-20) leads to the following *spatial domain* expression for $f(t)$:

$$f(t) = \sum_{n=-\infty}^{\infty} f(n\Delta T) \operatorname{sinc}\left[(t - n\Delta T)/n\Delta T\right] \quad (4.3-12)$$

where the sinc function is defined in Eq. (4.2-19). This result is not unexpected because the inverse Fourier transform of the box filter, $H(\mu)$, is a sinc function (see Example 4.1). Equation (4.3-12) shows that the perfectly reconstructed function is an infinite sum of sinc functions weighted by the sample values, and has the important property that the reconstructed function is identically equal to the sample values at multiple integer increments of ΔT . That is, for any $t = k\Delta T$, where k is an integer, $f(t)$ is equal to the k th sample $f(k\Delta T)$. This follows from Eq. (4.3-12) because $\operatorname{sinc}(0) = 1$ and $\operatorname{sinc}(m) = 0$ for any other integer value of m . Between sample points, values of $f(t)$ are *interpolations* formed by the sum of the sinc functions.

Equation (4.3-12) requires an infinite number of terms for the interpolations between samples. In practice, this implies that we have to look for approximations that are finite interpolations between samples. As we discussed in Section 2.4.4, the principal interpolation approaches used in image processing are nearest-neighbor, bilinear, and bicubic interpolation. We discuss the effects of interpolation on images in Section 4.5.4.

4.4 The Discrete Fourier Transform (DFT) of One Variable

One of the key goals of this chapter is the derivation of the *discrete Fourier transform* (DFT) starting from basic principles. The material up to this point may be viewed as the foundation of those basic principles, so now we have in place the necessary tools to derive the DFT.

4.4.1 Obtaining the DFT from the Continuous Transform of a Sampled Function

As discussed in Section 4.3.2, the Fourier transform of a sampled, band-limited function extending from $-\infty$ to ∞ is a *continuous, periodic* function that also extends from $-\infty$ to ∞ . In practice, we work with a finite number of samples, and the objective of this section is to derive the DFT corresponding to such sample sets.

Equation (4.3-5) gives the transform, $\tilde{F}(\mu)$ of sampled data in terms of the transform of the original function, but it does not give us an expression for $\tilde{F}(\mu)$ in terms of the sampled function $\tilde{f}(t)$ itself. We find such an expression directly from the definition of the Fourier transform in Eq. (4.2-16):

$$\tilde{F}(\mu) = \int_{-\infty}^{\infty} \tilde{f}(t) e^{-j2\pi\mu t} dt \quad (4.4-1)$$

By substituting Eq. (4.3-1) for $\tilde{f}(t)$, we obtain

$$\begin{aligned} \tilde{F}(\mu) &= \int_{-\infty}^{\infty} \tilde{f}(t) e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(t) \delta(t - n\Delta T) e^{-j2\pi\mu t} dt \\ &= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \delta(t - n\Delta T) e^{-j2\pi\mu t} dt \\ &= \sum_{n=-\infty}^{\infty} f_n e^{-j2\pi\mu n\Delta T} \end{aligned} \quad (4.4-2)$$

where the last step follows from Eq. (4.3-2). Although f_n is a discrete function, its Fourier $\tilde{F}(\mu)$ is continuous and infinitely periodic with period $1/\Delta T$, as we know from Eq. (4.3-5). Therefore, all we need to characterize $\tilde{F}(\mu)$ is one period, and sampling one period is the basis for the DFT.

Suppose that we want to obtain M equally spaced samples of $\tilde{F}(\mu)$ taken over the period $\mu = 0$ to $\mu = 1/\Delta T$. This is accomplished by taking the samples at the following frequencies:

$$\mu = \frac{m}{M\Delta T} \quad m = 0, 1, 2, \dots, M-1 \quad (4.4-3)$$

Substituting this result for μ into Eq. (4.4-2) and letting F_m denote the result yields

$$F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M} \quad m = 0, 1, 2, \dots, M-1 \quad (4.4-4)$$

This expression is the discrete Fourier transform we are seeking.[†] Given a set $\{f_n\}$ consisting of M samples of $f(t)$, Eq. (4.4-4) yields a sample set $\{F_m\}$ of M complex discrete values corresponding to the discrete Fourier transform of the input sample set. Conversely, given $\{F_m\}$, we can recover the sample set $\{f_n\}$ by using the *inverse discrete Fourier transform* (IDFT)

$$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m e^{j2\pi mn/M} \quad n = 0, 1, 2, \dots, M-1 \quad (4.4-5)$$

It is not difficult to show (Problem 4.8) that substituting Eq. (4.4-5) for f_n into Eq. (4.4-4) gives the identity $F_m \equiv F_m$. Similarly, substituting Eq. (4.4-4) into Eq. (4.4-5) for F_m yields $f_n \equiv f_n$. This implies that Eqs. (4.4-4) and (4.4-5) constitute a *discrete Fourier transform pair*. Furthermore, these identities indicate that the forward and inverse Fourier transforms exist for any set of samples whose values are finite. Note that neither expression depends explicitly on the sampling interval ΔT nor on the frequency intervals of Eq. (4.4-3). Therefore, the DFT pair is applicable to *any* finite set of discrete samples taken uniformly.

We used m and n in the preceding development to denote discrete variables because it is typical to do so for derivations. However, it is more intuitive, especially in two dimensions, to use the notation x and y for image coordinate variables and u and v for frequency variables, where these are understood to be *integers*.[‡] Then, Eqs. (4.4-4) and (4.4-5) become

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1 \quad (4.4-6)$$

and

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \dots, M-1 \quad (4.4-7)$$

where we used functional notation instead of subscripts for simplicity. Clearly, $F(u) \equiv F_m$ and $f(x) \equiv f_n$. From this point on, we use Eqs. (4.4-6) and (4.4-7) to denote the 1-D DFT pair. Some authors include the $1/M$ term in Eq. (4.4-6) instead of the way we show it in Eq. (4.4-7). That does not affect the proof that the two equations form a Fourier transform pair.

[†]Note from Fig. 4.6(b) that the interval $[0, 1/\Delta T]$ covers two back-to-back *half* periods of the transform. This means that the data in F_m requires re-ordering to obtain samples that are ordered from the lowest to the highest frequency of a period. This is the price paid for the notational convenience of taking the samples at $m = 0, 1, \dots, M-1$, instead of using samples on either side of the origin, which would require the use of negative notation. The procedure to order the transform data is discussed in Section 4.6.3.

[‡]We have been careful in using t for *continuous* spatial variables and μ for the corresponding *continuous* frequency variable. From this point on, we will use x and u to denote one-dimensional *discrete* spatial and frequency variables, respectively. When dealing with two-dimensional functions, we will use (t, z) and (μ, ν) to denote *continuous* spatial and frequency domain variables, respectively. Similarly, we will use (x, y) and (u, v) to denote their *discrete* counterparts.

It can be shown (Problem 4.9) that both the forward and inverse discrete transforms are infinitely periodic, with period M . That is,

$$F(u) = F(u + kM) \quad (4.4-8)$$

and

$$f(x) = f(x + kM) \quad (4.4-9)$$

where k is an integer.

The discrete equivalent of the convolution in Eq. (4.2-20) is

$$f(x) \star h(x) = \sum_{m=0}^{M-1} f(m)h(x-m) \quad (4.4-10)$$

for $x = 0, 1, 2, \dots, M-1$. Because in the preceding formulations the functions are periodic, their convolution also is periodic. Equation (4.4-10) gives one period of the periodic convolution. For this reason, the process inherent in this equation often is referred to as *circular convolution*, and is a direct result of the periodicity of the DFT and its inverse. This is in contrast with the convolution you studied in Section 3.4.2, in which values of the displacement, x , were determined by the requirement of sliding one function completely past the other, and were not fixed to the range $[0, M-1]$ as in circular convolution. We discuss this difference and its significance in Section 4.6.3 and in Fig. 4.28.

Finally, we point out that the convolution theorem given in Eqs. (4.2-21) and (4.2-22) is applicable also to discrete variables (Problem 4.10).

It is not obvious why the discrete function $f(x)$ should be periodic, considering that the continuous function from which it was sampled may not be. One informal way to reason this out is to keep in mind that sampling results in a periodic DFT. It is logical that $f(x)$, which is the inverse DFT, has to be periodic also for the DFT pair to exist.

4.4.2 Relationship Between the Sampling and Frequency Intervals

If $f(x)$ consists of M samples of a function $f(t)$ taken ΔT units apart, the duration of the record comprising the set $\{f(x)\}$, $x = 0, 1, 2, \dots, M-1$, is

$$T = M\Delta T \quad (4.4-11)$$

The corresponding spacing, Δu , in the discrete frequency domain follows from Eq. (4.4-3):

$$\Delta u = \frac{1}{M\Delta T} = \frac{1}{T} \quad (4.4-12)$$

The entire frequency range spanned by the M components of the DFT is

$$\Omega = M\Delta u = \frac{1}{\Delta T} \quad (4.4-13)$$

Thus, we see from Eqs. (4.4-12) and (4.4-13) that that the resolution in frequency, Δu , of the DFT depends on the duration T over which the continuous function, $f(t)$, is sampled, and the range of frequencies spanned by the DFT depends on the sampling interval ΔT . Observe that both expressions exhibit *inverse* relationships with respect to T and ΔT .

EXAMPLE 4.4:
The mechanics of
computing the
DFT.

■ Figure 4.11(a) shows four samples of a continuous function, $f(t)$, taken ΔT units apart. Figure 4.11(b) shows the sampled values in the x -domain. Note that the values of x are 0, 1, 2, and 3, indicating that we could be referring to any four samples of $f(t)$.

From Eq. (4.4-6),

$$\begin{aligned} F(0) &= \sum_{x=0}^3 f(x) = [f(0) + f(1) + f(2) + f(3)] \\ &= 1 + 2 + 4 + 4 = 11 \end{aligned}$$

The next value of $F(u)$ is

$$\begin{aligned} F(1) &= \sum_{x=0}^3 f(x) e^{-j2\pi(1)x/4} \\ &= 1e^0 + 2e^{-j\pi/2} + 4e^{-j\pi} + 4e^{-j3\pi/2} = -3 + 2j \end{aligned}$$

Similarly, $F(2) = -(1 + 0j)$ and $F(3) = -(3 + 2j)$. Observe that *all* values of $f(x)$ are used in computing *each* term of $F(u)$.

If instead we were given $F(u)$ and were asked to compute its inverse, we would proceed in the same manner, but using the inverse transform. For instance,

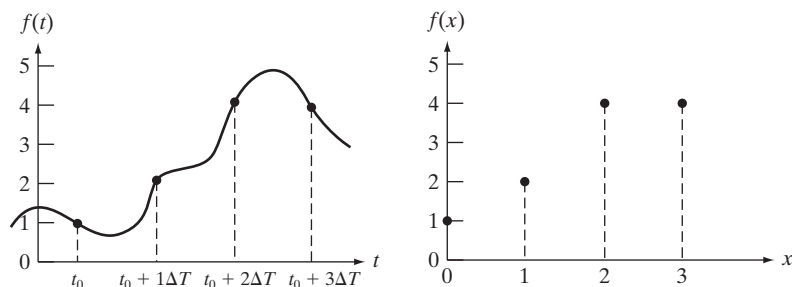
$$\begin{aligned} f(0) &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j2\pi u(0)} \\ &= \frac{1}{4} \sum_{u=0}^3 F(u) \\ &= \frac{1}{4} [11 - 3 + 2j - 1 - 3 - 2j] \\ &= \frac{1}{4} [4] = 1 \end{aligned}$$

which agrees with Fig. 4.11(b). The other values of $f(x)$ are obtained in a similar manner. ■

a b

FIGURE 4.11

(a) A function, and (b) samples in the x -domain. In (a), t is a continuous variable; in (b), x represents integer values.



4.5 Extension to Functions of Two Variables

In this section, we extend to two variables the concepts introduced in Sections 4.2 through 4.4.

4.5.1 The 2-D Impulse and Its Sifting Property

The impulse, $\delta(t, z)$, of two continuous variables, t and z , is defined as in Eq. (4.2-8):

$$\delta(t, z) = \begin{cases} \infty & \text{if } t = z = 0 \\ 0 & \text{otherwise} \end{cases} \quad (4.5-1a)$$

and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t, z) dt dz = 1 \quad (4.5-1b)$$

As in the 1-D case, the 2-D impulse exhibits the *sifting property* under integration,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \delta(t, z) dt dz = f(0, 0) \quad (4.5-2)$$

or, more generally for an impulse located at coordinates (t_0, z_0) ,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) \delta(t - t_0, z - z_0) dt dz = f(t_0, z_0) \quad (4.5-3)$$

As before, we see that the sifting property yields the value of the function $f(t, z)$ at the location of the impulse.

For discrete variables x and y , the 2-D discrete impulse is defined as

$$\delta(x, y) = \begin{cases} 1 & \text{if } x = y = 0 \\ 0 & \text{otherwise} \end{cases} \quad (4.5-4)$$

and its sifting property is

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \delta(x, y) = f(0, 0) \quad (4.5-5)$$

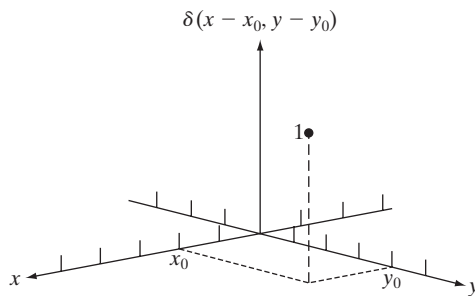
where $f(x, y)$ is a function of discrete variables x and y . For an impulse located at coordinates (x_0, y_0) (see Fig. 4.12) the sifting property is

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \delta(x - x_0, y - y_0) = f(x_0, y_0) \quad (4.5-6)$$

As before, the sifting property of a discrete impulse yields the value of the discrete function $f(x, y)$ at the location of the impulse.

FIGURE 4.12

Two-dimensional unit discrete impulse. Variables x and y are discrete, and δ is zero everywhere except at coordinates (x_0, y_0) .



4.5.2 The 2-D Continuous Fourier Transform Pair

Let $f(t, z)$ be a continuous function of two continuous variables, t and z . The two-dimensional, continuous Fourier transform pair is given by the expressions

$$F(\mu, \nu) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz \quad (4.5-7)$$

and

$$f(t, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mu, \nu) e^{j2\pi(\mu t + \nu z)} d\mu d\nu \quad (4.5-8)$$

where μ and ν are the frequency variables. When referring to images, t and z are interpreted to be continuous *spatial* variables. As in the 1-D case, the domain of the variables μ and ν defines the *continuous frequency domain*.

EXAMPLE 4.5: Obtaining the 2-D Fourier transform of a simple function.

■ Figure 4.13(a) shows a 2-D function analogous to the 1-D case in Example 4.1. Following a procedure similar to the one used in that example gives the result

$$\begin{aligned} F(\mu, \nu) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz \\ &= \int_{-T/2}^{T/2} \int_{-Z/2}^{Z/2} A e^{-j2\pi(\mu t + \nu z)} dt dz \\ &= ATZ \left[\frac{\sin(\pi\mu T)}{(\pi\mu T)} \right] \left[\frac{\sin(\pi\nu Z)}{(\pi\nu Z)} \right] \end{aligned}$$

The magnitude (spectrum) is given by the expression

$$|F(\mu, \nu)| = ATZ \left| \frac{\sin(\pi\mu T)}{(\pi\mu T)} \right| \left| \frac{\sin(\pi\nu Z)}{(\pi\nu Z)} \right|$$

Figure 4.13(b) shows a portion of the spectrum about the origin. As in the 1-D case, the locations of the zeros in the spectrum are inversely proportional to

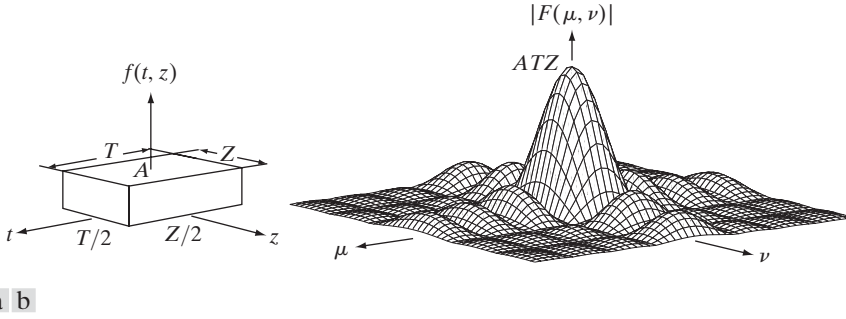


FIGURE 4.13 (a) A 2-D function, and (b) a section of its spectrum (not to scale). The block is longer along the t -axis, so the spectrum is more “contracted” along the μ -axis. Compare with Fig. 4.4.

the values of T and Z . Thus, the larger T and Z are, the more “contracted” the spectrum will become, and vice versa. ■

4.5.3 Two-Dimensional Sampling and the 2-D Sampling Theorem

In a manner similar to the 1-D case, sampling in two dimensions can be modeled using the sampling function (2-D impulse train):

$$s_{\Delta T \Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z) \quad (4.5-9)$$

where ΔT and ΔZ are the separations between samples along the t - and z -axis of the continuous function $f(t, z)$. Equation (4.5-9) describes a set of periodic impulses extending infinitely along the two axes (Fig. 4.14). As in the 1-D case illustrated in Fig. 4.5, multiplying $f(t, z)$ by $s_{\Delta T \Delta Z}(t, z)$ yields the sampled function.

Function $f(t, z)$ is said to be *band-limited* if its Fourier transform is 0 outside a rectangle established by the intervals $[-\mu_{\max}, \mu_{\max}]$ and $[-\nu_{\max}, \nu_{\max}]$; that is,

$$F(\mu, \nu) = 0 \quad \text{for } |\mu| \geq \mu_{\max} \text{ and } |\nu| \geq \nu_{\max} \quad (4.5-10)$$

The *two-dimensional sampling theorem* states that a continuous, band-limited function $f(t, z)$ can be recovered with no error from a set of its samples if the sampling intervals are

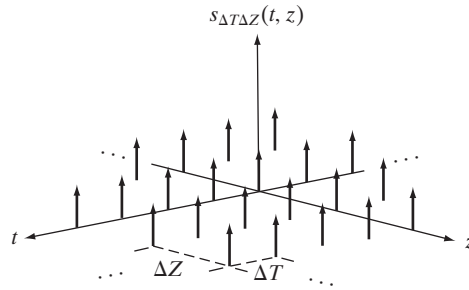
$$\Delta T < \frac{1}{2\mu_{\max}} \quad (4.5-11)$$

and

$$\Delta Z < \frac{1}{2\nu_{\max}} \quad (4.5-12)$$

or, expressed in terms of the sampling rate, if

FIGURE 4.14
Two-dimensional
impulse train.



$$\frac{1}{\Delta T} > 2\mu_{\max} \quad (4.5-13)$$

and

$$\frac{1}{\Delta Z} > 2\nu_{\max} \quad (4.5-14)$$

Stated another way, we say that no information is lost if a 2-D, band-limited, continuous function is represented by samples acquired at rates greater than twice the highest frequency content of the function in both the μ - and ν -directions.

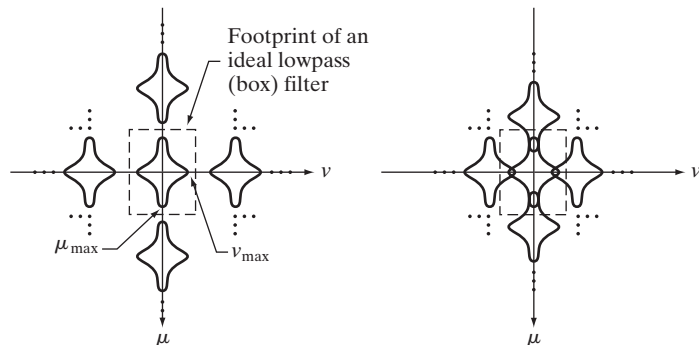
Figure 4.15 shows the 2-D equivalents of Figs. 4.6(b) and (d). A 2-D ideal box filter has the form illustrated in Fig. 4.13(a). The dashed portion of Fig. 4.15(a) shows the location of the filter to achieve the necessary isolation of a single period of the transform for reconstruction of a band-limited function from its samples, as in Section 4.3.3. From Section 4.3.4, we know that if the function is under-sampled the periods overlap, and it becomes impossible to isolate a single period, as Fig. 4.15(b) shows. Aliasing would result under such conditions.

4.5.4 Aliasing in Images

In this section, we extend the concept of aliasing to images and discuss several aspects related to image sampling and resampling.

a b

FIGURE 4.15
Two-dimensional
Fourier transforms
of (a) an over-
sampled, and
(b) under-sampled
band-limited
function.



Extension from 1-D aliasing

As in the 1-D case, a continuous function $f(t, z)$ of two continuous variables, t and z , can be band-limited in general only if it extends infinitely in both coordinate directions. The very act of limiting the duration of the function introduces corrupting frequency components extending to infinity in the frequency domain, as explained in Section 4.3.4. Because we cannot sample a function infinitely, aliasing is always present in digital images, just as it is present in sampled 1-D functions. There are two principal manifestations of aliasing in images: spatial aliasing and temporal aliasing. *Spatial aliasing* is due to under-sampling, as discussed in Section 4.3.4. *Temporal aliasing* is related to time intervals between images in a sequence of images. One of the most common examples of temporal aliasing is the “wagon wheel” effect, in which wheels with spokes in a sequence of images (for example, in a movie) appear to be rotating backwards. This is caused by the frame rate being too low with respect to the speed of wheel rotation in the sequence.

Our focus in this chapter is on spatial aliasing. The key concerns with spatial aliasing in images are the introduction of artifacts such as jaggedness in line features, spurious highlights, and the appearance of frequency patterns not present in the original image. The following example illustrates aliasing in images.

■ Suppose that we have an imaging system that is perfect, in the sense that it is noiseless and produces an exact digital image of what it sees, but the number of samples it can take is fixed at 96×96 pixels. If we use this system to digitize checkerboard patterns, it will be able to resolve patterns that are up to 96×96 squares, in which the size of each square is 1×1 pixels. In this limiting case, each pixel in the resulting image will correspond to one square in the pattern. We are interested in examining what happens when the detail (the size of the checkerboard squares) is less than one camera pixel; that is, when the imaging system is asked to digitize checkerboard patterns that have more than 96×96 squares in the field of view.

Figures 4.16(a) and (b) show the result of sampling checkerboards whose squares are of size 16 and 6 pixels on the side, respectively. These results are as expected. However, when the size of the squares is reduced to slightly less than one camera pixel a severely aliased image results, as Fig. 4.16(c) shows. Finally, reducing the size of the squares to slightly less than 0.5 pixels on the side yielded the image in Fig. 4.16(d). In this case, the aliased result looks like a normal checkerboard pattern. In fact, this image would result from sampling a checkerboard image whose squares were 12 pixels on the side. This last image is a good reminder that aliasing can create results that may be quite misleading. ■

EXAMPLE 4.6: Aliasing in images.

This example should not be construed as being unrealistic. Sampling a “perfect” scene under noiseless, distortion-free conditions is common when converting computer-generated models and vector drawings to digital images.

The effects of aliasing can be *reduced* by slightly defocusing the scene to be digitized so that high frequencies are attenuated. As explained in Section 4.3.4, anti-aliasing filtering has to be done at the “front-end,” *before* the image is sampled. There are no such things as after-the-fact software anti-aliasing filters that can be used to reduce the effects of aliasing caused by violations of the sampling theorem. Most commercial digital image manipulation packages do have a feature called “anti-aliasing.” However, as illustrated in Examples 4.7

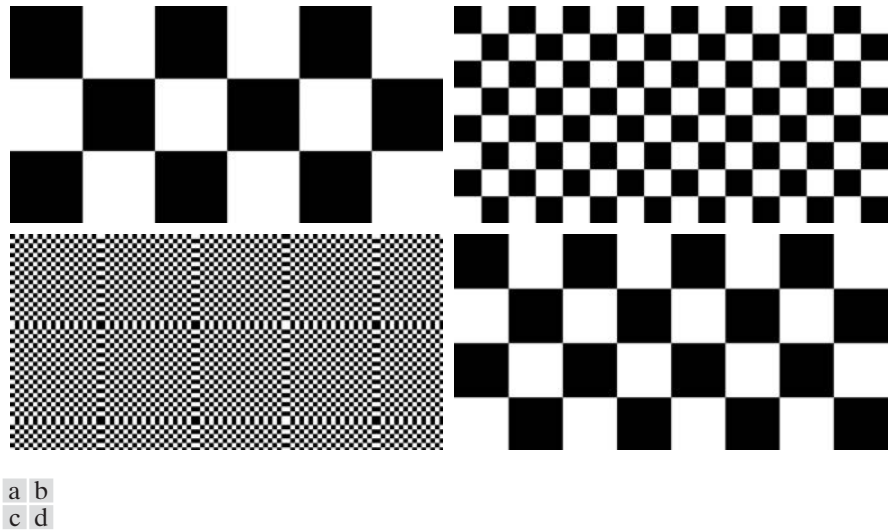


FIGURE 4.16 Aliasing in images. In (a) and (b), the lengths of the sides of the squares are 16 and 6 pixels, respectively, and aliasing is visually negligible. In (c) and (d), the sides of the squares are 0.9174 and 0.4798 pixels, respectively, and the results show significant aliasing. Note that (d) masquerades as a “normal” image.

and 4.8, this term is related to blurring a *digital* image to reduce additional aliasing artifacts caused by resampling. The term does not apply to reducing aliasing in the original sampled image. A significant number of commercial digital cameras have true anti-aliasing filtering built in, either in the lens or on the surface of the sensor itself. For this reason, it is difficult to illustrate aliasing using images obtained with such cameras.

Image interpolation and resampling

As in the 1-D case, perfect reconstruction of a band-limited image function from a set of its samples requires 2-D convolution in the spatial domain with a sinc function. As explained in Section 4.3.5, this theoretically perfect reconstruction requires interpolation using infinite summations which, in practice, forces us to look for approximations. One of the most common applications of 2-D interpolation in image processing is in image resizing (zooming and shrinking). Zooming may be viewed as over-sampling, while shrinking may be viewed as under-sampling. The key difference between these two operations and the sampling concepts discussed in previous sections is that zooming and shrinking are applied to *digital* images.

Interpolation was explained in Section 2.4.4. Our interest there was to illustrate the performance of nearest neighbor, bilinear, and bicubic interpolation. In this section, we give some additional examples with a focus on sampling and anti-aliasing issues. A special case of nearest neighbor interpolation that ties in nicely with over-sampling is zooming by *pixel replication*, which is applicable when we want to increase the size of an image an integer number of times. For

instance, to double the size of an image, we duplicate each column. This doubles the image size in the horizontal direction. Then, we duplicate each row of the enlarged image to double the size in the vertical direction. The same procedure is used to enlarge the image any integer number of times. The intensity-level assignment of each pixel is predetermined by the fact that new locations are exact duplicates of old locations.

Image shrinking is done in a manner similar to zooming. Under-sampling is achieved by row-column deletion (e.g., to shrink an image by one-half, we delete every other row and column). We can use the zooming grid analogy in Section 2.4.4 to visualize the concept of shrinking by a non-integer factor, except that we now expand the grid to fit over the original image, do intensity-level interpolation, and then shrink the grid back to its specified size. To reduce aliasing, it is a good idea to blur an image slightly before shrinking it (we discuss frequency domain blurring in Section 4.8). An alternate technique is to *super-sample* the original scene and then reduce (resample) its size by row and column deletion. This can yield sharper results than with smoothing, but it clearly requires access to the original scene. Clearly, if we have no access to the original scene (as typically is the case in practice) super-sampling is not an option.

The process of resampling an image without using band-limiting blurring is called *decimation*.

■ The effects of aliasing generally are worsened when the size of a digital image is reduced. Figure 4.17(a) is an image purposely created to illustrate the effects of aliasing (note the thinly-spaced parallel lines in all garments worn by the subject). There are no objectionable artifacts in Fig. 4.17(a), indicating that

EXAMPLE 4.7: Illustration of aliasing in resampled images.



a b c

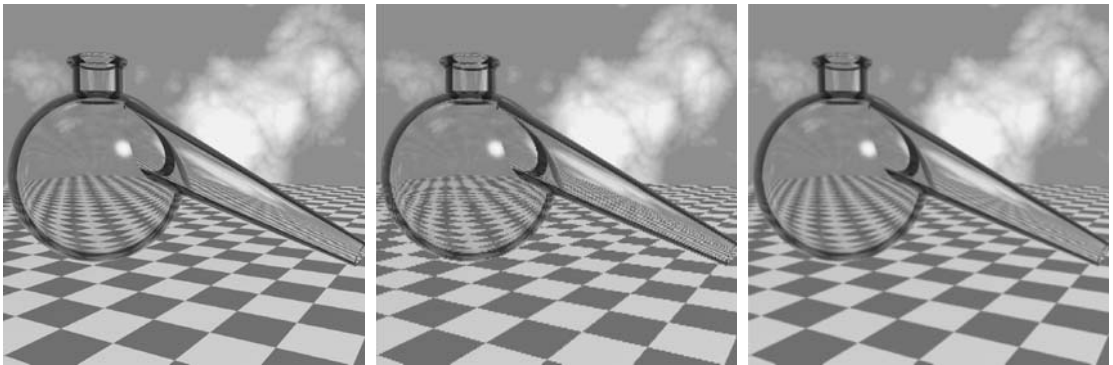
FIGURE 4.17 Illustration of aliasing on resampled images. (a) A digital image with negligible visual aliasing. (b) Result of resizing the image to 50% of its original size by pixel deletion. Aliasing is clearly visible. (c) Result of blurring the image in (a) with a 3×3 averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

the sampling rate used initially was sufficient to avoid visible aliasing. In Fig. 4.17(b), the image was reduced to 50% of its original size using row-column deletion. The effects of aliasing are quite visible in this image (see, for example the areas around the subject's knees). The digital “equivalent” of anti-aliasing filtering of continuous images is to attenuate the high frequencies of a digital image by smoothing it before resampling. Figure 4.17(c) shows the result of smoothing the image in Fig. 4.17(a) with a 3×3 averaging filter (see Section 3.5) before reducing its size. The improvement over Fig. 4.17(b) is evident. Images (b) and (c) were resized up to their original dimension by pixel replication to simplify comparisons. ■

When you work with images that have strong edge content, the effects of aliasing are seen as block-like image components, called *jaggies*. The following example illustrates this phenomenon.

EXAMPLE 4.8:
Illustration of
jaggies in image
shrinking.

■ Figure 4.18(a) shows a 1024×1024 digital image of a computer-generated scene in which aliasing is negligible. Figure 4.18(b) is the result of reducing the size of (a) by 75% to 256×256 pixels using bilinear interpolation and then using pixel replication to bring the image back to its original size in order to make the effects of aliasing (jaggies in this case) more visible. As in Example 4.7, the effects of aliasing can be made less objectionable by smoothing the image before resampling. Figure 4.18(c) is the result of using a 5×5 averaging filter prior to reducing the size of the image. As this figure shows, jaggies were reduced significantly. The size reduction and increase to the original size in Fig. 4.18(c) were done using the same approach used to generate Fig. 4.18(b). ■



a b c

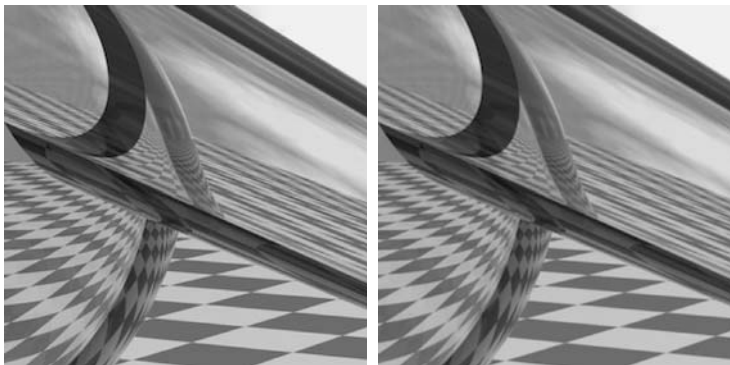
FIGURE 4.18 Illustration of jaggies. (a) A 1024×1024 digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a 5×5 averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)

■ In the previous two examples, we used pixel replication to zoom the small resampled images. This is not a preferred approach in general, as Fig. 4.19 illustrates. Figure 4.19(a) shows a 1024×1024 zoomed image generated by pixel replication from a 256×256 section out of the center of the image in Fig. 4.18(a). Note the “blocky” edges. The zoomed image in Fig. 4.19(b) was generated from the same 256×256 section, but using bilinear interpolation. The edges in this result are considerably smoother. For example, the edges of the bottle neck and the large checkerboard squares are not nearly as blocky in (b) as they are in (a).

EXAMPLE 4.9: Illustration of jaggies in image zooming.

Moiré patterns

Before leaving this section, we examine another type of artifact, called *moiré patterns*,[†] that sometimes result from sampling scenes with periodic or nearly periodic components. In optics, moiré patterns refer to beat patterns produced between two gratings of approximately equal spacing. These patterns are a common everyday occurrence. We see them, for example, in overlapping insect window screens and on the interference between TV raster lines and striped materials. In digital image processing, the problem arises routinely when scanning media print, such as newspapers and magazines, or in images with periodic components whose spacing is comparable to the spacing between samples. It is important to note that moiré patterns are more general than sampling artifacts. For instance, Fig. 4.20 shows the moiré effect using ink drawings that have not been digitized. Separately, the patterns are clean and void of interference. However, superimposing one pattern on the other creates



a b

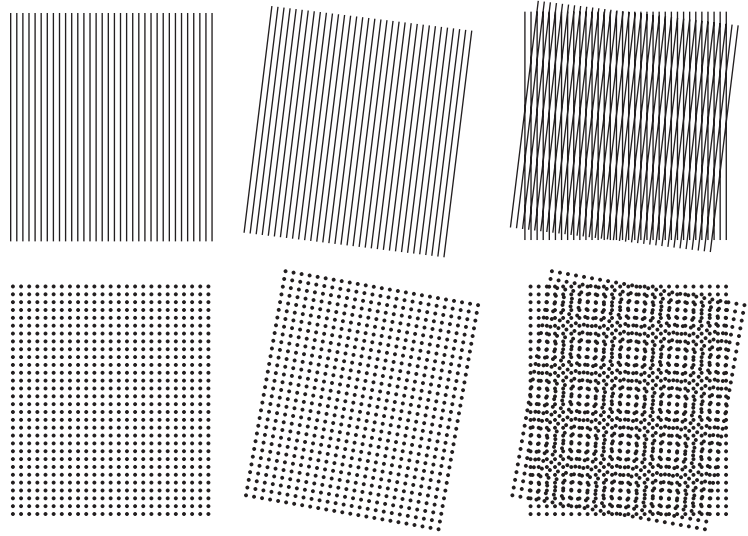
FIGURE 4.19 Image zooming. (a) A 1024×1024 digital image generated by pixel replication from a 256×256 image extracted from the middle of Fig. 4.18(a). (b) Image generated using bi-linear interpolation, showing a significant reduction in jaggies.

[†]The term *moiré* is a French word (not the name of a person) that appears to have originated with weavers who first noticed interference patterns visible on some fabrics; the term is rooted on the word *mohair*, a cloth made from Angola goat hairs.

a	b	c
d	e	f

FIGURE 4.20

Examples of the moiré effect. These are ink drawings, not digitized patterns. Superimposing one pattern on the other is equivalent mathematically to multiplying the patterns.



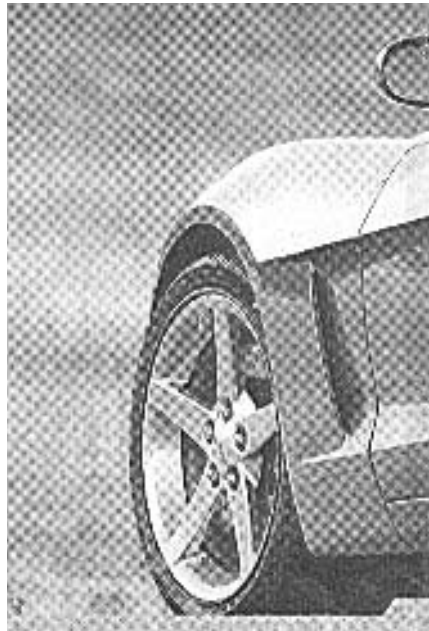
Color printing uses red, green, and blue dots to produce the sensation in the eye of continuous color.

a beat pattern that has frequencies not present in either of the original patterns. Note in particular the moiré effect produced by two patterns of dots, as this is the effect of interest in the following discussion.

Newspapers and other printed materials make use of so called *halftone dots*, which are black dots or ellipses whose sizes and various joining schemes are used to simulate gray tones. As a rule, the following numbers are typical: newspapers are printed using 75 halftone dots per inch (*dpi* for short), magazines use 133 dpi, and high-quality brochures use 175 dpi. Figure 4.21 shows

FIGURE 4.21

A newspaper image of size 246×168 pixels sampled at 75 dpi showing a moiré pattern. The moiré pattern in this image is the interference pattern created between the $\pm 45^\circ$ orientation of the halftone dots and the north-south orientation of the sampling grid used to digitize the image.



what happens when a newspaper image is sampled at 75 dpi. The sampling lattice (which is oriented vertically and horizontally) and dot patterns on the newspaper image (oriented at $\pm 45^\circ$) interact to create a uniform moiré pattern that makes the image look blotchy. (We discuss a technique in Section 4.10.2 for reducing moiré interference patterns.)

As a related point of interest, Fig. 4.22 shows a newspaper image sampled at 400 dpi to avoid moiré effects. The enlargement of the region surrounding the subject's left eye illustrates how halftone dots are used to create shades of gray. The dot size is inversely proportional to image intensity. In light areas, the dots are small or totally absent (see, for example, the white part of the eye). In light gray areas, the dots are larger, as shown below the eye. In darker areas, when dot size exceeds a specified value (typically 50%), dots are allowed to join along two specified directions to form an interconnected mesh (see, for example, the left part of the eye). In some cases the dots join along only one direction, as in the top right area below the eyebrow.

4.5.5 The 2-D Discrete Fourier Transform and Its Inverse

A development similar to the material in Sections 4.3 and 4.4 would yield the following 2-D *discrete Fourier transform* (DFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad (4.5-15)$$

where $f(x, y)$ is a digital image of size $M \times N$. As in the 1-D case, Eq. (4.5-15) must be evaluated for values of the discrete variables u and v in the ranges $u = 0, 1, 2, \dots, M-1$ and $v = 0, 1, 2, \dots, N-1$.[†]

Sometimes you will find in the literature the $1/MN$ constant in front of DFT instead of the IDFT. At times, the constant is expressed as $1/\sqrt{MN}$ and is included in front of the forward and inverse transforms, thus creating a more symmetric pair. Any of these formulations is correct, provided that you are consistent.

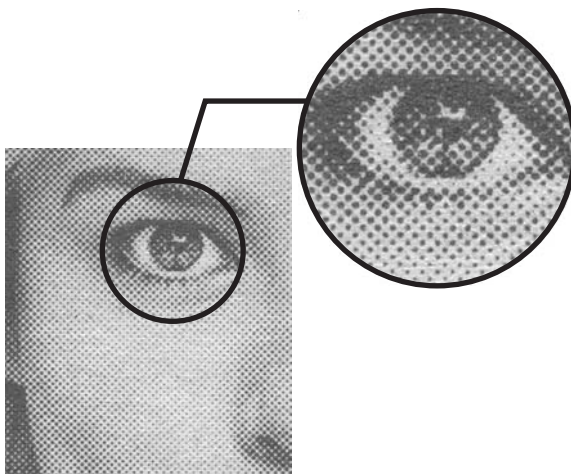


FIGURE 4.22

A newspaper image and an enlargement showing how halftone dots are arranged to render shades of gray.

[†]As mentioned in Section 4.4.1, keep in mind that in this chapter we use (t, z) and (μ, ν) to denote 2-D *continuous* spatial and frequency-domain variables. In the 2-D *discrete* case, we use (x, y) for spatial variables and (u, v) for frequency-domain variables.

Given the transform $F(u, v)$, we can obtain $f(x, y)$ by using the *inverse discrete Fourier transform* (IDFT):

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)} \quad (4.5-16)$$

for $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$. Equations (4.5-15) and (4.5-16) constitute the 2-D *discrete Fourier transform pair*. The rest of this chapter is based on properties of these two equations and their use for image filtering in the frequency domain.

4.6 Some Properties of the 2-D Discrete Fourier Transform

In this section, we introduce several properties of the 2-D discrete Fourier transform and its inverse.

4.6.1 Relationships Between Spatial and Frequency Intervals

The relationships between spatial sampling and the corresponding frequency-domain intervals are as explained in Section 4.4.2. Suppose that a continuous function $f(t, z)$ is sampled to form a digital image, $f(x, y)$, consisting of $M \times N$ samples taken in the t - and z -directions, respectively. Let ΔT and ΔZ denote the separations between samples (see Fig. 4.14). Then, the separations between the corresponding discrete, frequency domain variables are given by

$$\Delta u = \frac{1}{M\Delta T} \quad (4.6-1)$$

and

$$\Delta v = \frac{1}{N\Delta Z} \quad (4.6-2)$$

respectively. Note that the separations between samples in the frequency domain are inversely proportional both to the spacing between spatial samples and the number of samples.

4.6.2 Translation and Rotation

It can be shown by direct substitution into Eqs. (4.5-15) and (4.5-16) that the Fourier transform pair satisfies the following translation properties (Problem 4.16):

$$f(x, y) e^{j2\pi(u_0 x/M + v_0 y/N)} \Leftrightarrow F(u - u_0, v - v_0) \quad (4.6-3)$$

and

$$f(x - x_0, y - y_0) \Leftrightarrow F(u, v) e^{-j2\pi(x_0 u/M + y_0 v/N)} \quad (4.6-4)$$

That is, multiplying $f(x, y)$ by the exponential shown shifts the origin of the DFT to (u_0, v_0) and, conversely, multiplying $F(u, v)$ by the negative of that exponential shifts the origin of $f(x, y)$ to (x_0, y_0) . As we illustrate in Example 4.13, translation has no effect on the magnitude (spectrum) of $F(u, v)$.

Using the polar coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$$

results in the following transform pair:

$$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0) \quad (4.6-5)$$

which indicates that rotating $f(x, y)$ by an angle θ_0 rotates $F(u, v)$ by the same angle. Conversely, rotating $F(u, v)$ rotates $f(x, y)$ by the same angle.

4.6.3 Periodicity

As in the 1-D case, the 2-D Fourier transform and its inverse are infinitely periodic in the u and v directions; that is,

$$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N) \quad (4.6-6)$$

and

$$f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N) \quad (4.6-7)$$

where k_1 and k_2 are integers.

The periodicities of the transform and its inverse are important issues in the implementation of DFT-based algorithms. Consider the 1-D spectrum in Fig. 4.23(a). As explained in Section 4.4.1, the transform data in the interval from 0 to $M - 1$ consists of two back-to-back half periods meeting at point $M/2$. For display and filtering purposes, it is more convenient to have in this interval a complete period of the transform in which the data are contiguous, as in Fig. 4.23(b). It follows from Eq. (4.6-3) that

$$f(x) e^{j2\pi(u_0 x/M)} \Leftrightarrow F(u - u_0)$$

In other words, multiplying $f(x)$ by the exponential term shown shifts the data so that the origin, $F(0)$, is located at u_0 . If we let $u_0 = M/2$, the exponential term becomes $e^{j\pi x}$ which is equal to $(-1)^x$ because x is an integer. In this case,

$$f(x)(-1)^x \Leftrightarrow F(u - M/2)$$

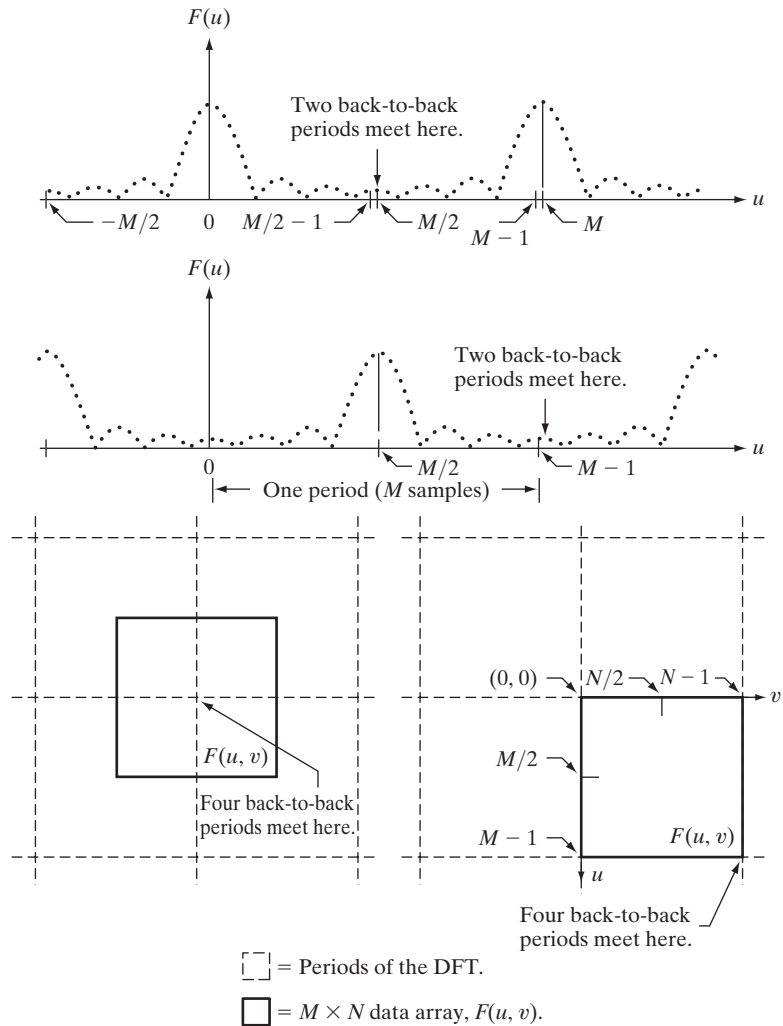
That is, multiplying $f(x)$ by $(-1)^x$ shifts the data so that $F(0)$ is at the *center* of the interval $[0, M - 1]$, which corresponds to Fig. 4.23(b), as desired.

In 2-D the situation is more difficult to graph, but the principle is the same, as Fig. 4.23(c) shows. Instead of two half periods, there are now four quarter periods meeting at the point $(M/2, N/2)$. The dashed rectangles correspond to

a
b
c d

FIGURE 4.23

Centering the Fourier transform.
(a) A 1-D DFT showing an infinite number of periods.
(b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$.
(c) A 2-D DFT showing an infinite number of periods. The solid area is the $M \times N$ data array, $F(u, v)$, obtained with Eq. (4.5-15). This array consists of four quarter periods.
(d) A Shifted DFT obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).



the infinite number of periods of the 2-D DFT. As in the 1-D case, visualization is simplified if we shift the data so that $F(0, 0)$ is at $(M/2, N/2)$. Letting $(u_0, v_0) = (M/2, N/2)$ in Eq. (4.6-3) results in the expression

$$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2) \quad (4.6-8)$$

Using this equation shifts the data so that $F(0, 0)$ is at the center of the *frequency rectangle* defined by the intervals $[0, M - 1]$ and $[0, N - 1]$, as desired. Figure 4.23(d) shows the result. We illustrate these concepts later in this section as part of Example 4.11 and Fig. 4.24.

4.6.4 Symmetry Properties

An important result from functional analysis is that any real *or* complex function, $w(x, y)$, can be expressed as the sum of an even and an odd part (*each of which can be real or complex*):

$$w(x, y) = w_e(x, y) + w_o(x, y) \quad (4.6-9)$$

where the even and odd parts are defined as

$$w_e(x, y) \triangleq \frac{w(x, y) + w(-x, -y)}{2} \quad (4.6-10a)$$

and

$$w_o(x, y) \triangleq \frac{w(x, y) - w(-x, -y)}{2} \quad (4.6-10b)$$

Substituting Eqs. (4.6-10a) and (4.6-10b) into Eq. (4.6-9) gives the identity $w(x, y) \equiv w(x, y)$, thus proving the validity of the latter equation. It follows from the preceding definitions that

$$w_e(x, y) = w_e(-x, -y) \quad (4.6-11a)$$

and that

$$w_o(x, y) = -w_o(-x, -y) \quad (4.6-11b)$$

Even functions are said to be *symmetric* and odd functions are *antisymmetric*. Because all indices in the DFT and IDFT are positive, when we talk about symmetry (antisymmetry) we are referring to symmetry (antisymmetry) about the *center point* of a sequence. In terms of Eq. (4.6-11), indices to the right of the center point of a 1-D array are considered positive, and those to the left are considered negative (similarly in 2-D). In our work, it is more convenient to think only in terms of nonnegative indices, in which case the definitions of evenness and oddness become:

$$w_e(x, y) = w_e(M - x, N - y) \quad (4.6-12a)$$

and

$$w_o(x, y) = -w_o(M - x, N - y) \quad (4.6-12b)$$

where, as usual, M and N are the number of rows and columns of a 2-D array.

To convince yourself that the samples of an odd function sum to zero, sketch one period of a 1-D sine wave about the origin or any other interval spanning one period.

We know from elementary mathematical analysis that the product of two even or two odd functions is even, and that the product of an even and an odd function is odd. In addition, the only way that a discrete function can be odd is if all its samples sum to zero. These properties lead to the important result that

$$\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} w_e(x, y) w_o(x, y) = 0 \quad (4.6-13)$$

for any two discrete even and odd functions w_e and w_o . In other words, because the argument of Eq. (4.6-13) is odd, the result of the summations is 0. The functions can be real or complex.

EXAMPLE 4.10:
Even and odd
functions.

■ Although evenness and oddness are visualized easily for continuous functions, these concepts are not as intuitive when dealing with discrete sequences. The following illustrations will help clarify the preceding ideas. Consider the 1-D sequence

$$\begin{aligned} f &= \{f(0) \quad f(1) \quad f(2) \quad f(3)\} \\ &= \{2 \quad 1 \quad 1 \quad 1\} \end{aligned}$$

in which $M = 4$. To test for evenness, the condition $f(x) = f(4 - x)$ must be satisfied; that is, we require that

$$f(0) = f(4), \quad f(2) = f(2), \quad f(1) = f(3), \quad f(3) = f(1)$$

Because $f(4)$ is outside the range being examined, and it can be any value, the value of $f(0)$ is immaterial in the test for evenness. We see that the next three conditions are satisfied by the values in the array, so the sequence is even. In fact, we conclude that *any* 4-point even sequence has to have the form

$$\{a \quad b \quad c \quad b\}$$

That is, only the second and last points must be equal in a 4-point even sequence.

An odd sequence has the interesting property that its first term, $w_0(0, 0)$, is always 0, a fact that follows directly from Eq. (4.6-10b). Consider the 1-D sequence

$$\begin{aligned} g &= \{g(0) \quad g(1) \quad g(2) \quad g(3)\} \\ &= \{0 \quad -1 \quad 0 \quad 1\} \end{aligned}$$

We easily can confirm that this is an odd sequence by noting that the terms in the sequence satisfy the condition $g(x) = -g(4 - x)$. For example, $g(1) = -g(3)$. Any 4-point odd sequence has the form

$$\{0 \quad -b \quad 0 \quad b\}$$

That is, when M is an even number, a 1-D odd sequence has the property that the points at locations 0 and $M/2$ always are zero. When M is odd, the first term still has to be 0, but the remaining terms form pairs with equal value but opposite sign.

The preceding discussion indicates that evenness and oddness of sequences depend also on the length of the sequences. For example, we already showed that the sequence $\{0 \quad -1 \quad 0 \quad 1\}$ is odd. However, the sequence $\{0 \quad -1 \quad 0 \quad 1 \quad 0\}$ is neither odd nor even, although the “basic” structure appears to be odd. This is an important issue in interpreting DFT results. We show later in this section that the DFTs of even and odd functions have some very important characteristics. Thus, it often is the case that understanding when a function is odd or even plays a key role in our ability to interpret image results based on DFTs.

The same basic considerations hold in 2-D. For example, the 6×6 2-D sequence

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

As an exercise, you should use Eq. (4.6-12b) to convince yourself that this 2-D sequence is odd.

is odd. However, adding another row and column of 0s would give a result that is neither odd nor even. Note that the inner structure of this array is a Sobel mask, as discussed in Section 3.6.4. We return to this mask in Example 4.15. ■

Armed with the preceding concepts, we can establish a number of important symmetry properties of the DFT and its inverse. A property used frequently is that the Fourier transform of a *real* function, $f(x, y)$, is *conjugate symmetric*:

$$F^*(u, v) = F(-u, -v) \quad (4.6-14)$$

If $f(x, y)$ is *imaginary*, its Fourier transform is *conjugate antisymmetric*: $F^*(-u, -v) = -F(u, v)$. The proof of Eq. (4.6-14) is as follows:

$$F^*(u, v) = \left[\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \right]^*$$

Conjugate symmetry also is called *hermitian symmetry*. The term *antihermitian* is used sometimes to refer to conjugate antisymmetry.

$$\begin{aligned}
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f^*(x, y) e^{j2\pi(ux/M + vy/N)} \\
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi([-u]x/M + [-v]y/N)} \\
&= F(-u, -v)
\end{aligned}$$

where the third step follows from the fact that $f(x, y)$ is real. A similar approach can be used to prove the conjugate antisymmetry exhibited by the transform of imaginary functions.

Table 4.1 lists symmetries and related properties of the DFT that are useful in digital image processing. Recall that the double arrows indicate Fourier transform pairs; that is, for any row in the table, the properties on the right are satisfied by the Fourier transform of the function having the properties listed on the left, and vice versa. For example, entry 5 reads: The DFT of a real function $f(x, y)$, in which (x, y) is replaced by $(-x, -y)$, is $F^*(u, v)$, where $F(u, v)$, the DFT of $f(x, y)$, is a complex function, and vice versa.

TABLE 4.1 Some symmetry properties of the 2-D DFT and its inverse. $R(u, v)$ and $I(u, v)$ are the real and imaginary parts of $F(u, v)$, respectively. The term *complex* indicates that a function has nonzero real and imaginary parts.

Spatial Domain [†]		Frequency Domain [†]
1)	$f(x, y)$ real	$\Leftrightarrow F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$\Leftrightarrow F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$\Leftrightarrow R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$\Leftrightarrow R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$\Leftrightarrow F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$\Leftrightarrow F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$\Leftrightarrow F^*(-u - v)$ complex
8)	$f(x, y)$ real and even	$\Leftrightarrow F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$\Leftrightarrow F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$\Leftrightarrow F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$\Leftrightarrow F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$\Leftrightarrow F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$\Leftrightarrow F(u, v)$ complex and odd

[†]Recall that x, y, u , and v are *discrete* (integer) variables, with x and u in the range $[0, M - 1]$, and y , and v in the range $[0, N - 1]$. To say that a complex function is *even* means that its real *and* imaginary parts are even, and similarly for an odd complex function.

■ With reference to the even and odd concepts discussed earlier and illustrated in Example 4.10, the following 1-D sequences and their transforms are short examples of the properties listed in Table 4.1. The numbers in parentheses on the right are the individual elements of $F(u)$, and similarly for $f(x)$ in the last two properties.

EXAMPLE 4.11:
1-D illustrations of properties from Table 4.1.

Property	$f(x)$	$F(u)$
3	$\{1 \ 2 \ 3 \ 4\}$	$\Leftrightarrow \{(10) \ (-2 + 2j) \ (-2) \ (-2 - 2j)\}$
4	$j\{1 \ 2 \ 3 \ 4\}$	$\Leftrightarrow \{(2.5j) \ (.5 - .5j) \ (-.5j) \ (-.5 - .5j)\}$
8	$\{2 \ 1 \ 1 \ 1\}$	$\Leftrightarrow \{(5) \ (1) \ (1) \ (1)\}$
9	$\{0 \ -1 \ 0 \ 1\}$	$\Leftrightarrow \{(0) \ (2j) \ (0) \ (-2j)\}$
10	$j\{2 \ 1 \ 1 \ 1\}$	$\Leftrightarrow \{(5j) \ (j) \ (j) \ (j)\}$
11	$j\{0 \ -1 \ 0 \ 1\}$	$\Leftrightarrow \{(0) \ (-2) \ (0) \ (2)\}$
12	$\{(4 + 4j) \ (3 + 2j) \ (0 + 2j) \ (3 + 2j)\}$	$\Leftrightarrow \{(10 + 10j) \ (4 + 2j) \ (-2 + 2j) \ (4 + 2j)\}$
13	$\{(0 + 0j) \ (1 + 1j) \ (0 + 0j) \ (-1 - j)\}$	$\Leftrightarrow \{(0 + 0j) \ (2 - 2j) \ (0 + 0j) \ (-2 + 2j)\}$

For example, in property 3 we see that a real function with elements $\{1 \ 2 \ 3 \ 4\}$ has Fourier transform whose real part, $\{10 \ -2 \ -2 \ -2\}$, is even and whose imaginary part, $\{0 \ 2 \ 0 \ -2\}$, is odd. Property 8 tells us that a real even function has a transform that is real and even also. Property 12 shows that an even complex function has a transform that is also complex and even. The other property examples are analyzed in a similar manner. ■

■ In this example, we prove several of the properties in Table 4.1 to develop familiarity with manipulating these important properties, and to establish a basis for solving some of the problems at the end of the chapter. We prove only the properties on the right given the properties on the left. The converse is proved in a manner similar to the proofs we give here.

EXAMPLE 4.12:
Proving several symmetry properties of the DFT from Table 4.1.

Consider property 3, which reads: If $f(x, y)$ is a real function, the real part of its DFT is even and the odd part is odd; similarly, if a DFT has real and imaginary parts that are even and odd, respectively, then its IDFT is a real function. We prove this property formally as follows. $F(u, v)$ is complex in general, so it can be expressed as the sum of a real and an imaginary part: $F(u, v) = R(u, v) + jI(u, v)$. Then, $F^*(u, v) = R(u, v) - jI(u, v)$. Also, $F(-u, -v) = R(-u, -v) + jI(-u, -v)$. But, as proved earlier, if $f(x, y)$ is real then $F^*(u, v) = F(-u, -v)$, which, based on the preceding two equations, means that $R(u, v) = R(-u, -v)$ and $I(u, v) = -I(-u, -v)$. In view of Eqs. (4.6-11a) and (4.6-11b), this proves that R is an even function and I is an odd function.

Next, we prove property 8. If $f(x, y)$ is real we know from property 3 that the real part of $F(u, v)$ is even, so to prove property 8 all we have to do is show that if $f(x, y)$ is real *and* even then the imaginary part of $F(u, v)$ is 0 (i.e., F is real). The steps are as follows:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

which we can write as

$$\begin{aligned}
F(u, v) &= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f_r(x, y)] e^{-j2\pi(ux/M + vy/N)} \\
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f_r(x, y)] e^{-j2\pi(ux/M)} e^{-j2\pi(vy/N)} \\
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\text{even}][\text{even} - j\text{odd}][\text{even} - j\text{odd}] \\
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\text{even}][\text{even} \cdot \text{even} - 2j\text{even} \cdot \text{odd} - \text{odd} \cdot \text{odd}] \\
&= \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\text{even} \cdot \text{even}] - 2j \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\text{even} \cdot \text{odd}] \\
&\quad - \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\text{even} \cdot \text{even}] \\
&= \text{real}
\end{aligned}$$

The fourth step follows from Euler's equation and the fact that the cos and sin are even and odd functions, respectively. We also know from property 8 that, in addition to being real, f is an even function. The only term in the penultimate line containing imaginary components is the second term, which is 0 according to Eq. (4.6-14). Thus, if f is real and even then F is real. As noted earlier, F is also even because f is real. This concludes the proof.

Finally, we prove the validity of property 6. From the definition of the DFT,

Note that we are not making a change of variable here. We are evaluating the DFT of $f(-x, -y)$, so we simply insert this function into the equation, as we would any other function.

$$\Im\{f(-x, -y)\} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(-x, -y) e^{-j2\pi(ux/M + vy/N)}$$

Because of periodicity, $f(-x, -y) = f(M - x, N - y)$. If we now define $m = M - x$ and $n = N - y$, then

$$\Im\{f(-x, -y)\} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) e^{-j2\pi(u[M-m]/M + v[N-n]/N)}$$

(To convince yourself that the summations are correct, try a 1-D transform and expand a few terms by hand.) Because $\exp[-j2\pi(\text{integer})] = 1$, it follows that

$$\begin{aligned}\Im\{f(-x, -y)\} &= \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) e^{j2\pi(um/M + vn/N)} \\ &= F(-u, -v)\end{aligned}$$

This concludes the proof. ■

4.6.5 Fourier Spectrum and Phase Angle

Because the 2-D DFT is complex in general, it can be expressed in polar form:

$$F(u, v) = |F(u, v)|e^{j\phi(u, v)} \quad (4.6-15)$$

where the magnitude

$$|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2} \quad (4.6-16)$$

is called the *Fourier* (or *frequency*) *spectrum*, and

$$\phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right] \quad (4.6-17)$$

is the *phase angle*. Recall from the discussion in Section 4.2.1 that the arctan must be computed using a four-quadrant arctangent, such as MATLAB's `atan2(Imag, Real)` function.

Finally, the *power spectrum* is defined as

$$\begin{aligned}P(u, v) &= |F(u, v)|^2 \\ &= R^2(u, v) + I^2(u, v)\end{aligned} \quad (4.6-18)$$

As before, R and I are the real and imaginary parts of $F(u, v)$ and all computations are carried out for the discrete variables $u = 0, 1, 2, \dots, M - 1$ and $v = 0, 1, 2, \dots, N - 1$. Therefore, $|F(u, v)|$, $\phi(u, v)$, and $P(u, v)$ are arrays of size $M \times N$.

The Fourier transform of a real function is conjugate symmetric [Eq. (4.6-14)], which implies that the spectrum has *even* symmetry about the origin:

$$|F(u, v)| = |F(-u, -v)| \quad (4.6-19)$$

The phase angle exhibits the following *odd* symmetry about the origin:

$$\phi(u, v) = -\phi(-u, -v) \quad (4.6-20)$$

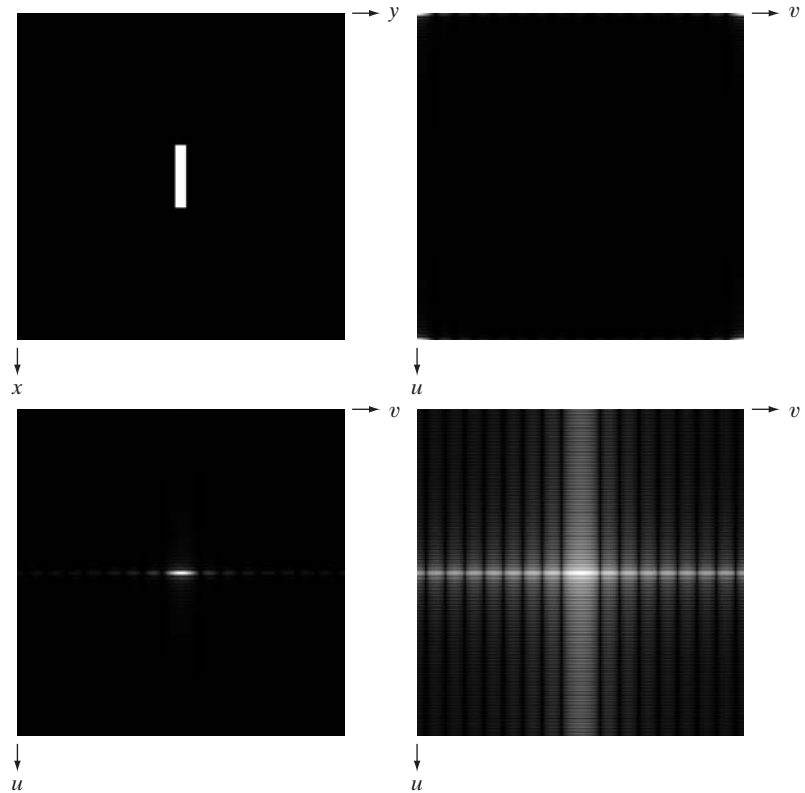
It follows from Eq. (4.5-15) that

$$F(0, 0) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

a b
c d

FIGURE 4.24

(a) Image. (b) Spectrum showing bright spots in the four corners. (c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.



which indicates that the zero-frequency term is proportional to the average value of $f(x, y)$. That is,

$$\begin{aligned} F(0, 0) &= MN \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \\ &= MN \bar{f}(x, y) \end{aligned} \quad (4.6-21)$$

where $\bar{f}$ denotes the average value of f . Then,

$$|F(0, 0)| = MN |\bar{f}(x, y)| \quad (4.6-22)$$

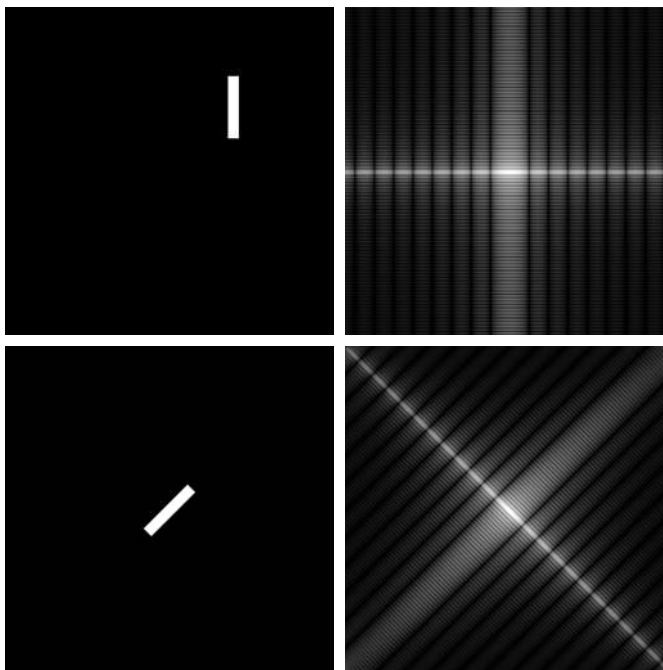
Because the proportionality constant MN usually is large, $|F(0, 0)|$ typically is the largest component of the spectrum by a factor that can be several orders of magnitude larger than other terms. Because frequency components u and v are zero at the origin, $F(0, 0)$ sometimes is called the *dc component* of the transform. This terminology is from electrical engineering, where “dc” signifies direct current (i.e., current of zero frequency).

EXAMPLE 4.13:
The 2-D Fourier
spectrum of a
simple function.

■ Figure 4.24(a) shows a simple image and Fig. 4.24(b) shows its spectrum, whose values were scaled to the range $[0, 255]$ and displayed in image form. The origins of both the spatial and frequency domains are at the top left. Two things are apparent in Fig. 4.22(b). As expected, the area around the origin of the

transform contains the highest values (and thus appears brighter in the image). However, note that the four corners of the spectrum contain similarly high values. The reason is the periodicity property discussed in the previous section. To center the spectrum, we simply multiply the image in (a) by $(-1)^{x+y}$ before computing the DFT, as indicated in Eq. (4.6-8). Figure 4.22(c) shows the result, which clearly is much easier to visualize (note the symmetry about the center point). Because the dc term dominates the values of the spectrum, the dynamic range of other intensities in the displayed image are compressed. To bring out those details, we perform a log transformation, as described in Section 3.2.2. Figure 4.24(d) shows the display of $(1 + \log|F(u, v)|)$. The increased rendition of detail is evident. Most spectra shown in this and subsequent chapters are scaled in this manner.

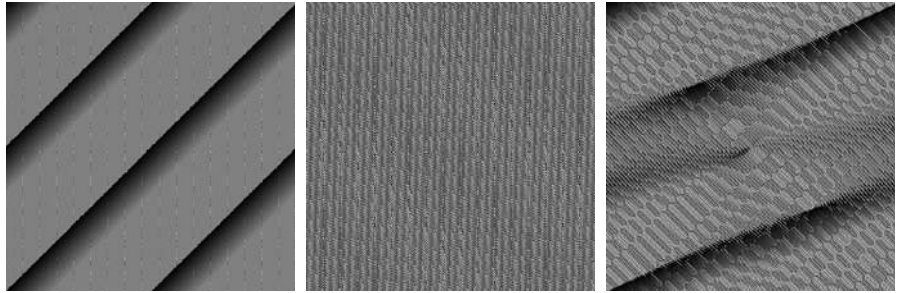
It follows from Eqs. (4.6-4) and (4.6-5) that the spectrum is insensitive to image translation (the absolute value of the exponential term is 1), but it rotates by the same angle of a rotated image. Figure 4.25 illustrates these properties. The spectrum in Fig. 4.25(b) is identical to the spectrum in Fig. 4.24(d). Clearly, the images in Figs. 4.24(a) and 4.25(a) are different, so if their Fourier spectra are the same then, based on Eq. (4.6-15), their phase angles must be different. Figure 4.26 confirms this. Figures 4.26(a) and (b) are the phase angle arrays (shown as images) of the DFTs of Figs. 4.24(a) and 4.25(a). Note the lack of similarity between the phase images, in spite of the fact that the only differences between their corresponding images is simple translation. In general, visual analysis of phase angle images yields little intuitive information. For instance, due to its 45° orientation, one would expect intuitively that the phase angle in



a b
c d

FIGURE 4.25

(a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum. (c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).



a b c

FIGURE 4.26 Phase angle array corresponding (a) to the image of the centered rectangle in Fig. 4.24(a), (b) to the translated image in Fig. 4.25(a), and (c) to the rotated image in Fig. 4.25(c).

Fig. 4.26(a) should correspond to the rotated image in Fig. 4.25(c), rather than to the image in Fig. 4.24(a). In fact, as Fig. 4.26(c) shows, the phase angle of the rotated image has a strong orientation that is much less than 45° . ■

The components of the spectrum of the DFT determine the amplitudes of the sinusoids that combine to form the resulting image. At any given frequency in the DFT of an image, a large amplitude implies a greater prominence of a sinusoid of that frequency in the image. Conversely, a small amplitude implies that less of that sinusoid is present in the image. Although, as Fig. 4.26 shows, the contribution of the phase components is less intuitive, it is just as important. The phase is a measure of displacement of the various sinusoids with respect to their origin. Thus, while the magnitude of the 2-D DFT is an array whose components determine the intensities in the image, the corresponding phase is an array of angles that carry much of the information about where discernable objects are located in the image. The following example clarifies these concepts further.

EXAMPLE 4.14: Further illustration of the properties of the Fourier spectrum and phase angle.

■ Figure 4.27(b) is the phase angle of the DFT of Fig. 4.27(a). There is no detail in this array that would lead us by visual analysis to associate it with features in its corresponding image (not even the symmetry of the phase angle is visible). However, the importance of the phase in determining shape characteristics is evident in Fig. 4.27(c), which was obtained by computing the inverse DFT of Eq. (4.6-15) using only phase information (i.e., with $|F(u, v)| = 1$ in the equation). Although the intensity information has been lost (remember, that information is carried by the spectrum) the key shape features in this image are unmistakably from Fig. 4.27(a).

Figure 4.27(d) was obtained using only the spectrum in Eq. (4.6-15) and computing the inverse DFT. This means setting the exponential term to 1, which in turn implies setting the phase angle to 0. The result is not unexpected. It contains only intensity information, with the dc term being the most dominant. There is no shape information in the image because the phase was set to zero.

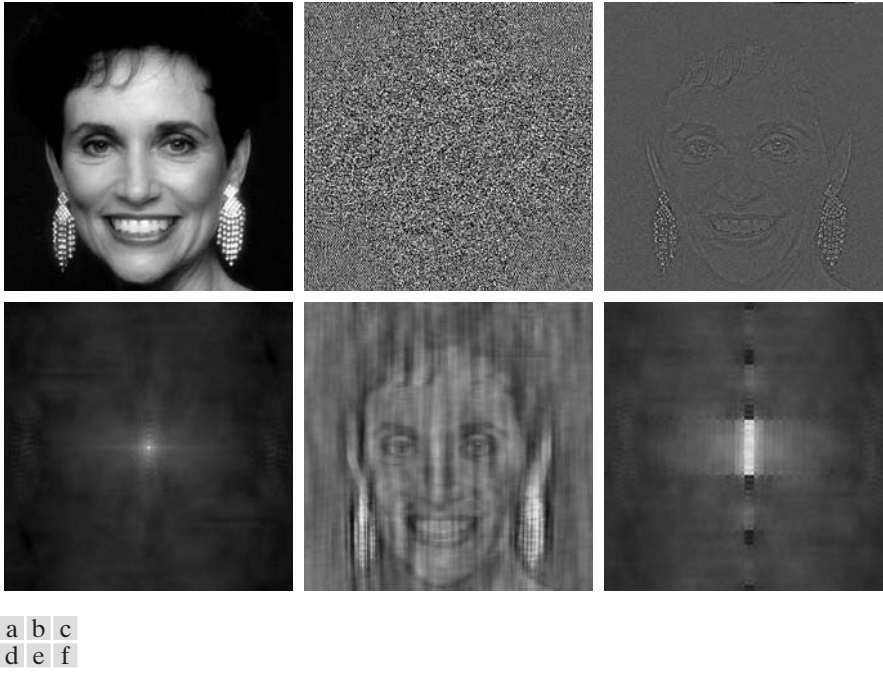


FIGURE 4.27 (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.

Finally, Figs. 4.27(e) and (f) show yet again the dominance of the phase in determining the feature content of an image. Figure 4.27(e) was obtained by computing the IDFT of Eq. (4.6-15) using the spectrum of the rectangle in Fig. 4.24(a) and the phase angle corresponding to the woman. The shape of the woman clearly dominates this result. Conversely, the rectangle dominates Fig. 4.27(f), which was computed using the spectrum of the woman and the phase angle of the rectangle. ■

4.6.6 The 2-D Convolution Theorem

Extending Eq. (4.4-10) to two variables results in the following expression for 2-D circular convolution:

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n) \quad (4.6-23)$$

for $x = 0, 1, 2, \dots, M - 1$ and $y = 0, 1, 2, \dots, N - 1$. As in Eq. (4.4-10), Eq. (4.6-23) gives one period of a 2-D periodic sequence. The 2-D convolution theorem is given by the expressions

$$f(x, y) \star h(x, y) \Leftrightarrow F(u, v) H(u, v) \quad (4.6-24)$$

and, conversely,

$$f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v) \quad (4.6-25)$$

where F and H are obtained using Eq. (4.5-15) and, as before, the double arrow is used to indicate that the left and right sides of the expressions constitute a Fourier transform pair. Our interest in the remainder of this chapter is in Eq. (4.6-24), which states that the inverse DFT of the product $F(u, v)H(u, v)$ yields $f(x, y) \star h(x, y)$, the 2-D spatial convolution of f and h . Similarly, the DFT of the spatial convolution yields the product of the transforms in the frequency domain. Equation (4.6-24) is the foundation of linear filtering and, as explained in Section 4.7, is the basis for all the filtering techniques discussed in this chapter.

Because we are dealing here with discrete quantities, computation of the Fourier transforms is carried out with a DFT algorithm. If we elect to compute the spatial convolution using the IDFT of the product of the two transforms, then the periodicity issues discussed in Section 4.6.3 must be taken into account. We give a 1-D example of this and then extend the conclusions to two variables. The left column of Fig. 4.28 implements convolution of two functions, f and h , using the 1-D equivalent of Eq. (3.4-2) which, because the two functions are of same size, is written as

$$f(x) \star h(x) = \sum_{m=0}^{399} f(x)h(x - m)$$

This equation is identical to Eq. (4.4-10), *but* the requirement on the displacement x is that it be sufficiently large to cause the flipped (rotated) version of h to slide completely past f . In other words, the procedure consists of (1) mirroring h about the origin (i.e., rotating it by 180°) [Fig. 4.28(c)], (2) translating the mirrored function by an amount x [Fig. 4.28(d)], and (3) for *each* value x of translation, computing the *entire* sum of products in the right side of the preceding equation. In terms of Fig. 4.28 this means multiplying the function in Fig. 4.28(a) by the function in Fig. 4.28(d) for *each* value of x . The displacement x ranges over all values required to completely slide h across f . Figure 4.28(e) shows the convolution of these two functions. Note that convolution is a function of the displacement variable, x , and that the range of x required in this example to completely slide h past f is from 0 to 799.

If we use the DFT and the convolution theorem to obtain the same result as in the left column of Fig. 4.28, we must take into account the periodicity inherent in the expression for the DFT. This is equivalent to convolving the two periodic functions in Figs. 4.28(f) and (g). The convolution procedure is the same as we just discussed, but the two functions now are periodic. Proceeding with these two functions as in the previous paragraph would yield the result in Fig. 4.28(j) which obviously is incorrect. Because we are convolving two periodic functions, the convolution itself is periodic. The closeness of the periods in Fig. 4.28 is such that they interfere with each other to cause what is commonly referred to as *wraparound error*. According to the convolution theorem, if we had computed the DFT of the two 400-point functions, f and h , multiplied the

We discuss efficient ways to compute the DFT in Section 4.11.

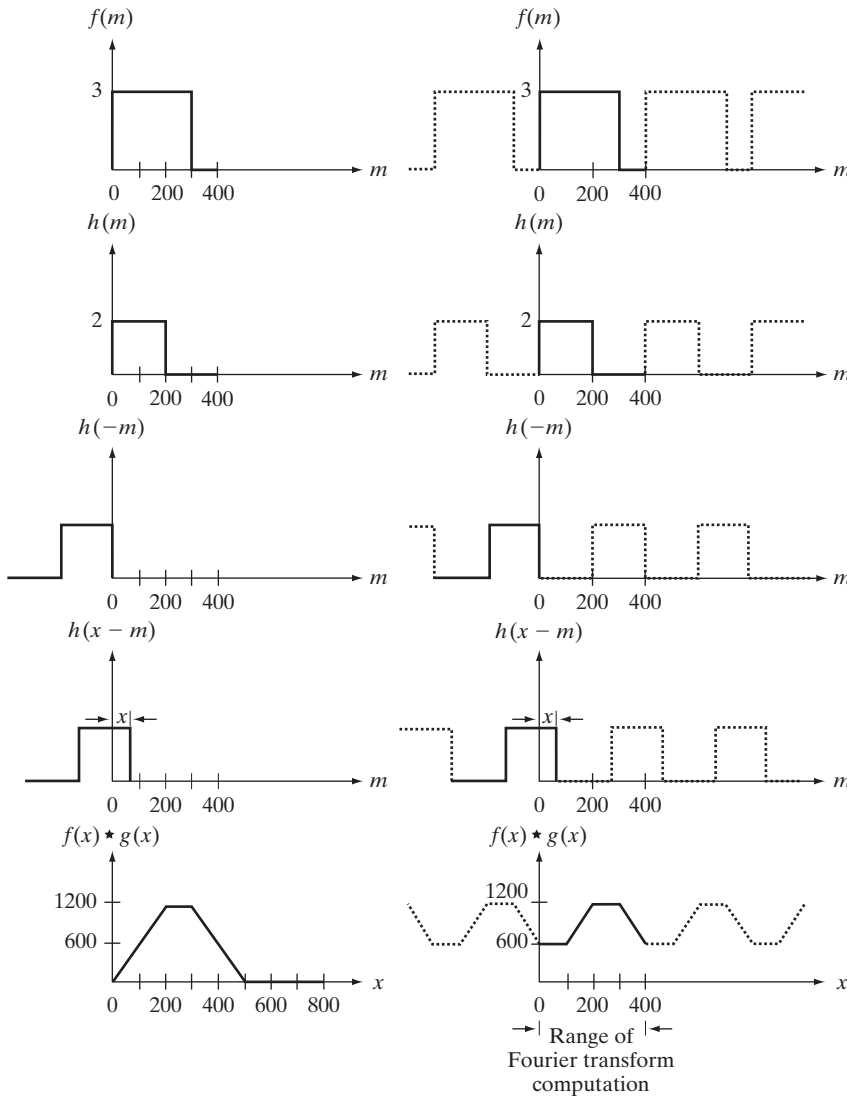


FIGURE 4.28 Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

two transforms, and then computed the inverse DFT, we would have obtained the erroneous 400-point segment of the convolution shown in Fig. 4.28(j).

Fortunately, the solution to the wraparound error problem is simple. Consider two functions, $f(x)$ and $h(x)$ composed of A and B samples, respectively. It can be shown (Brigham [1988]) that if we append zeros to both functions so that they have the same length, denoted by P , then wraparound is avoided by choosing

$$P \geq A + B - 1 \quad (4.6-26)$$

In our example, each function has 400 points, so the minimum value we could use is $P = 799$, which implies that we would append 399 zeros to the trailing edge of each function. This process is called *zero padding*. As an exercise, you

The zeros could be appended also to the beginning of the functions, or they could be divided between the beginning and end of the functions. It is simpler to append them at the end.

should convince yourself that if the periods of the functions in Figs. 4.28(f) and (g) were lengthened by appending to each period at least 399 zeros, the result would be a periodic convolution in which *each* period is identical to the correct result in Fig. 4.28(e). Using the DFT via the convolution theorem would result in a 799-point spatial function identical to Fig. 4.28(e). The conclusion, then, is that to obtain the same convolution result between the “straight” representation of the convolution equation approach in Chapter 3, and the DFT approach, functions in the latter must be padded prior to computing their transforms.

Visualizing a similar example in 2-D would be more difficult, but we would arrive at the same conclusion regarding wraparound error and the need for appending zeros to the functions. Let $f(x, y)$ and $h(x, y)$ be two image arrays of sizes $A \times B$ and $C \times D$ pixels, respectively. Wraparound error in their circular convolution can be avoided by padding these functions with zeros, as follows:

$$f_p(x, y) = \begin{cases} f(x, y) & 0 \leq x \leq A-1 \quad \text{and} \quad 0 \leq y \leq B-1 \\ 0 & A \leq x \leq P \quad \text{or} \quad B \leq y \leq Q \end{cases} \quad (4.6-27)$$

and

$$h_p(x, y) = \begin{cases} h(x, y) & 0 \leq x \leq C-1 \quad \text{and} \quad 0 \leq y \leq D-1 \\ 0 & C \leq x \leq P \quad \text{or} \quad D \leq y \leq Q \end{cases} \quad (4.6-28)$$

with

$$P \geq A + C - 1 \quad (4.6-29)$$

and

$$Q \geq B + D - 1 \quad (4.6-30)$$

The resulting padded images are of size $P \times Q$. If both arrays are of the same size, $M \times N$, then we require that

$$P \geq 2M - 1 \quad (4.6-31)$$

and

$$Q \geq 2N - 1 \quad (4.6-32)$$

We give an example in Section 4.7.2 showing the effects of wraparound error on images. As rule, DFT algorithms tend to execute faster with arrays of even size, so it is good practice to select P and Q as the smallest even integers that satisfy the preceding equations. If the two arrays are of the same size, this means that P and Q are selected as twice the array size.

The two functions in Figs. 4.28(a) and (b) conveniently become zero before the end of the sampling interval. If one or both of the functions were not zero at

the end of the interval, then a discontinuity would be created when zeros were appended to the function to eliminate wraparound error. This is analogous to multiplying a function by a box, which in the frequency domain would imply convolution of the original transform with a sinc function (see Example 4.1). This, in turn, would create so-called *frequency leakage*, caused by the high-frequency components of the sinc function. Leakage produces a blocky effect on images. Although leakage never can be totally eliminated, it can be reduced significantly by multiplying the sampled function by another function that tapers smoothly to near zero at both ends of the sampled record to dampen the sharp transitions (and thus the high frequency components) of the box. This approach, called *windowing* or *apodizing*, is an important consideration when fidelity in image reconstruction (as in high-definition graphics) is desired. If you are faced with the need for windowing, a good approach is to use a 2-D Gaussian function (see Section 4.8.3). One advantage of this function is that its Fourier transform is Gaussian also, thus producing low leakage.

A simple apodizing function is a triangle, centered on the data record, which tapers to 0 at both ends of the record. This is called the *Bartlett* window. Other common windows are the *Hamming* and the *Hann* windows. We can even use a Gaussian function. We return to the issue of windowing in Section 5.11.5.

4.6.7 Summary of 2-D Discrete Fourier Transform Properties

Table 4.2 summarizes the principal DFT definitions introduced in this chapter. Separability is discussed in Section 4.11.1 and obtaining the inverse using a forward transform algorithm is discussed in Section 4.11.2. Correlation is discussed in Chapter 12.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
4) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$
5) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$

TABLE 4.2
Summary of DFT definitions and corresponding expressions.

(Continued)

TABLE 4.2
(Continued)

Name	Expression(s)
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1M, v) = F(u, v + k_2N)$ $= F(u + k_1M, v + k_2N)$ $f(x, y) = f(x + k_1M, y) = f(x, y + k_2N)$ $= f(x + k_1M, y + k_2N)$
9) Convolution	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$
10) Correlation	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n)h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.
12) Obtaining the inverse Fourier transform using a forward transform algorithm.	$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v)e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.2.

Table 4.3 summarizes some important DFT pairs. Although our focus is on discrete functions, the last two entries in the table are Fourier transform pairs that can be derived only for continuous variables (note the use of continuous variable notation). We include them here because, with proper interpretation, they are quite useful in digital image processing. The differentiation pair can

TABLE 4.3
Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the closed-form, continuous expressions.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$
3) Translation (general)	$f(x, y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M + vy_0/N)}$
4) Translation to center of the frequency rectangle, ($M/2, N/2$)	$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$
6) Convolution theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$ $f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$

(Continued)

TABLE 4.3
(Continued)

Name	DFT Pairs
7) Correlation theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v) H(u, v)$ $f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$
9) Rectangle	$\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $j \frac{1}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$
11) Cosine	$\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $\frac{1}{2} [\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) Differentiation (The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$ $\frac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) Gaussian	$A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

[†]Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

be used to derive the frequency-domain equivalent of the Laplacian defined in Eq. (3.6-3) (Problem 4.26). The Gaussian pair is discussed in Section 4.7.4.

Tables 4.1 through 4.3 provide a summary of properties useful when working with the DFT. Many of these properties are key elements in the development of the material in the rest of this chapter, and some are used in subsequent chapters.

4.7 The Basics of Filtering in the Frequency Domain

In this section, we lay the groundwork for all the filtering techniques discussed in the remainder of the chapter.

4.7.1 Additional Characteristics of the Frequency Domain

We begin by observing in Eq. (4.5-15) that *each* term of $F(u, v)$ contains *all* values of $f(x, y)$, modified by the values of the exponential terms. Thus, with the exception of trivial cases, it usually is impossible to make direct associations between specific components of an image and its transform. However, some general statements can be made about the relationship between the frequency

components of the Fourier transform and spatial features of an image. For instance, because frequency is directly related to spatial rates of change, it is not difficult intuitively to associate frequencies in the Fourier transform with patterns of intensity variations in an image. We showed in Section 4.6.5 that the slowest varying frequency component ($u = v = 0$) is proportional to the average intensity of an image. As we move away from the origin of the transform, the low frequencies correspond to the slowly varying intensity components of an image. In an image of a room, for example, these might correspond to smooth intensity variations on the walls and floor. As we move further away from the origin, the higher frequencies begin to correspond to faster and faster intensity changes in the image. These are the edges of objects and other components of an image characterized by abrupt changes in intensity.

Filtering techniques in the frequency domain are based on modifying the Fourier transform to achieve a specific objective and then computing the inverse DFT to get us back to the image domain, as introduced in Section 2.6.7. It follows from Eq. (4.6-15) that the two components of the transform to which we have access are the transform magnitude (spectrum) and the phase angle. Section 4.6.5 covered the basic properties of these two components of the transform. We learned there that visual analysis of the phase component generally is not very useful. The spectrum, however, provides some useful guidelines as to gross characteristics of the image from which the spectrum was generated. For example, consider Fig. 4.29(a), which is a scanning electron microscope image of an integrated circuit, magnified approximately 2500 times. Aside from the interesting construction of the device itself, we note two principal features: strong edges that run approximately at $\pm 45^\circ$ and two white, oxide protrusions resulting from thermally-induced failure. The Fourier spectrum in Fig. 4.29(b) shows prominent components along the $\pm 45^\circ$ directions that correspond to the edges just mentioned. Looking carefully along the vertical axis, we see a vertical component

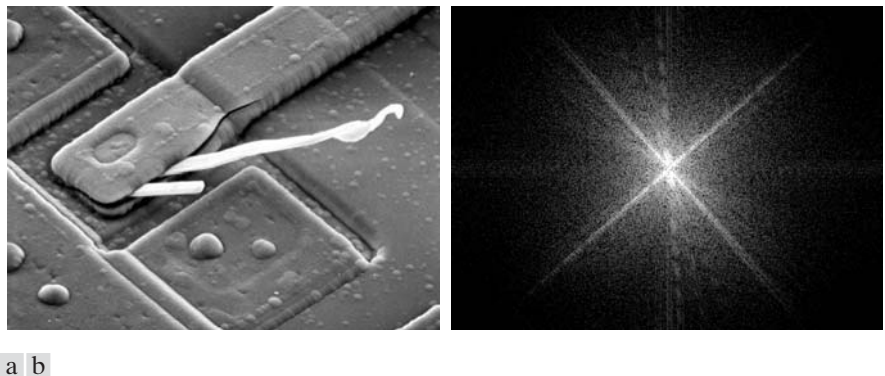


FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

that is off-axis slightly to the left. This component was caused by the edges of the oxide protrusions. Note how the angle of the frequency component with respect to the vertical axis corresponds to the inclination (with respect to the horizontal axis) of the long white element, and note also the zeros in the vertical frequency component, corresponding to the narrow vertical span of the oxide protrusions.

These are typical of the types of associations that can be made in general between the frequency and spatial domains. As we show later in this chapter, even these types of gross associations, coupled with the relationships mentioned previously between frequency content and rate of change of intensity levels in an image, can lead to some very useful results. In the next section, we show the effects of modifying various frequency ranges in the transform of Fig. 4.29(a).

4.7.2 Frequency Domain Filtering Fundamentals

Filtering in the frequency domain consists of modifying the Fourier transform of an image and then computing the inverse transform to obtain the processed result. Thus, given a digital image, $f(x, y)$, of size $M \times N$, the basic filtering equation in which we are interested has the form:

$$g(x, y) = \mathfrak{F}^{-1}[H(u, v)F(u, v)] \quad (4.7-1)$$

where $\mathfrak{F}^{-1}$ is the IDFT, $F(u, v)$ is the DFT of the input image, $f(x, y)$, $H(u, v)$ is a *filter function* (also called simply the *filter*, or the *filter transfer function*), and $g(x, y)$ is the filtered (output) image. Functions F , H , and g are arrays of size $M \times N$, the same as the input image. The product $H(u, v)F(u, v)$ is formed using array multiplication, as defined in Section 2.6.1. The filter function modifies the transform of the input image to yield a processed output, $g(x, y)$. Specification of $H(u, v)$ is simplified considerably by using functions that are symmetric about their center, which requires that $F(u, v)$ be centered also. As explained in Section 4.6.3, this is accomplished by multiplying the input image by $(-1)^{x+y}$ prior to computing its transform.[†]

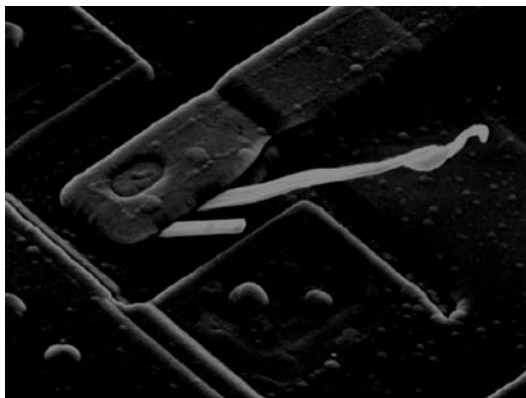
We are now in a position to consider the filtering process in some detail. One of the simplest filters we can construct is a filter $H(u, v)$ that is 0 at the center of the transform and 1 elsewhere. This filter would reject the dc term and “pass” (i.e., leave unchanged) all other terms of $F(u, v)$ when we form the product $H(u, v)F(u, v)$. We know from Eq. (4.6-21) that the dc term is responsible for the average intensity of an image, so setting it to zero will reduce the average intensity of the output image to zero. Figure 4.30 shows the result of this operation using Eq. (4.7-1). As expected, the image became much darker. (An average of zero

If H is real and symmetric and f is real (as is typically the case), then the IDFT in Eq. (4.7-1) should yield real quantities in theory. In practice, the inverse generally contains parasitic complex terms from round-off and other computational inaccuracies. Thus, it is customary to take the real part of the IDFT to form g .

[†]Many software implementations of the 2-D DFT (e.g., MATLAB) do not center the transform. This implies that filter functions must be arranged to correspond to the same data format as the uncentered transform (i.e., with the origin at the top left). The net result is that filters are more difficult to generate and display. We use centering in our discussions to aid in visualization, which is crucial in developing a clear understanding of filtering concepts. Either method can be used in practice, as long as consistency is maintained.

FIGURE 4.30

Result of filtering the image in Fig. 4.29(a) by setting to 0 the term $F(M/2, N/2)$ in the Fourier transform.



implies the existence of negative intensities. Therefore, although it illustrates the principle, Fig. 4.30 is not a true representation of the original, as all negative intensities were clipped (set to 0) for display purposes.)

As noted earlier, low frequencies in the transform are related to slowly varying intensity components in an image, such as the walls of a room or a cloudless sky in an outdoor scene. On the other hand, high frequencies are caused by sharp transitions in intensity, such as edges and noise. Therefore, we would expect that a filter $H(u, v)$ that attenuates high frequencies while passing low frequencies (appropriately called a *lowpass filter*) would blur an image, while a filter with the opposite property (called a *highpass filter*) would enhance sharp detail, but cause a reduction in contrast in the image. Figure 4.31 illustrates these effects. Note the similarity between Figs. 4.31(e) and Fig. 4.30. The reason is that the highpass filter shown eliminates the dc term, resulting in the same basic effect that led to Fig. 4.30. Adding a small constant to the filter does not affect sharpening appreciably, but it does prevent elimination of the dc term and thus preserves tonality, as Fig. 4.31(f) shows.

Equation (4.7-1) involves the product of two functions in the frequency domain which, by the convolution theorem, implies convolution in the spatial domain. We know from the discussion in Section 4.6.6 that if the functions in question are not padded we can expect wraparound error. Consider what happens when we apply Eq. (4.7-1) without padding. Figure 4.32(a) shows a simple image, and Fig. 4.32(b) is the result of lowpass filtering the image with a Gaussian lowpass filter of the form shown in Fig. 4.31(a). As expected, the image is blurred. However, the blurring is not uniform; the top white edge is blurred, but the side white edges are not. Padding the input image according to Eqs. (4.6-31) and (4.6-32) before applying Eq. (4.7-1) results in the filtered image in Fig. 4.32(c). This result is as expected.

Figure 4.33 illustrates the reason for the discrepancy between Figs. 4.32(b) and (c). The dashed areas in Fig. 4.33 correspond to the image in Fig. 4.32(a). Figure 4.33(a) shows the periodicity implicit in the use of the DFT, as explained in Section 4.6.3. Imagine convolving the *spatial* representation of the blurring filter with this image. When the filter is passing through the top of the

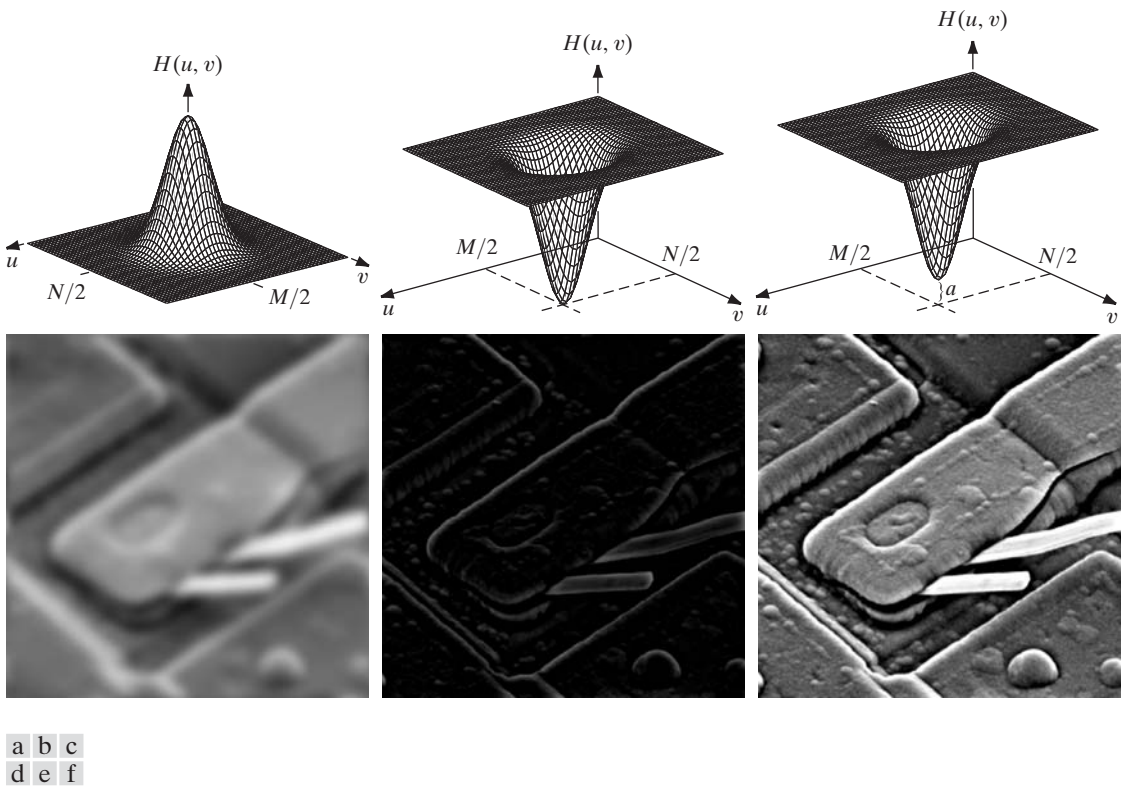


FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).



FIGURE 4.32 (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with padding. Compare the light area of the vertical edges in (b) and (c).

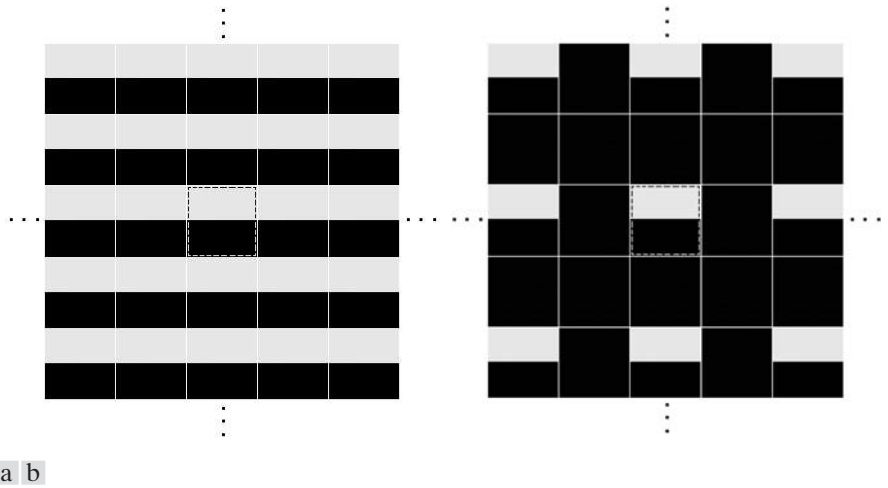
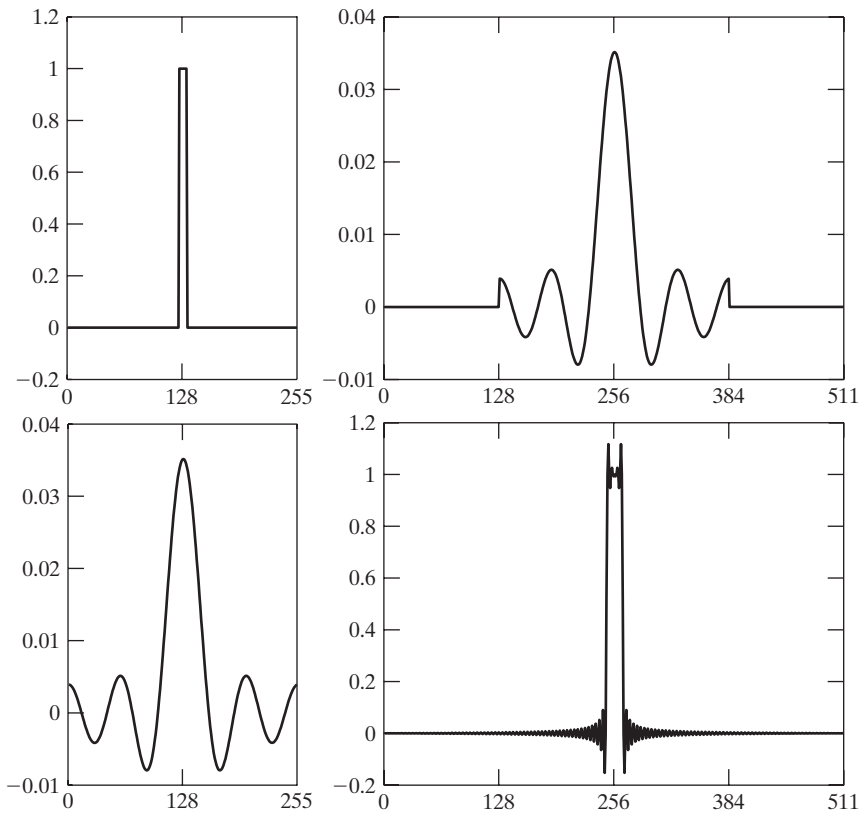


FIGURE 4.33 2-D image periodicity inherent in using the DFT. (a) Periodicity without image padding. (b) Periodicity after padding with 0s (black). The dashed areas in the center correspond to the image in Fig. 4.32(a). (The thin white lines in both images are superimposed for clarity; they are not part of the data.)

dashed image, it will encompass part of the image and also part of the bottom of the periodic image right above it. When a dark and a light region reside under the filter, the result is a mid-gray, blurred output. However, when the filter is passing through the top right side of the image, the filter will encompass only light areas in the image and its right neighbor. The average of a constant is the same constant, so filtering will have no effect in this area, giving the result in Fig. 4.32(b). Padding the image with 0s creates a uniform border around the periodic sequence, as Fig. 4.33(b) shows. Convoluting the blurring function with the padded “mosaic” of Fig. 4.33(b) gives the correct result in Fig. 4.32(c). You can see from this example that failure to pad an image can lead to erroneous results. If the purpose of filtering is only for rough visual analysis, the padding step is skipped sometimes.

Thus far, the discussion has centered on padding the input image, but Eq. (4.7-1) also involves a filter that can be specified either in the spatial or in the frequency domain. However, padding is done in the spatial domain, which raises an important question about the relationship between *spatial* padding and filters specified directly in the frequency domain.

At first glance, one could conclude that the way to handle padding of a frequency domain filter is to construct the filter to be of the same size as the image, compute the IDFT of the filter to obtain the corresponding spatial filter, pad that filter in the spatial domain, and then compute its DFT to return to the frequency domain. The 1-D example in Fig. 4.34 illustrates the pitfalls in this approach. Figure 4.34(a) shows a 1-D ideal lowpass filter in the frequency domain. The filter is real and has even symmetry, so we know from property 8 in Table 4.1 that its IDFT will be real and symmetric also. Figure 4.34(b) shows the result of multiplying the elements of the frequency domain filter



a c
b d

FIGURE 4.34

(a) Original filter specified in the (centered) frequency domain. (b) Spatial representation obtained by computing the IDFT of (a). (c) Result of padding (b) to twice its length (note the discontinuities). (d) Corresponding filter in the frequency domain obtained by computing the DFT of (c). Note the ringing caused by the discontinuities in (c). (The curves appear continuous because the points were joined to simplify visual analysis.)

by $(-1)^u$ and computing its IDFT to obtain the corresponding spatial filter. The extremes of this spatial function are not zero so, as Fig. 4.34(c) shows, zero-padding the function created two discontinuities (padding the two ends of the function is the same as padding one end, as long as the total number of zeros used is the same).

To get back to the frequency domain, we compute the DFT of the spatial, padded filter. Figure 4.34(d) shows the result. The discontinuities in the spatial filter created ringing in its frequency domain counterpart, as you would expect from the results in Example 4.1. Viewed another way, we know from that example that the Fourier transform of a box function is a sinc function with frequency components extending to infinity, and we would expect the same behavior from the inverse transform of a box. That is, the spatial representation of an ideal (box) frequency domain filter has components extending to infinity. Therefore, any spatial truncation of the filter to implement zero-padding will introduce discontinuities, which will then in general result in ringing in the frequency domain (truncation can be avoided in this case if it is done at zero crossings, but we are interested in general procedures, and not all filters have zero crossings).

What the preceding results tell us is that, because we cannot work with an infinite number of components, we cannot use an ideal frequency domain filter [as in

See the end of Section 4.3.3 regarding the definition of an ideal filter.

Fig. 4.34(a)] and simultaneously use zero padding to avoid wraparound error. A decision on which limitation to accept is required. Our objective is to work with specified filter shapes in the frequency domain (including ideal filters) without having to be concerned with truncation issues. One approach is to zero-pad images and then create filters in the frequency domain to be of the same size as the padded images (remember, images and filters must be of the same size when using the DFT). Of course, this will result in wraparound error because no padding is used for the filter, but in practice this error is mitigated significantly by the separation provided by the padding of the image, and it is preferable to ringing. Smooth filters (such as those in Fig. 4.31) present even less of a problem. Specifically, then, the approach we will follow in this chapter in order to work with filters of a specified shape directly in the frequency domain is to pad images to size $P \times Q$ and construct filters of the same dimensions. As explained earlier, P and Q are given by Eqs. (4.6-29) and (4.6-30).

We conclude this section by analyzing the phase angle of the filtered transform. Because the DFT is a complex array, we can express it in terms of its real and imaginary parts:

$$F(u, v) = R(u, v) + jI(u, v) \quad (4.7-2)$$

Equation (4.7-1) then becomes

$$g(x, y) = \mathfrak{F}^{-1}[H(u, v)R(u, v) + jH(u, v)I(u, v)] \quad (4.7-3)$$

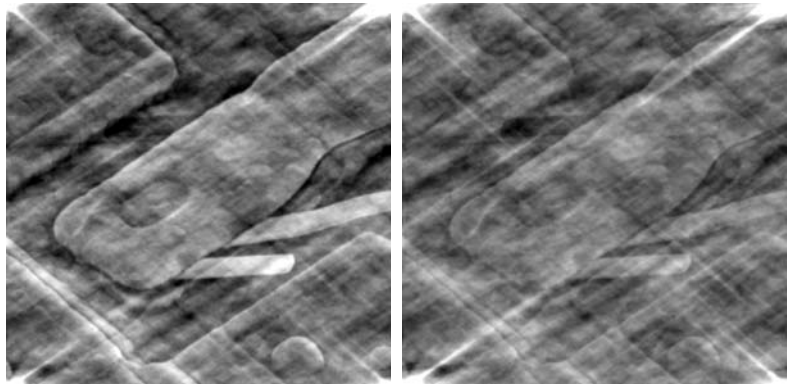
The phase angle is not altered by filtering in the manner just described because $H(u, v)$ cancels out when the ratio of the imaginary and real parts is formed in Eq. (4.6-17). Filters that affect the real and imaginary parts equally, and thus have no effect on the phase, are appropriately called *zero-phase-shift* filters. These are the only types of filters considered in this chapter.

Even small changes in the phase angle can have dramatic (usually undesirable) effects on the filtered output. Figure 4.35 illustrates the effect of something as simple as a scalar change. Figure 4.35(a) shows an image resulting from multiplying the angle array in Eq. (4.6-15) by 0.5, without changing

a b

FIGURE 4.35

(a) Image resulting from multiplying by 0.5 the phase angle in Eq. (4.6-15) and then computing the IDFT. (b) The result of multiplying the phase by 0.25. The spectrum was not changed in either of the two cases.



$|F(u, v)|$, and then computing the IDFT. The basic shapes remain unchanged, but the intensity distribution is quite distorted. Figure 4.35(b) shows the result of multiplying the phase by 0.25. The image is almost unrecognizable.

4.7.3 Summary of Steps for Filtering in the Frequency Domain

The material in the previous two sections can be summarized as follows:

1. Given an input image $f(x, y)$ of size $M \times N$, obtain the padding parameters P and Q from Eqs. (4.6-31) and (4.6-32). Typically, we select $P = 2M$ and $Q = 2N$.
2. Form a padded image, $f_p(x, y)$, of size $P \times Q$ by appending the necessary number of zeros to $f(x, y)$.
3. Multiply $f_p(x, y)$ by $(-1)^{x+y}$ to center its transform.
4. Compute the DFT, $F(u, v)$, of the image from step 3.
5. Generate a real, symmetric filter function, $H(u, v)$, of size $P \times Q$ with center at coordinates $(P/2, Q/2)$.[†] Form the product $G(u, v) = H(u, v)F(u, v)$ using array multiplication; that is, $G(i, k) = H(i, k)F(i, k)$.
6. Obtain the processed image:

$$g_p(x, y) = \left\{ \text{real} \left[\mathfrak{S}^{-1} [G(u, v)] \right] \right\} (-1)^{x+y}$$

where the real part is selected in order to ignore parasitic complex components resulting from computational inaccuracies, and the subscript p indicates that we are dealing with padded arrays.

7. Obtain the final processed result, $g(x, y)$, by extracting the $M \times N$ region from the top, left quadrant of $g_p(x, y)$.

As noted earlier, centering helps in visualizing the filtering process and in generating the filter functions themselves, but centering is not a fundamental requirement.

Figure 4.36 illustrates the preceding steps. The legend in the figure explains the source of each image. If it were enlarged, Fig. 4.36(c) would show black dots interleaved in the image because negative intensities are clipped to 0 for display. Note in Fig. 4.36(h) the characteristic dark border exhibited by lowpass filtered images processed using zero padding.

4.7.4 Correspondence Between Filtering in the Spatial and Frequency Domains

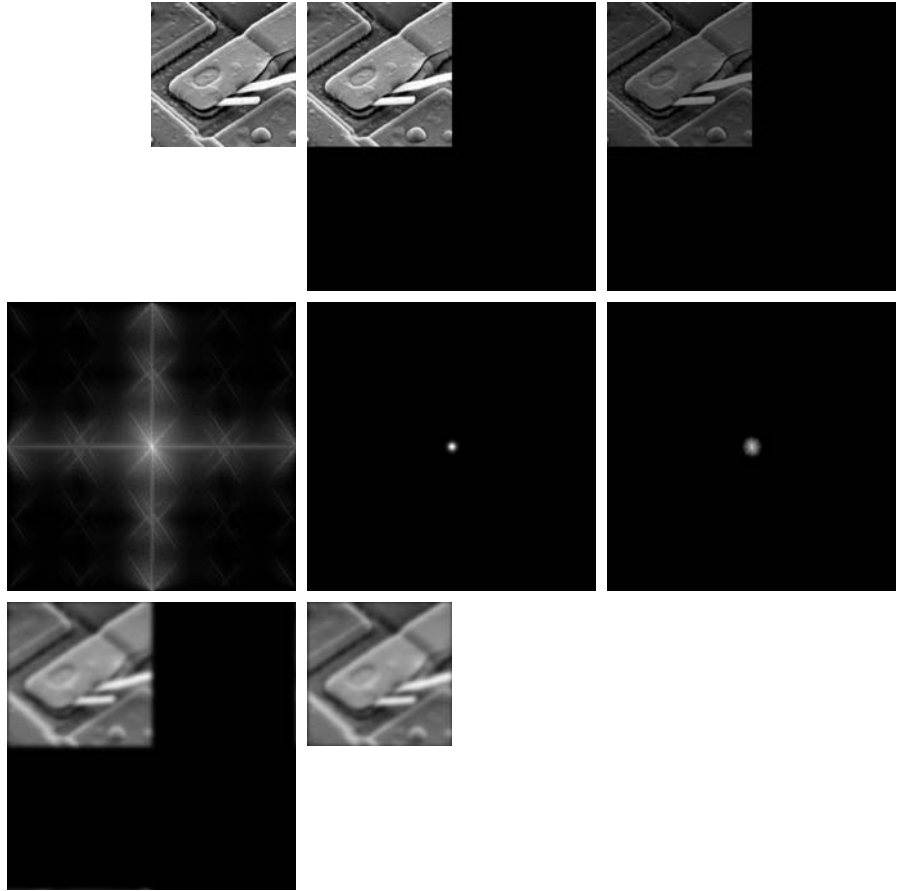
The link between filtering in the spatial and frequency domains is the convolution theorem. In Section 4.7.2, we defined filtering in the frequency domain as the multiplication of a filter function, $H(u, v)$, times $F(u, v)$, the Fourier transform of the input image. Given a filter $H(u, v)$, suppose that we want to find its equivalent representation in the spatial domain. If we let $f(x, y) = \delta(x, y)$, it follows from Table 4.3 that $F(u, v) = 1$. Then, from Eq. (4.7-1), the filtered output is $\mathfrak{S}^{-1}\{H(u, v)\}$. But this is the inverse transform of the frequency domain filter, which is the corresponding filter in the

[†]If $H(u, v)$ is to be generated from a *given* spatial filter, $h(x, y)$, then we form $h_p(x, y)$ by padding the spatial filter to size $P \times Q$, multiply the expanded array by $(-1)^{x+y}$, and compute the DFT of the result to obtain a centered $H(u, v)$. Example 4.15 illustrates this procedure.

a	b	c
d	e	f
g	h	

FIGURE 4.36

(a) An $M \times N$ image, f .
 (b) Padded image, f_p of size $P \times Q$.
 (c) Result of multiplying f_p by $(-1)^{x+y}$.
 (d) Spectrum of F_p . (e) Centered Gaussian lowpass filter, H , of size $P \times Q$.
 (f) Spectrum of the product HF_p .
 (g) g_p , the product of $(-1)^{x+y}$ and the real part of the IDFT of HF_p .
 (h) Final result, g , obtained by cropping the first M rows and N columns of g_p .



spatial domain. Conversely, it follows from a similar analysis and the convolution theorem that, given a spatial filter, we obtain its frequency domain representation by taking the forward Fourier transform of the spatial filter. Therefore, the two filters form a Fourier transform pair:

$$h(x, y) \Leftrightarrow H(u, v) \quad (4.7-4)$$

where $h(x, y)$ is a spatial filter. Because this filter can be obtained from the response of a frequency domain filter to an impulse, $h(x, y)$ sometimes is referred to as the *impulse response* of $H(u, v)$. Also, because all quantities in a discrete implementation of Eq. (4.7-4) are finite, such filters are called *finite impulse response* (FIR) filters. These are the only types of linear spatial filters considered in this book.

We introduced spatial convolution in Section 3.4.1 and discussed its implementation in connection with Eq. (3.4-2), which involved convolving functions of different sizes. When we speak of spatial convolution in terms of the

convolution theorem and the DFT, it is implied that we are convolving periodic functions, as explained in Fig. 4.28. For this reason, as explained earlier, Eq. (4.6-23) is referred to as *circular convolution*. Furthermore, convolution in the context of the DFT involves functions of the same size, whereas in Eq. (3.4-2) the functions typically are of different sizes.

In practice, we prefer to implement convolution filtering using Eq. (3.4-2) with small filter masks because of speed and ease of implementation in hardware and/or firmware. However, filtering concepts are more intuitive in the frequency domain. One way to take advantage of the properties of both domains is to specify a filter in the frequency domain, compute its IDFT, and then use the resulting, full-size spatial filter as a *guide* for constructing smaller spatial filter masks (more formal approaches are mentioned in Section 4.11.4). This is illustrated next. Later in this section, we illustrate also the converse, in which a small spatial filter is given and we obtain its full-size frequency domain representation. This approach is useful for analyzing the behavior of small spatial filters in the frequency domain. Keep in mind during the following discussion that the Fourier transform and its inverse are linear processes (Problem 4.14), so the discussion is limited to linear filtering.

In the following discussion, we use Gaussian filters to illustrate how frequency domain filters can be used as guides for specifying the coefficients of some of the small masks discussed in Chapter 3. Filters based on Gaussian functions are of particular interest because, as noted in Table 4.3, both the forward and inverse Fourier transforms of a Gaussian function are real Gaussian functions. We limit the discussion to 1-D to illustrate the underlying principles. Two-dimensional Gaussian filters are discussed later in this chapter.

Let $H(u)$ denote the 1-D frequency domain Gaussian filter:

$$H(u) = A e^{-u^2/2\sigma^2} \quad (4.7-5)$$

where σ is the standard deviation of the Gaussian curve. The corresponding filter in the spatial domain is obtained by taking the inverse Fourier transform of $H(u)$ (Problem 4.31):

$$h(x) = \sqrt{2\pi}\sigma A e^{-2\pi^2\sigma^2 x^2} \quad (4.7-6)$$

These equations[†] are important for two reasons: (1) They are a Fourier transform pair, both components of which are Gaussian and *real*. This facilitates analysis because we do not have to be concerned with complex numbers. In addition, Gaussian curves are intuitive and easy to manipulate. (2) The functions behave reciprocally. When $H(u)$ has a broad profile (large value of σ),

[†]As mentioned in Table 4.3, closed forms for the forward and inverse Fourier transforms of Gaussians are valid only for continuous functions. To use discrete formulations we simply sample the continuous Gaussian transforms. Our use of discrete variables here implies that we are dealing with sampled transforms.

$h(x)$ has a narrow profile, and vice versa. In fact, as σ approaches infinity, $H(u)$ tends toward a constant function and $h(x)$ tends toward an impulse, which implies no filtering in the frequency and spatial domains, respectively.

Figures 4.37(a) and (b) show plots of a Gaussian lowpass filter in the frequency domain and the corresponding lowpass filter in the spatial domain. Suppose that we want to use the shape of $h(x)$ in Fig. 4.37(b) as a *guide* for specifying the coefficients of a small spatial mask. The key similarity between the two filters is that all their values are positive. Thus, we conclude that we can implement lowpass filtering in the spatial domain by using a mask with all positive coefficients (as we did in Section 3.5.1). For reference, Fig. 4.37(b) shows two of the masks discussed in that section. Note the reciprocal relationship between the width of the filters, as discussed in the previous paragraph. The narrower the frequency domain filter, the more it will attenuate the low frequencies, resulting in increased blurring. In the spatial domain, this means that a larger mask must be used to increase blurring, as illustrated in Example 3.13.

More complex filters can be constructed using the basic Gaussian function of Eq. (4.7-5). For example, we can construct a highpass filter as the *difference* of Gaussians:

$$H(u) = Ae^{-u^2/2\sigma_1^2} - Be^{-u^2/2\sigma_2^2} \quad (4.7-7)$$

with $A \geq B$ and $\sigma_1 > \sigma_2$. The corresponding filter in the spatial domain is

$$h(x) = \sqrt{2\pi}\sigma_1 Ae^{-2\pi^2\sigma_1^2x^2} - \sqrt{2\pi}\sigma_2 Be^{-2\pi^2\sigma_2^2x^2} \quad (4.7-8)$$

Figures 4.37(c) and (d) show plots of these two equations. We note again the reciprocity in width, but the most important feature here is that $h(x)$ has a positive center term with negative terms on either side. The small masks shown in

a c
b d

FIGURE 4.37

(a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.

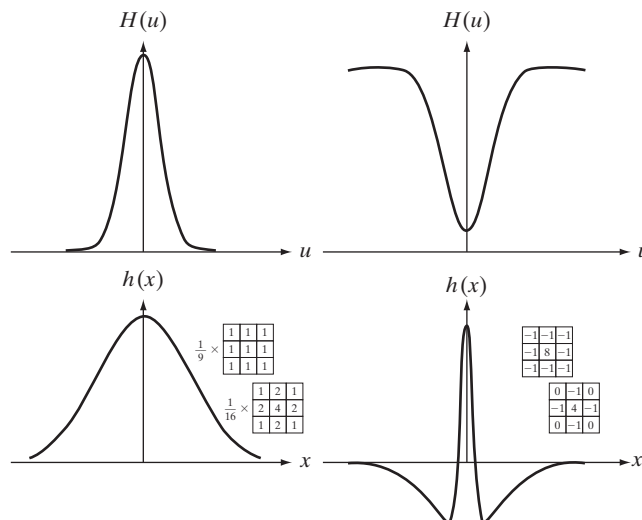


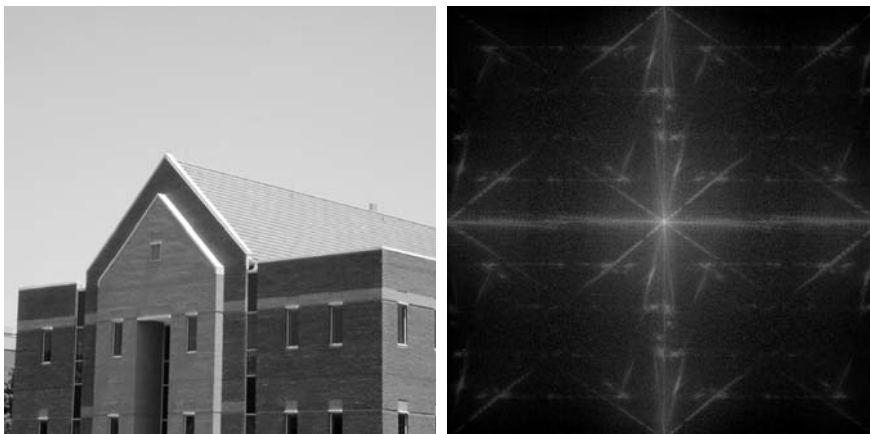
Fig. 4.37(d) “capture” this property. These two masks were used in Chapter 3 as sharpening filters, which we now know are highpass filters.

Although we have gone through significant effort to get here, be assured that it is impossible to truly understand filtering in the frequency domain without the foundation we have just established. In practice, the frequency domain can be viewed as a “laboratory” in which we take advantage of the correspondence between frequency content and image appearance. As is demonstrated numerous times later in this chapter, some tasks that would be exceptionally difficult or impossible to formulate directly in the spatial domain become almost trivial in the frequency domain. Once we have selected a specific filter via experimentation in the frequency domain, the actual implementation of the method usually is done in the spatial domain. One approach is to specify small spatial masks that attempt to capture the “essence” of the full filter function in the spatial domain, as we explained in Fig. 4.37. A more formal approach is to design a 2-D digital filter by using approximations based on mathematical or statistical criteria. We touch on this point again in Section 4.11.4.

■ In this example, we start with a spatial mask and show how to generate its corresponding filter in the frequency domain. Then, we compare the filtering results obtained using frequency domain and spatial techniques. This type of analysis is useful when one wishes to compare the performance of given spatial masks against one or more “full” filter candidates in the frequency domain, or to gain deeper understanding about the performance of a mask. To keep matters simple, we use the 3×3 Sobel vertical edge detector from Fig. 3.41(e). Figure 4.38(a) shows a 600×600 pixel image, $f(x, y)$, that we wish to filter, and Fig. 4.38(b) shows its spectrum.

Figure 4.39(a) shows the Sobel mask, $h(x, y)$ (the perspective plot is explained below). Because the input image is of size 600×600 pixels and the filter is of size 3×3 we avoid wraparound error by padding f and h to size

EXAMPLE 4.15:
Obtaining a frequency domain filter from a small spatial mask.



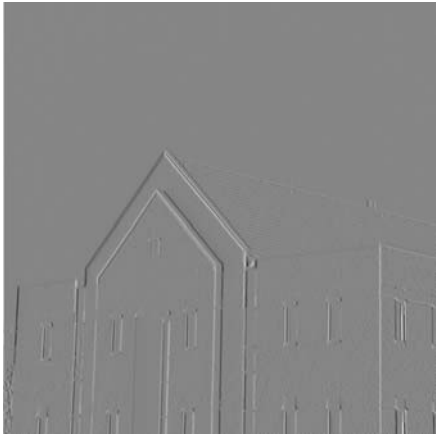
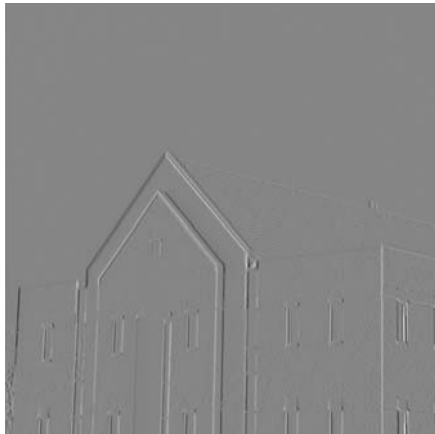
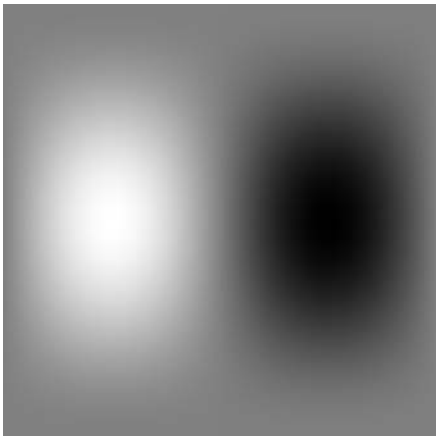
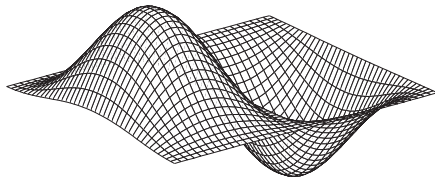
a b

FIGURE 4.38
(a) Image of a building, and
(b) its spectrum.

a b
c d

FIGURE 4.39
(a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.

-1	0	1
-2	0	2
-1	0	1



602×602 pixels, according to Eqs. (4.6-29) and (4.6-30). The Sobel mask exhibits odd symmetry, provided that it is embedded in an array of zeros of even size (see Example 4.10). To maintain this symmetry, we place $h(x, y)$ so that its center is at the center of the 602×602 padded array. This is an important aspect of filter generation. If we preserve the odd symmetry with respect to the padded array in forming $h_p(x, y)$, we know from property 9 in Table 4.1 that $H(u, v)$ will be purely imaginary. As we show at the end of this example, this will yield results that are identical to filtering the image spatially using $h(x, y)$. If the symmetry were not preserved, the results would no longer be same.

The procedure used to generate $H(u, v)$ is: (1) multiply $h_p(x, y)$ by $(-1)^{x+y}$ to center the frequency domain filter; (2) compute the forward DFT of the result in (1); (3) set the real part of the resulting DFT to 0 to account for parasitic real parts (we know that $H(u, v)$ has to be purely imaginary); and (4) multiply the result by $(-1)^{u+v}$. This last step reverses the multiplication of $H(u, v)$ by $(-1)^{u+v}$, which is implicit when $h(x, y)$ was moved to the center of $h_p(x, y)$. Figure 4.39(a) shows a perspective plot of $H(u, v)$, and Fig. 4.39(b) shows

$H(u, v)$ as an image. As, expected, the function is odd, thus the antisymmetry about its center. Function $H(u, v)$ is used as any other frequency domain filter in the procedure outlined in Section 4.7.3.

Figure 4.39(c) is the result of using the filter just obtained in the procedure outlined in Section 4.7.3 to filter the image in Fig. 4.38(a). As expected from a derivative filter, edges are enhanced and all the constant intensity areas are reduced to zero (the grayish tone is due to scaling for display). Figure 4.39(d) shows the result of filtering the same image in the spatial domain directly, using $h(x, y)$ in the procedure outlined in Section 3.6.4. The results are identical. ■

4.8 Image Smoothing Using Frequency Domain Filters

The remainder of this chapter deals with various filtering techniques in the frequency domain. We begin with lowpass filters. Edges and other sharp intensity transitions (such as noise) in an image contribute significantly to the high-frequency content of its Fourier transform. Hence, smoothing (blurring) is achieved in the frequency domain by high-frequency attenuation; that is, by *lowpass* filtering. In this section, we consider three types of lowpass filters: ideal, Butterworth, and Gaussian. These three categories cover the range from very sharp (ideal) to very smooth (Gaussian) filtering. The Butterworth filter has a parameter called the *filter order*. For high order values, the Butterworth filter approaches the ideal filter. For lower order values, the Butterworth filter is more like a Gaussian filter. Thus, the Butterworth filter may be viewed as providing a transition between two “extremes.” All filtering in this section follows the procedure outlined in Section 4.7.3, so all filter functions, $H(u, v)$, are understood to be discrete functions of size $P \times Q$; that is, the discrete frequency variables are in the range $u = 0, 1, 2, \dots, P - 1$ and $v = 0, 1, 2, \dots, Q - 1$.

4.8.1 Ideal Lowpass Filters

A 2-D lowpass filter that passes without attenuation all frequencies within a circle of radius D_0 from the origin and “cuts off” all frequencies outside this circle is called an *ideal lowpass filter* (ILPF); it is specified by the function

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases} \quad (4.8-1)$$

where D_0 is a positive constant and $D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the frequency rectangle; that is,

$$D(u, v) = \left[(u - P/2)^2 + (v - Q/2)^2 \right]^{1/2} \quad (4.8-2)$$

where, as before, P and Q are the padded sizes from Eqs. (4.6-31) and (4.6-32). Figure 4.40(a) shows a perspective plot of $H(u, v)$ and Fig. 4.40(b) shows the filter displayed as an image. As mentioned in Section 4.3.3, the name *ideal* indicates that all frequencies on or inside a circle of radius D_0 are passed

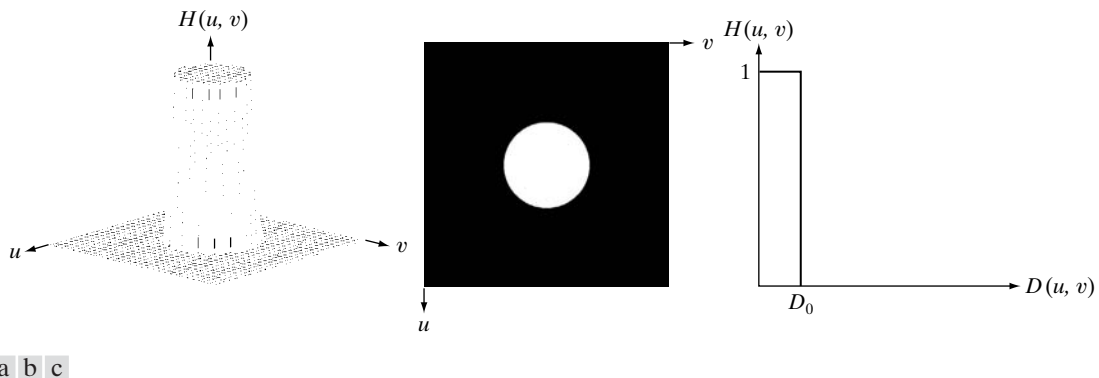


FIGURE 4.40 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

without attenuation, whereas all frequencies outside the circle are completely attenuated (filtered out). The ideal lowpass filter is radially symmetric about the origin, which means that the filter is completely defined by a radial cross section, as Fig. 4.40(c) shows. Rotating the cross section by 360° yields the filter in 2-D.

For an ILPF cross section, the point of transition between $H(u, v) = 1$ and $H(u, v) = 0$ is called the *cutoff frequency*. In the case of Fig. 4.40, for example, the cutoff frequency is D_0 . The sharp cutoff frequencies of an ILPF cannot be realized with electronic components, although they certainly can be simulated in a computer. The effects of using these “nonphysical” filters on a digital image are discussed later in this section.

The lowpass filters introduced in this chapter are compared by studying their behavior as a function of the same cutoff frequencies. One way to establish a set of standard cutoff frequency loci is to compute circles that enclose specified amounts of total image power P_T . This quantity is obtained by summing the components of the power spectrum of the padded images at each point (u, v) , for $u = 0, 1, \dots, P - 1$ and $v = 0, 1, \dots, Q - 1$; that is,

$$P_T = \sum_{u=0}^{P-1} \sum_{v=0}^{Q-1} P(u, v) \quad (4.8-3)$$

where $P(u, v)$ is given in Eq. (4.6-18). If the DFT has been centered, a circle of radius D_0 with origin at the center of the frequency rectangle encloses α percent of the power, where

$$\alpha = 100 \left[\sum_u \sum_v P(u, v) / P_T \right] \quad (4.8-4)$$

and the summation is taken over values of (u, v) that lie inside the circle or on its boundary.

Figures 4.41(a) and (b) show a test pattern image and its spectrum. The circles superimposed on the spectrum have radii of 10, 30, 60, 160, and 460 pixels, respectively. These circles enclose α percent of the image power, for $\alpha = 87.0, 93.1, 95.7, 97.8$, and 99.2% , respectively. The spectrum falls off rapidly, with 87% of the total power being enclosed by a relatively small circle of radius 10.

■ Figure 4.42 shows the results of applying ILPFs with cutoff frequencies at the radii shown in Fig. 4.41(b). Figure 4.42(b) is useless for all practical purposes, unless the objective of blurring is to eliminate all detail in the image, except the “blobs” representing the largest objects. The severe blurring in this image is a clear indication that most of the sharp detail information in the picture is contained in the 13% power removed by the filter. As the filter radius increases, less and less power is removed, resulting in less blurring. Note that the images in Figs. 4.42(c) through (e) are characterized by “ringing,” which becomes finer in texture as the amount of high frequency content removed decreases. Ringing is visible even in the image [Fig. 4.42(e)] in which only 2% of the total power was removed. This ringing behavior is a characteristic of ideal filters, as you will see shortly. Finally, the result for $\alpha = 99.2$ shows very slight blurring in the noisy squares but, for the most part, this image is quite close to the original. This indicates that little edge information is contained in the upper 0.8% of the spectrum power in this particular case.

It is clear from this example that ideal lowpass filtering is not very practical. However, it is useful to study their behavior as part of our development of

EXAMPLE 4.16:
Image smoothing
using an ILPF.

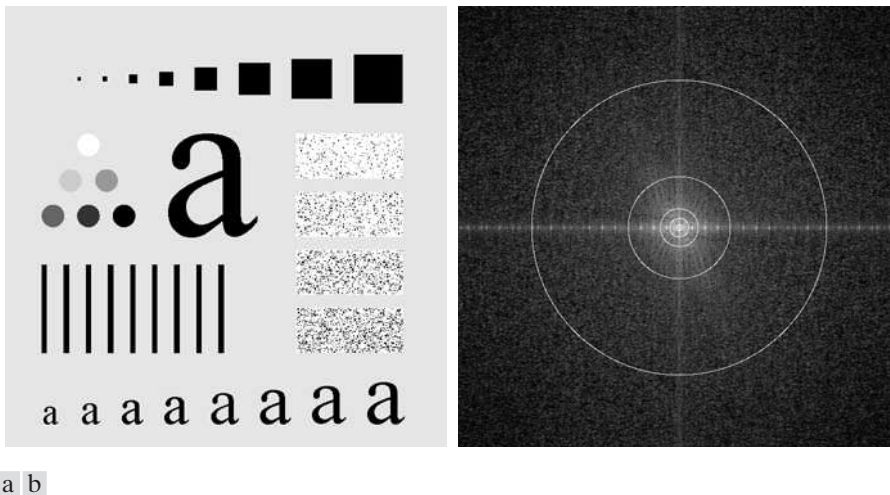


FIGURE 4.41 (a) Test pattern of size 688×688 pixels, and (b) its Fourier spectrum. The spectrum is double the image size due to padding but is shown in half size so that it fits in the page. The superimposed circles have radii equal to 10, 30, 60, 160, and 460 with respect to the full-size spectrum image. These radii enclose 87.0, 93.1, 95.7, 97.8, and 99.2% of the padded image power, respectively.

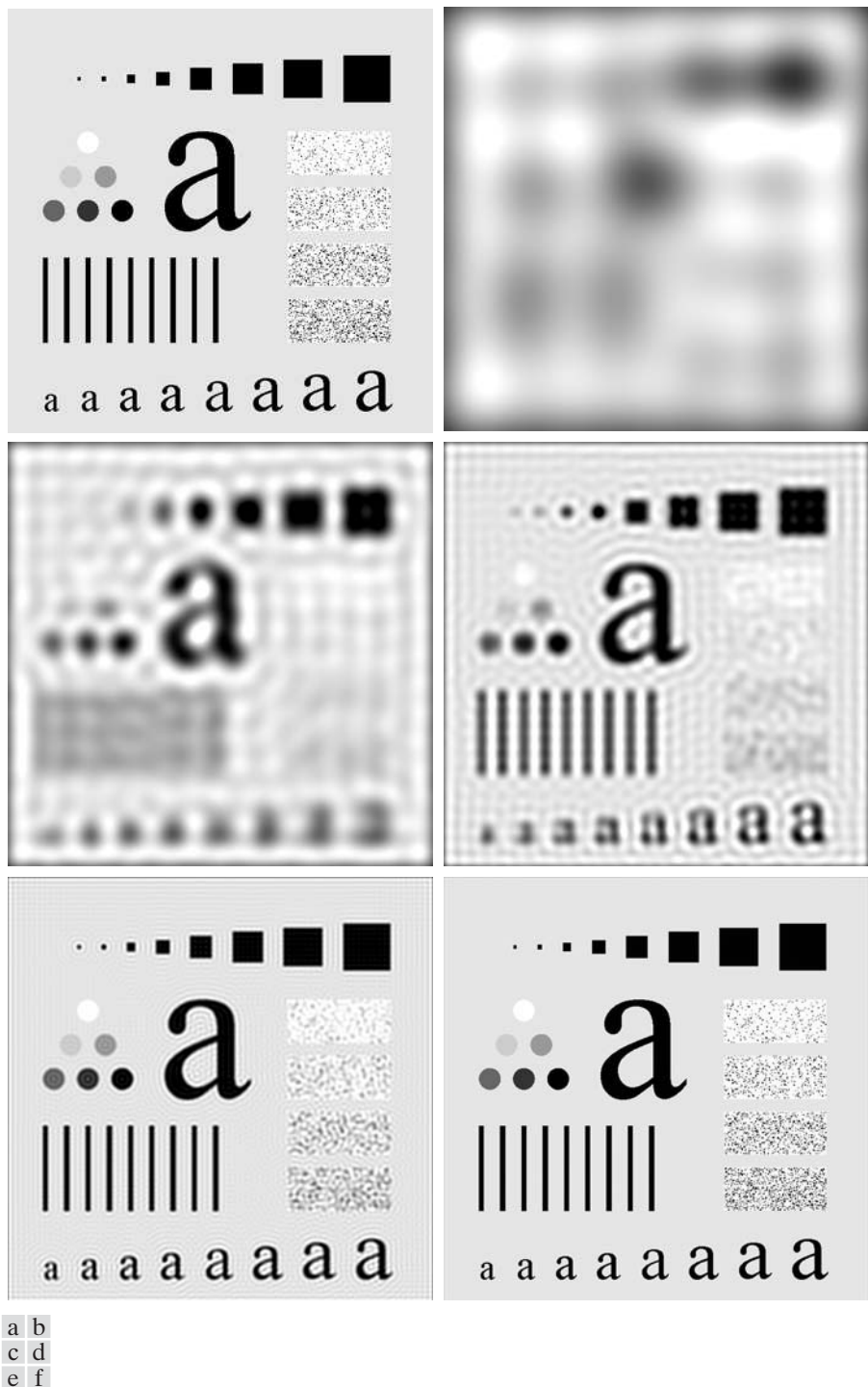


FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

filtering concepts. Also, as shown in the discussion that follows, some interesting insight is gained by attempting to explain the ringing property of ILPFs in the spatial domain. ■

The blurring and ringing properties of ILPFs can be explained using the convolution theorem. Figure 4.43(a) shows the spatial representation, $h(x, y)$, of an ILPF of radius 10, and Fig. 4.43(b) shows the intensity profile of a line passing through the center of the image. Because a cross section of the ILPF in the frequency domain looks like a box filter, it is not unexpected that a cross section of the corresponding spatial filter has the shape of a sinc function. Filtering in the spatial domain is done by convolving $h(x, y)$ with the image. Imagine each pixel in the image being a discrete impulse whose strength is proportional to the intensity of the image at that location. Convolution of a sinc with an impulse copies the sinc at the location of the impulse. The center lobe of the sinc is the principal cause of blurring, while the outer, smaller lobes are mainly responsible for ringing. Convolution of the sinc with every pixel in the image provides a nice model for explaining the behavior of ILPFs. Because the “spread” of the sinc function is inversely proportional to the radius of $H(u, v)$, the larger D_0 becomes, the more the spatial sinc approaches an impulse which, in the limit, causes no blurring at all when convolved with the image. This type of reciprocal behavior should be routine to you by now. In the next two sections, we show that it is possible to achieve blurring with little or no ringing, which is an important objective in lowpass filtering.

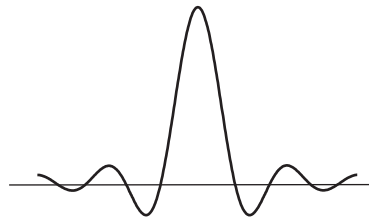
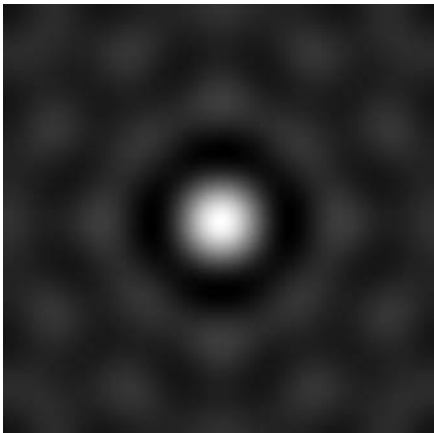
4.8.2 Butterworth Lowpass Filters

The transfer function of a Butterworth lowpass filter (BLPF) of order n , and with cutoff frequency at a distance D_0 from the origin, is defined as

$$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}} \quad (4.8-5)$$

where $D(u, v)$ is given by Eq. (4.8-2). Figure 4.44 shows a perspective plot, image display, and radial cross sections of the BLPF function.

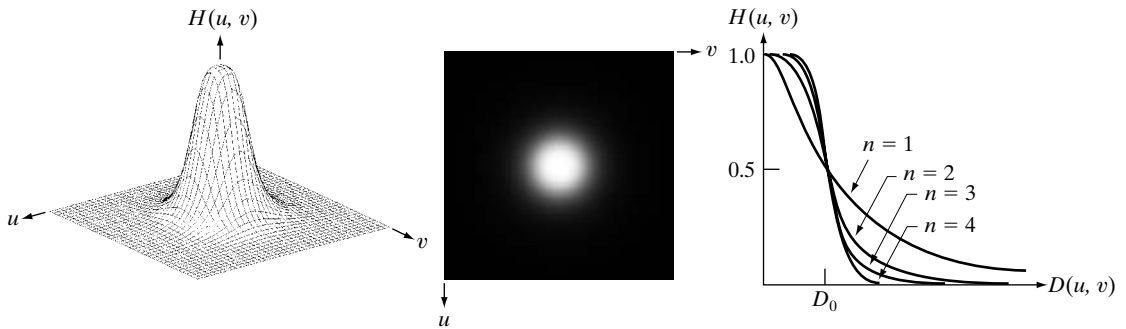
The transfer function of the Butterworth lowpass filter normally is written as the square root of our expression. However, our interest here is in the basic *form* of the filter, so we exclude the square root for computational convenience.



a b

FIGURE 4.43

(a) Representation in the spatial domain of an ILPF of radius 5 and size 1000×1000 . (b) Intensity profile of a horizontal line passing through the center of the image.



a b c

FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Unlike the ILPF, the BLPF transfer function does not have a sharp discontinuity that gives a clear cutoff between passed and filtered frequencies. For filters with smooth transfer functions, defining a cutoff frequency locus at points for which $H(u, v)$ is down to a certain fraction of its maximum value is customary. In Eq. (4.8-5), (down 50% from its maximum value of 1) when $D(u, v) = D_0$.

EXAMPLE 4.17:
Image smoothing
with a
Butterworth
lowpass filter.

■ Figure 4.45 shows the results of applying the BLPF of Eq. (4.8-5) to Fig. 4.45(a), with $n = 2$ and D_0 equal to the five radii in Fig. 4.41(b). Unlike the results in Fig. 4.42 for the ILPF, we note here a smooth transition in blurring as a function of increasing cutoff frequency. Moreover, no ringing is visible in any of the images processed with this particular BLPF, a fact attributed to the filter's smooth transition between low and high frequencies. ■

A BLPF of order 1 has no ringing in the spatial domain. Ringing generally is imperceptible in filters of order 2, but can become significant in filters of higher order. Figure 4.46 shows a comparison between the *spatial* representation of BLPFs of various orders (using a cutoff frequency of 5 in all cases). Shown also is the intensity profile along a horizontal scan line through the center of each filter. These filters were obtained and displayed using the same procedure used to generate Fig. 4.43. To facilitate comparisons, additional enhancing with a gamma transformation [see Eq. (3.2-3)] was applied to the images of Fig. 4.46. The BLPF of order 1 [Fig. 4.46(a)] has neither ringing nor negative values. The filter of order 2 does show mild ringing and small negative values, but they certainly are less pronounced than in the ILPF. As the remaining images show, ringing in the BLPF becomes significant for higher-order filters. A Butterworth filter of order 20 exhibits characteristics similar to those of the ILPF (in the limit, both filters are identical). BLPFs of order 2 are a good compromise between effective lowpass filtering and acceptable ringing.

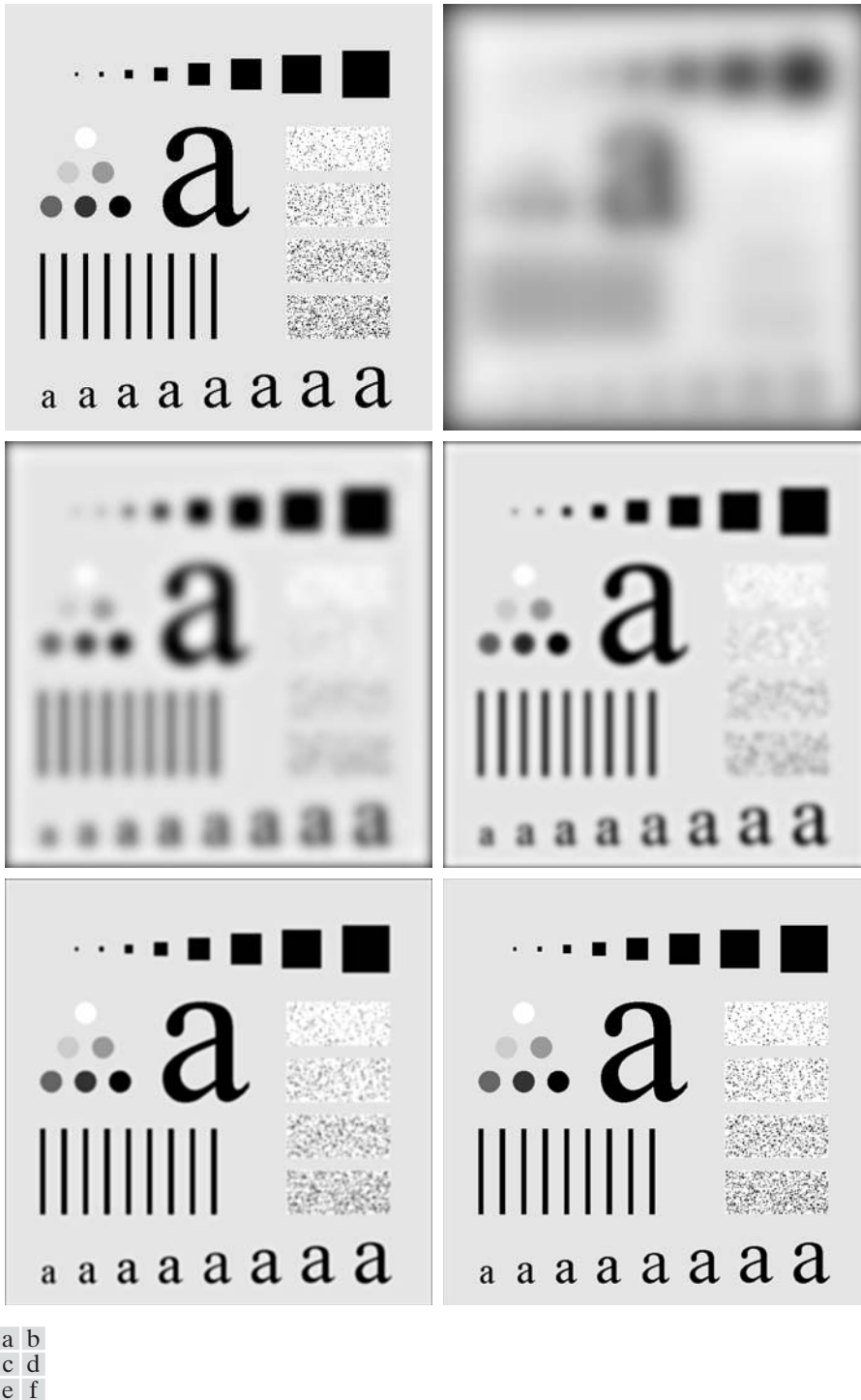


FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

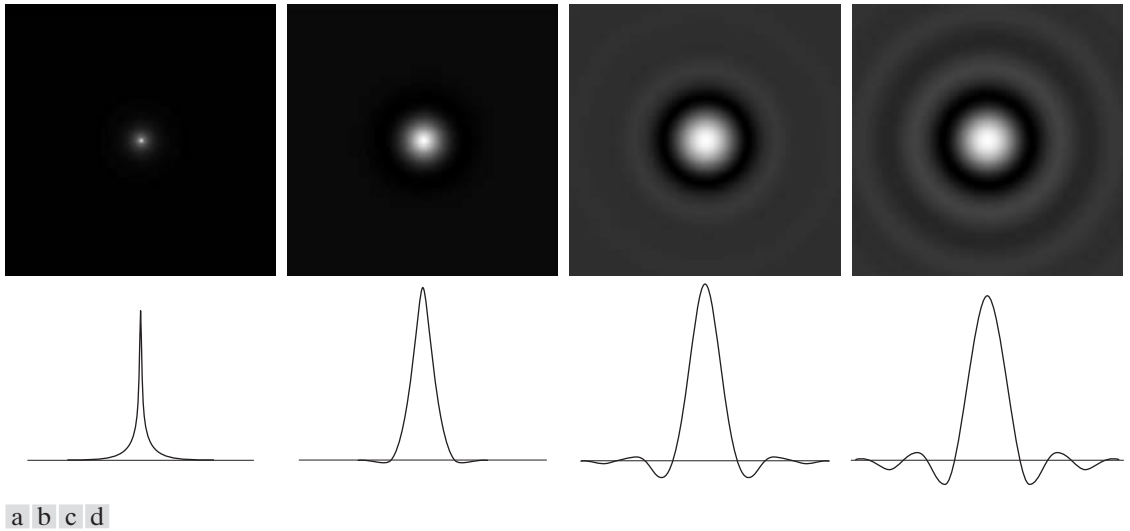


FIGURE 4.46 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding intensity profiles through the center of the filters (the size in all cases is 1000×1000 and the cutoff frequency is 5). Observe how ringing increases as a function of filter order.

4.8.3 Gaussian Lowpass Filters

Gaussian lowpass filters (GLPFs) of one dimension were introduced in Section 4.7.4 as an aid in exploring some important relationships between the spatial and frequency domains. The form of these filters in two dimensions is given by

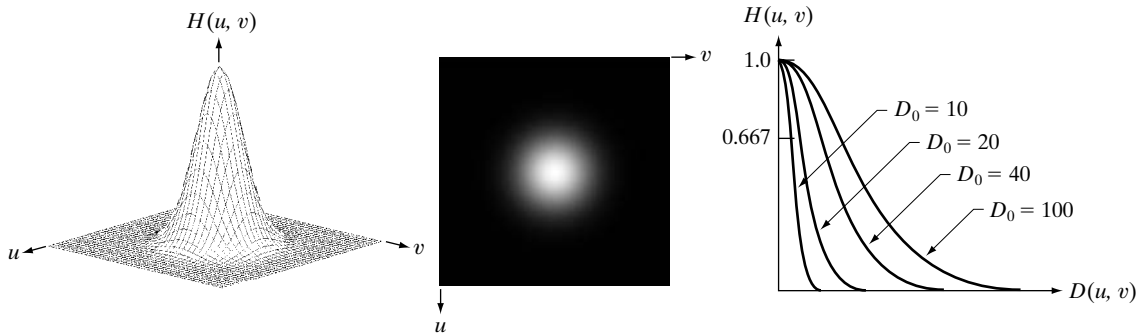
$$H(u, v) = e^{-D^2(u, v)/2\sigma^2} \quad (4.8-6)$$

where, as in Eq. (4.8-2), $D(u, v)$ is the distance from the center of the frequency rectangle. Here we do not use a multiplying constant as in Section 4.7.4 in order to be consistent with the filters discussed in the present section, whose highest value is 1. As before, σ is a measure of spread about the center. By letting $\sigma = D_0$, we can express the filter using the notation of the other filters in this section:

$$H(u, v) = e^{-D^2(u, v)/2D_0^2} \quad (4.8-7)$$

where D_0 is the cutoff frequency. When $D(u, v) = D_0$, the GLPF is down to 0.607 of its maximum value.

As Table 4.3 shows, the inverse Fourier transform of the GLPF is Gaussian also. This means that a spatial Gaussian filter, obtained by computing the IDFT of Eq. (4.8-7), will have no ringing. Figure 4.47 shows a perspective plot, image display, and radial cross sections of a GLPF function, and Table 4.4 summarizes the lowpass filters discussed in this section.



a b c

FIGURE 4.47 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .

TABLE 4.4

Lowpass filters. D_0 is the cutoff frequency and n is the order of the Butterworth filter.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$	$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$	$H(u, v) = e^{-D^2(u, v)/2D_0^2}$

■ Figure 4.48 shows the results of applying the GLPF of Eq. (4.8-7) to Fig. 4.48(a), with D_0 equal to the five radii in Fig. 4.41(b). As in the case of the BLPF of order 2 (Fig. 4.45), we note a smooth transition in blurring as a function of increasing cutoff frequency. The GLPF achieved slightly less smoothing than the BLPF of order 2 for the same value of cutoff frequency, as can be seen, for example, by comparing Figs. 4.45(c) and 4.48(c). This is expected, because the profile of the GLPF is not as “tight” as the profile of the BLPF of order 2. However, the results are quite comparable, and we are assured of no ringing in the case of the GLPF. This is an important characteristic in practice, especially in situations (e.g., medical imaging) in which any type of artifact is unacceptable. In cases where tight control of the transition between low and high frequencies about the cutoff frequency are needed, then the BLPF presents a more suitable choice. The price of this additional control over the filter profile is the possibility of ringing. ■

EXAMPLE 4.18: Image smoothing with a Gaussian lowpass filter.

4.8.4 Additional Examples of Lowpass Filtering

In the following discussion, we show several practical applications of lowpass filtering in the frequency domain. The first example is from the field of machine perception with application to character recognition; the second is from the printing and publishing industry; and the third is related to processing

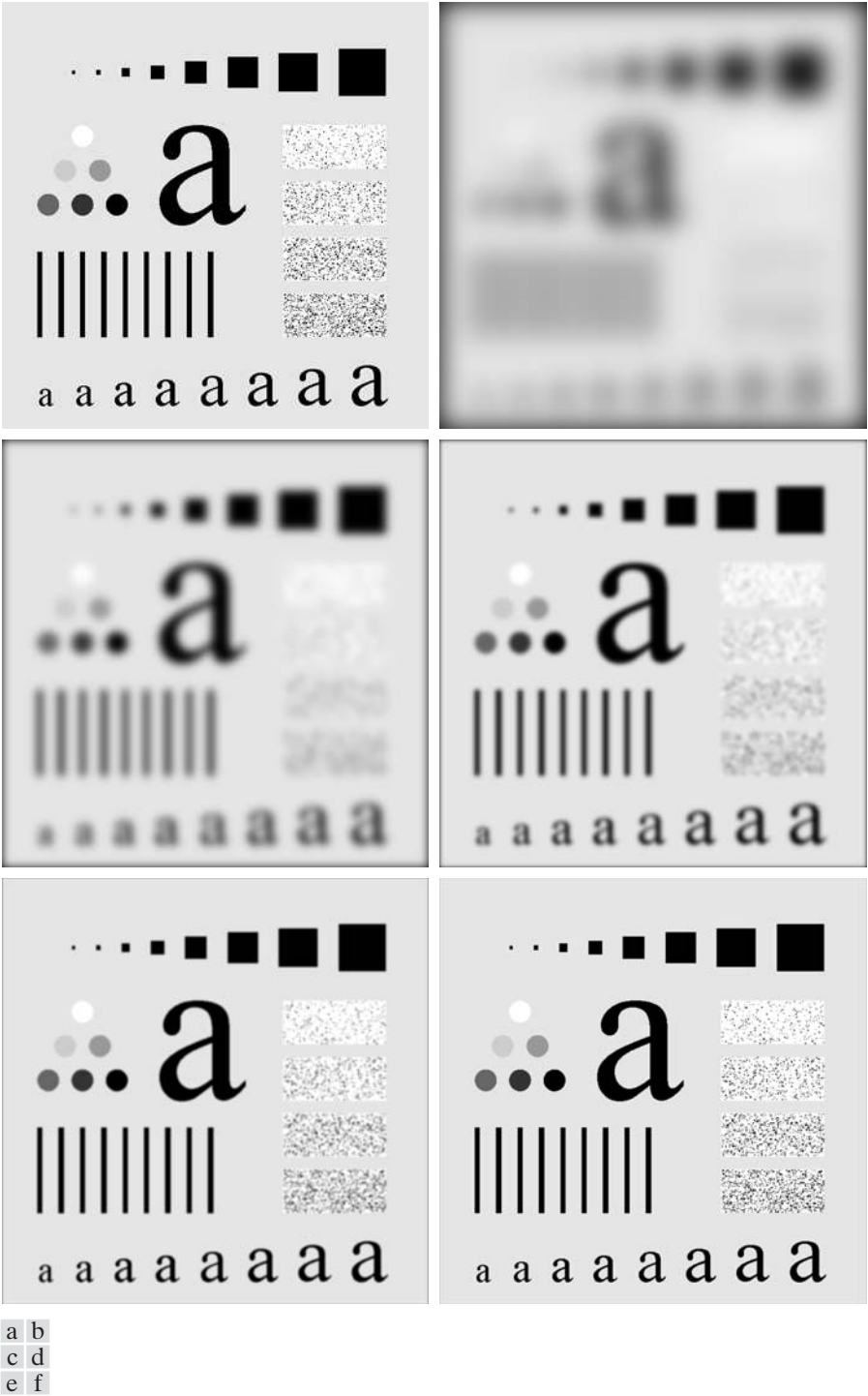


FIGURE 4.48 (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.

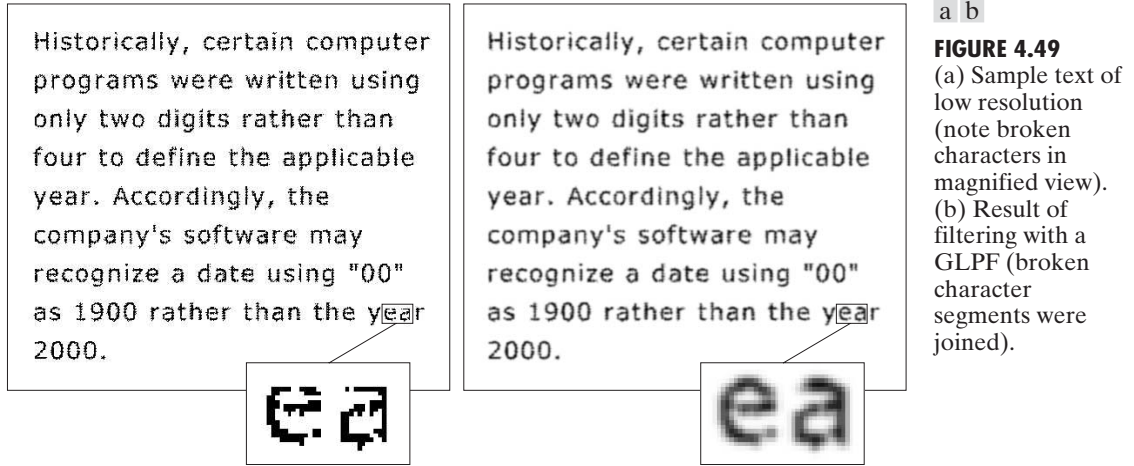


FIGURE 4.49
(a) Sample text of low resolution (note broken characters in magnified view). (b) Result of filtering with a GLPF (broken character segments were joined).

satellite and aerial images. Similar results can be obtained using the lowpass spatial filtering techniques discussed in Section 3.5.

Figure 4.49 shows a sample of text of poor resolution. One encounters text like this, for example, in fax transmissions, duplicated material, and historical records. This particular sample is free of additional difficulties like smudges, creases, and torn sections. The magnified section in Fig. 4.49(a) shows that the characters in this document have distorted shapes due to lack of resolution, and many of the characters are broken. Although humans fill these gaps visually without difficulty, machine recognition systems have real difficulties reading broken characters. One approach for handling this problem is to bridge small gaps in the input image by blurring it. Figure 4.49(b) shows how well characters can be “repaired” by this simple process using a Gaussian lowpass filter with $D_0 = 80$. The images are of size 444×508 pixels.

Lowpass filtering is a staple in the printing and publishing industry, where it is used for numerous preprocessing functions, including unsharp masking, as discussed in Section 3.6.3. “Cosmetic” processing is another use of lowpass filtering prior to printing. Figure 4.50 shows an application of lowpass filtering for producing a smoother, softer-looking result from a sharp original. For human faces, the typical objective is to reduce the sharpness of fine skin lines and small blemishes. The magnified sections in Figs. 4.50(b) and (c) clearly show a significant reduction in fine skin lines around the eyes in this case. In fact, the smoothed images look quite soft and pleasing.

Figure 4.51 shows two applications of lowpass filtering on the same image, but with totally different objectives. Figure 4.51(a) is an 808×754 very high resolution radiometer (VHRR) image showing part of the Gulf of Mexico (dark) and Florida (light), taken from a NOAA satellite (note the horizontal sensor scan lines). The boundaries between bodies of water were caused by loop currents. This image is illustrative of remotely sensed images in which sensors have the tendency to produce pronounced scan lines along the direction in which the scene is being scanned (see Example 4.24 for an illustration of a

We discuss unsharp masking in the frequency domain in Section 4.9.5



a b c

FIGURE 4.50 (a) Original image (784×732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

physical cause). Lowpass filtering is a crude but simple way to reduce the effect of these lines, as Fig. 4.51(b) shows (we consider more effective approaches in Sections 4.10 and 5.4.1). This image was obtained using a GLPF with $D_0 = 50$. The reduction in the effect of the scan lines can simplify the detection of features such as the interface boundaries between ocean currents.

Figure 4.51(c) shows the result of significantly more aggressive Gaussian lowpass filtering with $D_0 = 20$. Here, the objective is to blur out as much detail as possible while leaving large features recognizable. For instance, this type of filtering could be part of a preprocessing stage for an image analysis system that searches for features in an image bank. An example of such features could be lakes of a given size, such as Lake Okeechobee in the lower eastern region of Florida, shown as a nearly round dark region in Fig. 4.51(c). Lowpass filtering helps simplify the analysis by averaging out features smaller than the ones of interest.

4.9 Image Sharpening Using Frequency Domain Filters

In the previous section, we showed that an image can be smoothed by attenuating the high-frequency components of its Fourier transform. Because edges and other abrupt changes in intensities are associated with high-frequency components, image sharpening can be achieved in the frequency domain by highpass filtering, which attenuates the low-frequency components without disturbing high-frequency information in the Fourier transform. As in Section

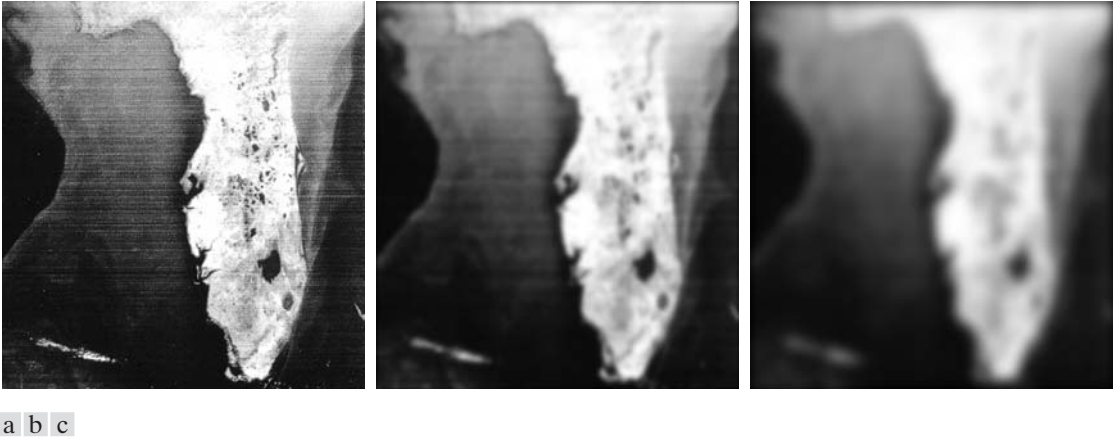


FIGURE 4.51 (a) Image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

4.8, we consider only zero-phase-shift filters that are radially symmetric. All filtering in this section is based on the procedure outlined in Section 4.7.3, so all filter functions, $H(u, v)$, are understood to be discrete functions of size $P \times Q$; that is, the discrete frequency variables are in the range $u = 0, 1, 2, \dots, P - 1$ and $v = 0, 1, 2, \dots, Q - 1$.

A highpass filter is obtained from a given lowpass filter using the equation

$$H_{\text{HP}}(u, v) = 1 - H_{\text{LP}}(u, v) \quad (4.9-1)$$

where $H_{\text{LP}}(u, v)$ is the transfer function of the lowpass filter. That is, when the lowpass filter attenuates frequencies, the highpass filter passes them, and vice versa.

In this section, we consider ideal, Butterworth, and Gaussian highpass filters. As in the previous section, we illustrate the characteristics of these filters in both the frequency and spatial domains. Figure 4.52 shows typical 3-D plots, image representations, and cross sections for these filters. As before, we see that the Butterworth filter represents a transition between the sharpness of the ideal filter and the broad smoothness of the Gaussian filter. Figure 4.53, discussed in the sections that follow, illustrates what these filters look like in the spatial domain. The spatial filters were obtained and displayed by using the procedure used to generate Figs. 4.43 and 4.46.

4.9.1 Ideal Highpass Filters

A 2-D *ideal highpass filter* (IHPF) is defined as

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases} \quad (4.9-2)$$

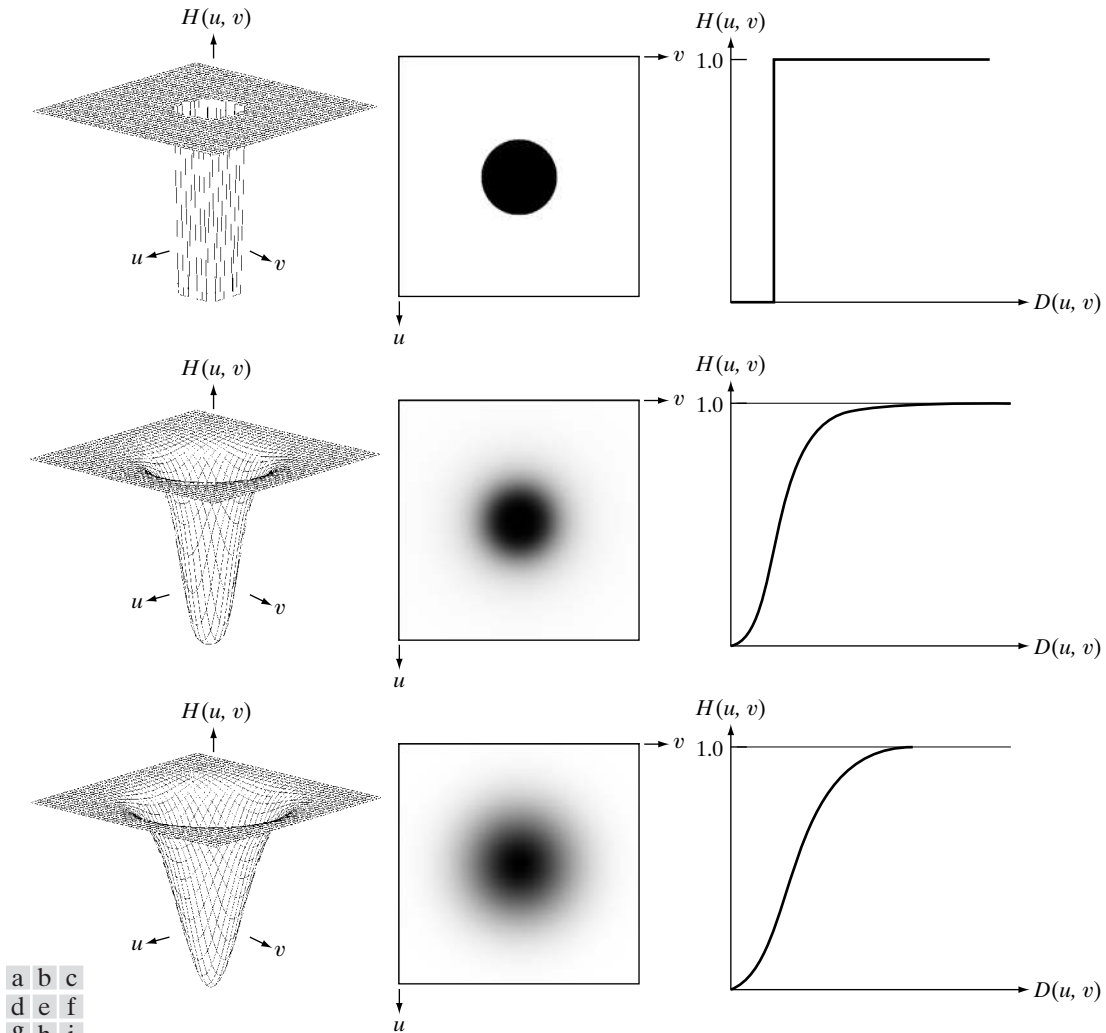


FIGURE 4.52 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

where D_0 is the cutoff frequency and $D(u, v)$ is given by Eq. (4.8-2). This expression follows directly from Eqs. (4.8-1) and (4.9-1). As intended, the IHPF is the opposite of the ILPF in the sense that it sets to zero all frequencies inside a circle of radius D_0 while passing, without attenuation, all frequencies outside the circle. As in the case of the ILPF, the IHPF is not physically realizable. However, we consider it here for completeness and, as before, because its properties can be used to explain phenomena such as ringing in the spatial domain. The discussion will be brief.

Because of the way in which they are related [Eq. (4.9-1)], we can expect IHPFs to have the same ringing properties as ILPFs. This is demonstrated

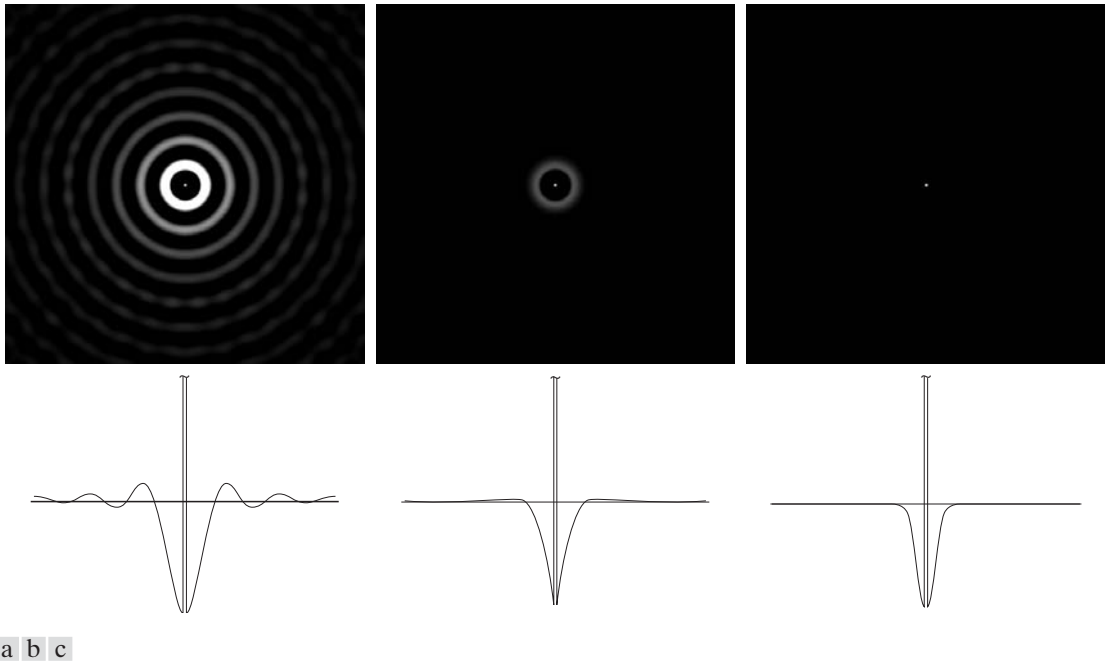


FIGURE 4.53 Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

clearly in Fig. 4.54, which consists of various IHPF results using the original image in Fig. 4.41(a) with D_0 set to 30, 60, and 160 pixels, respectively. The ringing in Fig. 4.54(a) is so severe that it produced distorted, thickened object boundaries (e.g., look at the large letter “a”). Edges of the top three circles do not show well because they are not as strong as the other edges in the image (the intensity of these three objects is much closer to the background intensity,

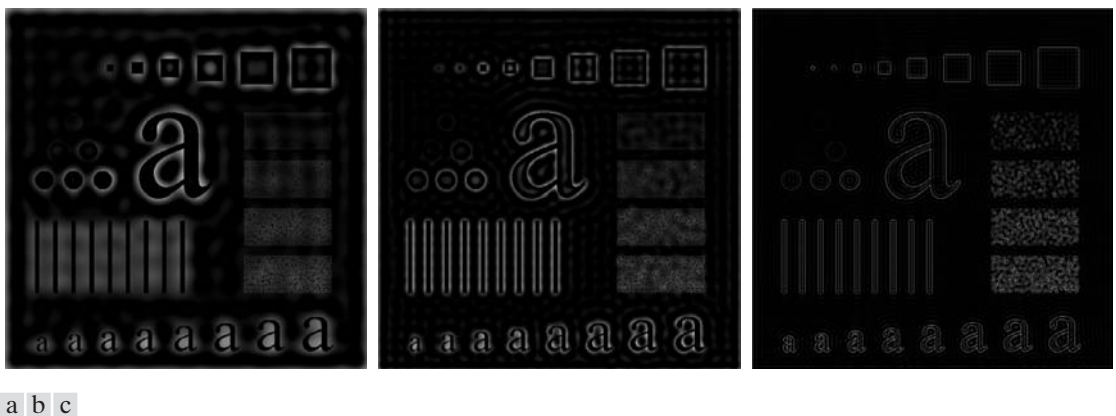


FIGURE 4.54 Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with $D_0 = 30, 60,$ and 160 .

giving discontinuities of smaller magnitude). Looking at the “spot” size of the spatial representation of the IHPF in Fig. 4.53(a) and keeping in mind that filtering in the spatial domain is convolution of the spatial filter with the image helps explain why the smaller objects and lines appear almost solid white. Look in particular at the three small squares in the top row and the thin, vertical bars in Fig. 4.54(a). The situation improved somewhat with $D_0 = 60$. Edge distortion is quite evident still, but now we begin to see filtering on the smaller objects. Due to the now familiar inverse relationship between the frequency and spatial domains, we know that the spot size of this filter is smaller than the spot of the filter with $D_0 = 30$. The result for $D_0 = 160$ is closer to what a highpass-filtered image should look like. Here, the edges are much cleaner and less distorted, and the smaller objects have been filtered properly. Of course, the constant background in all images is zero in these highpass-filtered images because highpass filtering is analogous to differentiation in the spatial domain.

4.9.2 Butterworth Highpass Filters

A 2-D *Butterworth highpass filter* (BHPF) of order n and cutoff frequency D_0 is defined as

$$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}} \quad (4.9-3)$$

where $D(u, v)$ is given by Eq. (4.8-2). This expression follows directly from Eqs. (4.8-5) and (4.9-1). The middle row of Fig. 4.52 shows an image and cross section of the BHPF function.

As with lowpass filters, we can expect Butterworth highpass filters to behave smoother than IHPFs. Figure 4.55 shows the performance of a BHPF, of

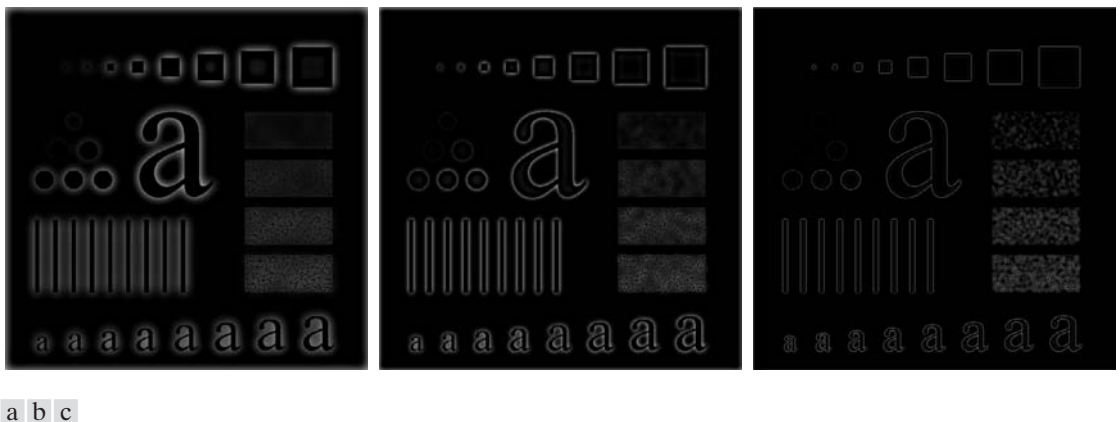


FIGURE 4.55 Results of highpass filtering the image in Fig. 4.41(a) using a BHPF of order 2 with $D_0 = 30, 60$, and 160, corresponding to the circles in Fig. 4.41(b). These results are much smoother than those obtained with an IHPF.

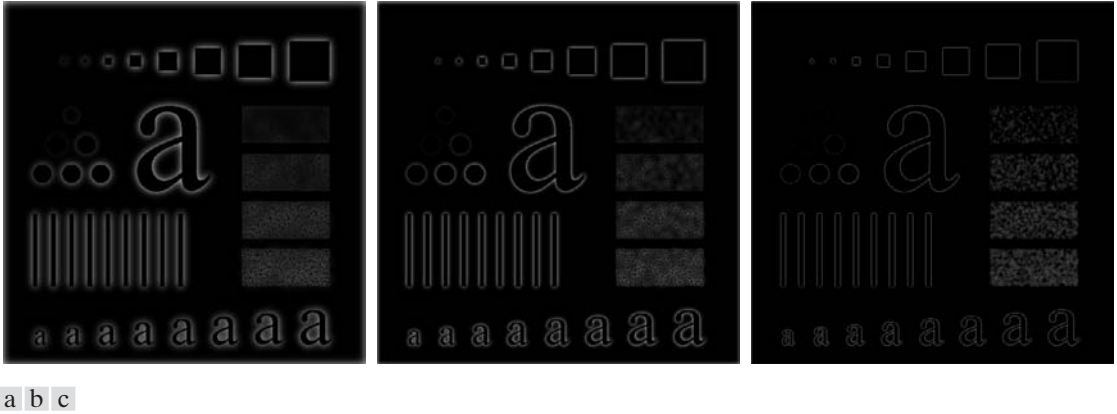


FIGURE 4.56 Results of highpass filtering the image in Fig. 4.41(a) using a GHPF with $D_0 = 30, 60$, and 160 , corresponding to the circles in Fig. 4.41(b). Compare with Figs. 4.54 and 4.55.

order 2 and with D_0 set to the same values as in Fig. 4.54. The boundaries are much less distorted than in Fig. 4.54, even for the smallest value of cutoff frequency. Because the spot sizes in the center areas of the IHPF and the BHPF are similar [see Figs. 4.53(a) and (b)], the performance of the two filters on the smaller objects is comparable. The transition into higher values of cutoff frequencies is much smoother with the BHPF.

4.9.3 Gaussian Highpass Filters

The transfer function of the Gaussian highpass filter (GHPF) with cutoff frequency locus at a distance D_0 from the center of the frequency rectangle is given by

$$H(u, v) = 1 - e^{-D^2(u,v)/2D_0^2} \quad (4.9-4)$$

where $D(u, v)$ is given by Eq. (4.8-2). This expression follows directly from Eqs. (4.8-7) and (4.9-1). The third row in Fig. 4.52 shows a perspective plot, image, and cross section of the GHPF function. Following the same format as for the BHPF, we show in Fig. 4.56 comparable results using GHPFs. As expected, the results obtained are more gradual than with the previous two filters. Even the filtering of the smaller objects and thin bars is cleaner with the Gaussian filter. Table 4.5 contains a summary of the highpass filters discussed in this section.

TABLE 4.5

Highpass filters. D_0 is the cutoff frequency and n is the order of the Butterworth filter.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$	$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$	$H(u, v) = 1 - e^{-D^2(u,v)/2D_0^2}$

EXAMPLE 4.19:

Using highpass filtering and thresholding for image enhancement.

The value $D_0 = 50$ is approximately 2.5% of the short dimension of the padded image. The idea is for D_0 to be close to the origin so low frequencies are attenuated, but not completely eliminated. A range of 2% to 5% of the short dimension is a good starting point.

■ Figure 4.57(a) is a 1026×962 image of a thumb print in which smudges (a typical problem) are evident. A key step in automated fingerprint recognition is enhancement of print ridges and the reduction of smudges. Enhancement is useful also in human interpretation of prints. In this example, we use highpass filtering to enhance the ridges and reduce the effects of smudging. Enhancement of the ridges is accomplished by the fact that they contain high frequencies, which are unchanged by a highpass filter. On the other hand, the filter reduces low frequency components, which correspond to slowly varying intensities in the image, such as the background and smudges. Thus, enhancement is achieved by reducing the effect of all features except those with high frequencies, which are the features of interest in this case.

Figure 4.57(b) is the result of using a Butterworth highpass filter of order 4 with a cutoff frequency of 50. As expected, the highpass-filtered image lost its gray tones because the dc term was reduced to 0. The net result is that dark tones typically predominate in highpass-filtered images, thus requiring additional processing to enhance details of interest. A simple approach is to threshold the filtered image. Figure 4.57(c) shows the result of setting to black all negative values and to white all positive values in the filtered image. Note how the ridges are clear and the effect of the smudges has been reduced considerably. In fact, ridges that are barely visible in the top, right section of the image in Fig. 4.57(a) are nicely enhanced in Fig. 4.57(c). ■

4.9.4 The Laplacian in the Frequency Domain

In Section 3.6.2, we used the Laplacian for image enhancement in the spatial domain. In this section, we revisit the Laplacian and show that it yields equivalent results using frequency domain techniques. It can be shown (Problem 4.26) that the Laplacian can be implemented in the frequency domain using the filter

$$H(u, v) = -4\pi^2(u^2 + v^2) \quad (4.9-5)$$

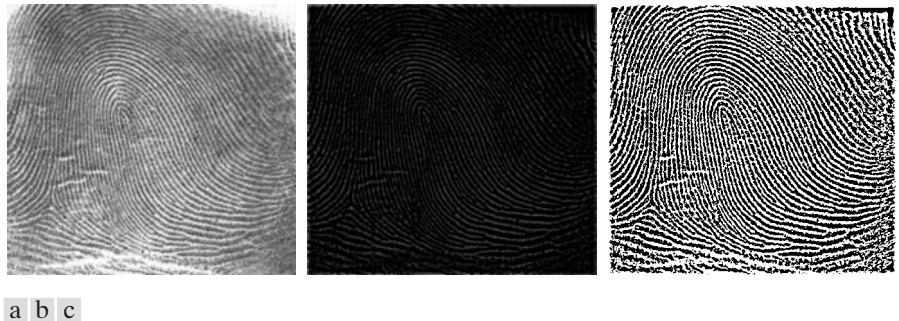


FIGURE 4.57 (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

or, with respect to the center of the frequency rectangle, using the filter

$$\begin{aligned} H(u, v) &= -4\pi^2[(u - P/2)^2 + (v - Q/2)^2] \\ &= -4\pi^2 D^2(u, v) \end{aligned} \quad (4.9-6)$$

where $D(u, v)$ is the distance function given in Eq. (4.8-2). Then, the Laplacian image is obtained as:

$$\nabla^2 f(x, y) = \mathfrak{F}^{-1}\{H(u, v)F(u, v)\} \quad (4.9-7)$$

where $F(u, v)$ is the DFT of $f(x, y)$. As explained in Section 3.6.2, enhancement is achieved using the equation:

$$g(x, y) = f(x, y) + c\nabla^2 f(x, y) \quad (4.9-8)$$

Here, $c = -1$ because $H(u, v)$ is negative. In Chapter 3, $f(x, y)$ and $\nabla^2 f(x, y)$ had comparable values. However, computing $\nabla^2 f(x, y)$ with Eq. (4.9-7) introduces DFT scaling factors that can be several orders of magnitude larger than the maximum value of f . Thus, the differences between f and its Laplacian must be brought into comparable ranges. The easiest way to handle this problem is to normalize the values of $f(x, y)$ to the range $[0, 1]$ (before computing its DFT) and divide $\nabla^2 f(x, y)$ by its maximum value, which will bring it to the approximate range $[-1, 1]$ (recall that the Laplacian has negative values). Equation (4.9-8) can then be applied.

In the frequency domain, Eq. (4.9-8) is written as

$$\begin{aligned} g(x, y) &= \mathfrak{F}^{-1}\{F(u, v) - H(u, v)F(u, v)\} \\ &= \mathfrak{F}^{-1}\{[1 - H(u, v)]F(u, v)\} \\ &= \mathfrak{F}^{-1}\{[1 + 4\pi^2 D^2(u, v)]F(u, v)\} \end{aligned} \quad (4.9-9)$$

Although this result is elegant, it has the same scaling issues just mentioned, compounded by the fact that the normalizing factor is not as easily computed. For this reason, Eq. (4.9-8) is the preferred implementation in the frequency domain, with $\nabla^2 f(x, y)$ computed using Eq. (4.9-7) and scaled using the approach mentioned in the previous paragraph.

■ Figure 4.58(a) is the same as Fig. 3.38(a), and Fig. 4.58(b) shows the result of using Eq. (4.9-8), in which the Laplacian was computed in the frequency domain using Eq. (4.9-7). Scaling was done as described in connection with that equation. We see by comparing Figs. 4.58(b) and 3.38(e) that the frequency domain and spatial results are identical visually. Observe that the results in these two figures correspond to the Laplacian mask in Fig. 3.37(b), which has a -8 in the center (Problem 4.26). ■

EXAMPLE 4.20: Image sharpening in the frequency domain using the Laplacian.

a b

FIGURE 4.58

(a) Original, blurry image.
 (b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.38(e).



4.9.5 Unsharp Masking, Highboost Filtering, and High-Frequency-Emphasis Filtering

In this section, we discuss frequency domain formulations of the unsharp masking and high-boost filtering image sharpening techniques introduced in Section 3.6.3. Using frequency domain methods, the mask defined in Eq. (3.6-8) is given by

$$g_{\text{mask}}(x, y) = f(x, y) - f_{\text{LP}}(x, y) \quad (4.9-10)$$

with

$$f_{\text{LP}}(x, y) = \mathfrak{F}^{-1}[H_{\text{LP}}(u, v)F(u, v)] \quad (4.9-11)$$

where $H_{\text{LP}}(u, v)$ is a lowpass filter and $F(u, v)$ is the Fourier transform of $f(x, y)$. Here, $f_{\text{LP}}(x, y)$ is a smoothed image analogous to $\bar{f}(x, y)$ in Eq. (3.6-8). Then, as in Eq. (3.6-9),

$$g(x, y) = f(x, y) + k * g_{\text{mask}}(x, y) \quad (4.9-12)$$

This expression defines unsharp masking when $k = 1$ and highboost filtering when $k > 1$. Using the preceding results, we can express Eq. (4.9-12) entirely in terms of frequency domain computations involving a lowpass filter:

$$g(x, y) = \mathfrak{F}^{-1}\{[1 + k * [1 - H_{\text{LP}}(u, v)]]F(u, v)\} \quad (4.9-13)$$

Using Eq. (4.9-1), we can express this result in terms of a highpass filter:

$$g(x, y) = \mathfrak{F}^{-1}\{[1 + k * H_{\text{HP}}(u, v)]F(u, v)\} \quad (4.9-14)$$

The expression contained within the square brackets is called a *high-frequency-emphasis filter*. As noted earlier, highpass filters set the dc term to zero, thus reducing the average intensity in the filtered image to 0. The high-frequency-emphasis filter does not have this problem because of the 1 that is added to the highpass filter. The constant, k , gives control over the proportion of high frequencies that influence the final result. A slightly more general formulation of high-frequency-emphasis filtering is the expression

$$g(x, y) = \mathfrak{F}^{-1}\{[k_1 + k_2 * H_{HP}(u, v)]F(u, v)\} \quad (4.9-15)$$

where $k_1 \geq 0$ gives controls of the offset from the origin [see Fig. 4.31(c)] and $k_2 \geq 0$ controls the contribution of high frequencies.

■ Figure 4.59(a) shows a 416×596 chest X-ray with a narrow range of intensity levels. The objective of this example is to enhance the image using high-frequency-emphasis filtering. X-rays cannot be focused in the same manner that optical lenses are focused, and the resulting images generally tend to be slightly blurred. Because the intensities in this particular image are biased toward the dark end of the gray scale, we also take this opportunity to give an example of how spatial domain processing can be used to complement frequency-domain filtering.

Figure 4.59(b) shows the result of highpass filtering using a Gaussian filter with $D_0 = 40$ (approximately 5% of the short dimension of the padded image). As expected, the filtered result is rather featureless, but it shows faintly the principal edges in the image. Figure 4.59(c) shows the advantage of high-emphasis filtering, where we used Eq. (4.9-15) with $k_1 = 0.5$ and $k_2 = 0.75$. Although the image is still dark, the gray-level tonality due to the low-frequency components was not lost.

As discussed in Section 3.3.1, an image characterized by intensity levels in a narrow range of the gray scale is an ideal candidate for histogram equalization. As Fig. 4.59(d) shows, this was indeed an appropriate method to further enhance the image. Note the clarity of the bone structure and other details that simply are not visible in any of the other three images. The final enhanced image is a little noisy, but this is typical of X-ray images when their gray scale is expanded. The result obtained using a combination of high-frequency emphasis and histogram equalization is superior to the result that would be obtained by using either method alone. ■

EXAMPLE 4.21:
Image enhancement using high-frequency-emphasis filtering.

Artifacts such as ringing are unacceptable in medical imaging. Thus, it is good practice to avoid using filters that have the potential for introducing artifacts in the processed image. Because spatial and frequency domain Gaussian filters are Fourier transform pairs, these filters produce smooth results that are void of artifacts.

4.9.6 Homomorphic Filtering

The illumination-reflectance model introduced in Section 2.3.4 can be used to develop a frequency domain procedure for improving the appearance of an image by simultaneous intensity range compression and contrast enhancement. From the discussion in that section, an image $f(x, y)$ can be expressed as the product of its illumination, $i(x, y)$, and reflectance, $r(x, y)$, components:

$$f(x, y) = i(x, y)r(x, y) \quad (4.9-16)$$

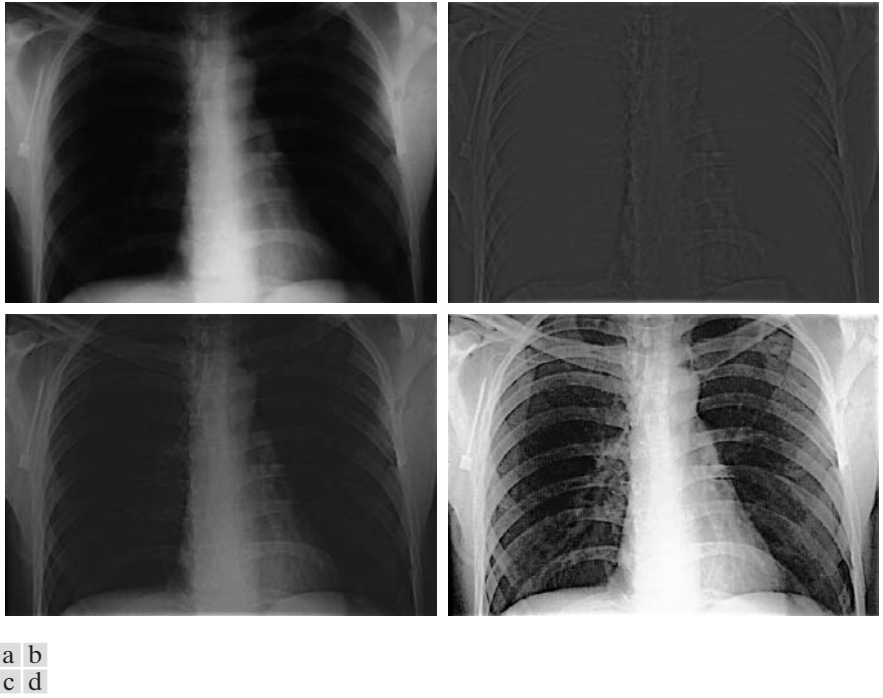


FIGURE 4.59 (a) A chest X-ray image. (b) Result of highpass filtering with a Gaussian filter. (c) Result of high-frequency-emphasis filtering using the same filter. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

This equation cannot be used directly to operate on the frequency components of illumination and reflectance because the Fourier transform of a product is not the product of the transforms:

$$\mathfrak{F}[f(x, y)] \neq \mathfrak{F}[i(x, y)]\mathfrak{F}[r(x, y)] \quad (4.9-17)$$

However, suppose that we define

$$\begin{aligned} z(x, y) &= \ln f(x, y) \\ &= \ln i(x, y) + \ln r(x, y) \end{aligned} \quad (4.9-18)$$

Then,

$$\begin{aligned} \mathfrak{F}\{z(x, y)\} &= \mathfrak{F}\{\ln f(x, y)\} \\ &= \mathfrak{F}\{\ln i(x, y)\} + \mathfrak{F}\{\ln r(x, y)\} \end{aligned} \quad (4.9-19)$$

or

$$Z(u, v) = F_i(u, v) + F_r(u, v) \quad (4.9-20)$$

If an image $f(x, y)$ with intensities in the range $[0, L - 1]$ has any 0 values, a 1 must be added to every element of the image to avoid having to deal with $\ln(0)$. The 1 is then subtracted at the end of the filtering process.

where $F_i(u, v)$ and $F_r(u, v)$ are the Fourier transforms of $\ln i(x, y)$ and $\ln r(x, y)$, respectively.

We can filter $Z(u, v)$ using a filter $H(u, v)$ so that

$$\begin{aligned} S(u, v) &= H(u, v)Z(u, v) \\ &= H(u, v)F_i(u, v) + H(u, v)F_r(u, v) \end{aligned} \quad (4.9-21)$$

The filtered image in the spatial domain is

$$\begin{aligned} s(x, y) &= \mathfrak{F}^{-1}\{S(u, v)\} \\ &= \mathfrak{F}^{-1}\{H(u, v)F_i(u, v)\} + \mathfrak{F}^{-1}\{H(u, v)F_r(u, v)\} \end{aligned} \quad (4.9-22)$$

By defining

$$i'(x, y) = \mathfrak{F}^{-1}\{H(u, v)F_i(u, v)\} \quad (4.9-23)$$

and

$$r'(x, y) = \mathfrak{F}^{-1}\{H(u, v)F_r(u, v)\} \quad (4.9-24)$$

we can express Eq. (4.9-23) in the form

$$s(x, y) = i'(x, y) + r'(x, y) \quad (4.9-25)$$

Finally, because $z(x, y)$ was formed by taking the natural logarithm of the input image, we reverse the process by taking the exponential of the filtered result to form the output image:

$$\begin{aligned} g(x, y) &= e^{s(x, y)} \\ &= e^{i'(x, y)}e^{r'(x, y)} \\ &= i_0(x, y)r_0(x, y) \end{aligned} \quad (4.9-26)$$

where

$$i_0(x, y) = e^{i'(x, y)} \quad (4.9-27)$$

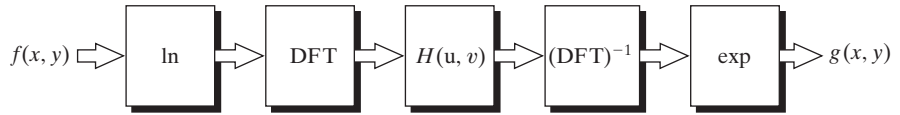
and

$$r_0(x, y) = e^{r'(x, y)} \quad (4.9-28)$$

are the illumination and reflectance components of the output (processed) image.

FIGURE 4.60

Summary of steps in homomorphic filtering.



The filtering approach just derived is summarized in Fig. 4.60. This method is based on a special case of a class of systems known as *homomorphic systems*. In this particular application, the key to the approach is the separation of the illumination and reflectance components achieved in the form shown in Eq. (4.9-20). The *homomorphic filter function* $H(u, v)$ then can operate on these components separately, as indicated by Eq. (4.9-21).

The illumination component of an image generally is characterized by slow spatial variations, while the reflectance component tends to vary abruptly, particularly at the junctions of dissimilar objects. These characteristics lead to associating the low frequencies of the Fourier transform of the logarithm of an image with illumination and the high frequencies with reflectance. Although these associations are rough approximations, they can be used to advantage in image filtering, as illustrated in Example 4.22.

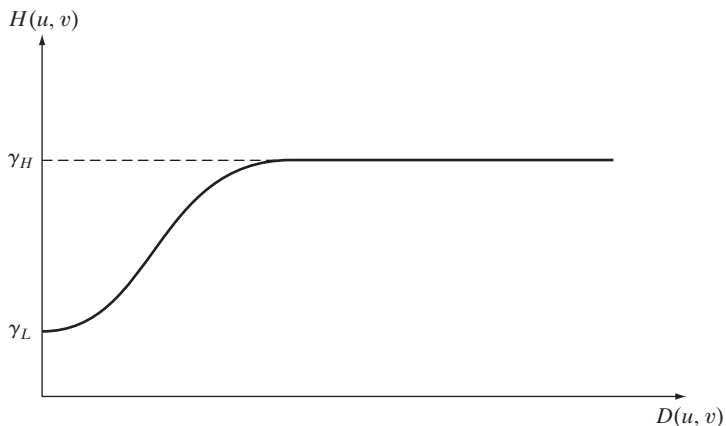
A good deal of control can be gained over the illumination and reflectance components with a homomorphic filter. This control requires specification of a filter function $H(u, v)$ that affects the low- and high-frequency components of the Fourier transform in different, controllable ways. Figure 4.61 shows a cross section of such a filter. If the parameters γ_L and γ_H are chosen so that $\gamma_L < 1$ and $\gamma_H > 1$, the filter function in Fig. 4.61 tends to attenuate the contribution made by the low frequencies (illumination) and amplify the contribution made by high frequencies (reflectance). The net result is simultaneous dynamic range compression and contrast enhancement.

The shape of the function in Fig. 4.61 can be approximated using the basic form of a highpass filter. For example, using a slightly modified form of the Gaussian highpass filter yields the function

$$H(u, v) = (\gamma_H - \gamma_L) \left[1 - e^{-c[D^2(u, v)/D_0^2]} \right] + \gamma_L \quad (4.9-29)$$

FIGURE 4.61

Radial cross section of a circularly symmetric homomorphic filter function. The vertical axis is at the center of the frequency rectangle and $D(u, v)$ is the distance from the center.



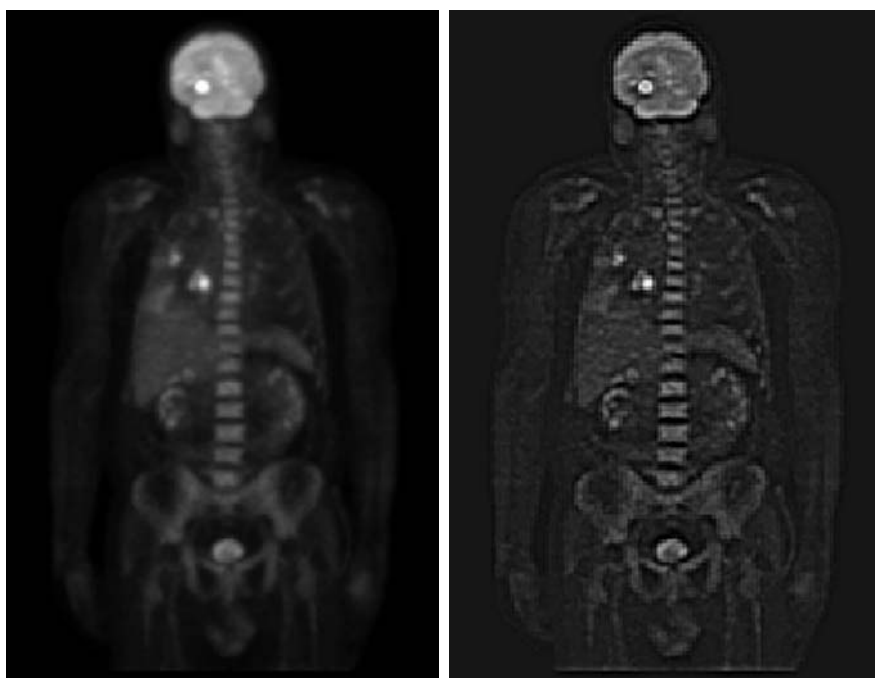
where $D(u, v)$ is defined in Eq. (4.8-2) and the constant c controls the sharpness of the slope of the function as it transitions between γ_L and γ_H . This filter is similar to the high-emphasis filter discussed in the previous section.

■ Figure 4.62(a) shows a full body PET (Positron Emission Tomography) scan of size 1162×746 pixels. The image is slightly blurry and many of its low-intensity features are obscured by the high intensity of the “hot spots” dominating the dynamic range of the display. (These hot spots were caused by a tumor in the brain and one in the lungs.) Figure 4.62(b) was obtained by homomorphic filtering Fig. 4.62(a) using the filter in Eq. (4.9-29) with $\gamma_L = 0.25$, $\gamma_H = 2$, $c = 1$, and $D_0 = 80$. A cross section of this filter looks just like Fig. 4.61, with a slightly steeper slope.

Note in Fig. 4.62(b) how much sharper the hot spots, the brain, and the skeleton are in the processed image, and how much more detail is visible in this image. By reducing the effects of the dominant illumination components (the hot spots), it became possible for the dynamic range of the display to allow lower intensities to become much more visible. Similarly, because the high frequencies are enhanced by homomorphic filtering, the reflectance components of the image (edge information) were sharpened considerably. The enhanced image in Fig. 4.62(b) is a significant improvement over the original.

EXAMPLE 4.22:
Image enhancement using homomorphic filtering.

Recall that filtering uses image padding, so the filter is of size $P \times Q$.



a b

FIGURE 4.62
(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI PET Systems.)

4.10 Selective Filtering

The filters discussed in the previous two sections operate over the entire frequency rectangle. There are applications in which it is of interest to process specific bands of frequencies or small regions of the frequency rectangle. Filters in the first category are called *bandreject* or *bandpass filters*, respectively. Filters in the second category are called *notch filters*.

4.10.1 Bandreject and Bandpass Filters

These types of filters are easy to construct using the concepts from the previous two sections. Table 4.6 shows expressions for ideal, Butterworth, and Gaussian bandreject filters, where $D(u, v)$ is the distance from the center of the frequency rectangle, as given in Eq. (4.8-2), D_0 is the radial center of the band, and W is the width of the band. Figure 4.63(a) shows a Gaussian bandreject filter in image form, where black is 0 and white is 1.

A bandpass filter is obtained from a bandreject filter in the same manner that we obtained a highpass filter from a lowpass filter:

$$H_{BP}(u, v) = 1 - H_{BR}(u, v) \quad (4.10-1)$$

Figure 4.63(b) shows a Gaussian bandpass filter in image form.

4.10.2 Notch Filters

Notch filters are the most useful of the selective filters. A notch filter rejects (or passes) frequencies in a predefined neighborhood about the center of the frequency rectangle. Zero-phase-shift filters must be symmetric about the origin, so a notch with center at (u_0, v_0) must have a corresponding notch at location $(-u_0, -v_0)$. Notch reject filters are constructed as products of highpass filters whose centers have been translated to the centers of the notches. The general form is:

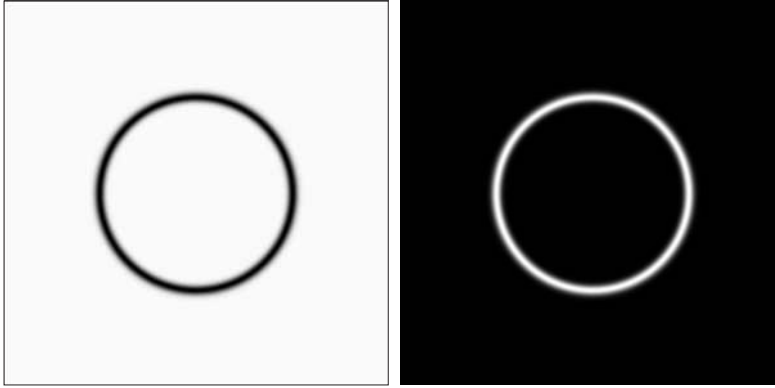
$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v) \quad (4.10-2)$$

where $H_k(u, v)$ and $H_{-k}(u, v)$ are highpass filters whose centers are at (u_k, v_k) and $(-u_k, -v_k)$, respectively. These centers are specified with respect to the

TABLE 4.6

Bandreject filters. W is the width of the band, D is the distance $D(u, v)$ from the center of the filter, D_0 is the cutoff frequency, and n is the order of the Butterworth filter. We show D instead of $D(u, v)$ to simplify the notation in the table.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u, v) = \frac{1}{1 + \left[\frac{DW}{D^2 - D_0^2} \right]^{2n}}$	$H(u, v) = 1 - e^{-\left[\frac{D^2 - D_0^2}{DW} \right]^2}$



a b

FIGURE 4.63

(a) Bandreject Gaussian filter.
 (b) Corresponding bandpass filter.
 The thin black border in (a) was added for clarity; it is not part of the data.

center of the frequency rectangle, $(M/2, N/2)$. The distance computations for each filter are thus carried out using the expressions

$$D_k(u, v) = [(u - M/2 - u_k)^2 + (v - N/2 - v_k)^2]^{1/2} \quad (4.10-3)$$

and

$$D_{-k}(u, v) = [(u - M/2 + u_k)^2 + (v - N/2 + v_k)^2]^{1/2} \quad (4.10-4)$$

For example, the following is a Butterworth notch reject filter of order n , containing three notch pairs:

$$H_{NR}(u, v) = \prod_{k=1}^3 \left[\frac{1}{1 + [D_{0k}/D_k(u, v)]^{2n}} \right] \left[\frac{1}{1 + [D_{0k}/D_{-k}(u, v)]^{2n}} \right] \quad (4.10-5)$$

where D_k and D_{-k} are given by Eqs. (4.10-3) and (4.10-4). The constant D_{0k} is the same for each pair of notches, but it can be different for different pairs. Other notch reject filters are constructed in the same manner, depending on the highpass filter chosen. As with the filters discussed earlier, a *notch pass filter* is obtained from a notch reject filter using the expression

$$H_{NP}(u, v) = 1 - H_{NR}(u, v) \quad (4.10-6)$$

As the next three examples show, one of the principal applications of notch filtering is for selectively modifying local regions of the DFT. This type of processing typically is done interactively, working directly on DFTs obtained without padding. The advantages of working interactively with actual DFTs (as opposed to having to “translate” from padded to actual frequency values) outweigh any wraparound errors that may result from not using padding in the filtering process. Also, as we show in Section 5.4.4, even more powerful notch filtering techniques than those discussed here are based on unpadded DFTs. To get an idea of how DFT values change as a function of padding, see Problem 4.22.

EXAMPLE 4.23:

Reduction of
moiré patterns
using notch
filtering.

■ Figure 4.64(a) is the scanned newspaper image from Fig. 4.21, showing a prominent moiré pattern, and Fig. 4.64(b) is its spectrum. We know from Table 4.3 that the Fourier transform of a pure sine, which is a periodic function, is a pair of conjugate symmetric impulses. The symmetric “impulse-like” bursts in Fig. 4.64(b) are a result of the near periodicity of the moiré pattern. We can attenuate these bursts by using notch filtering.

a	b
c	d

FIGURE 4.64

(a) Sampled newspaper image showing a moiré pattern.
(b) Spectrum.
(c) Butterworth notch reject filter multiplied by the Fourier transform.
(d) Filtered image.

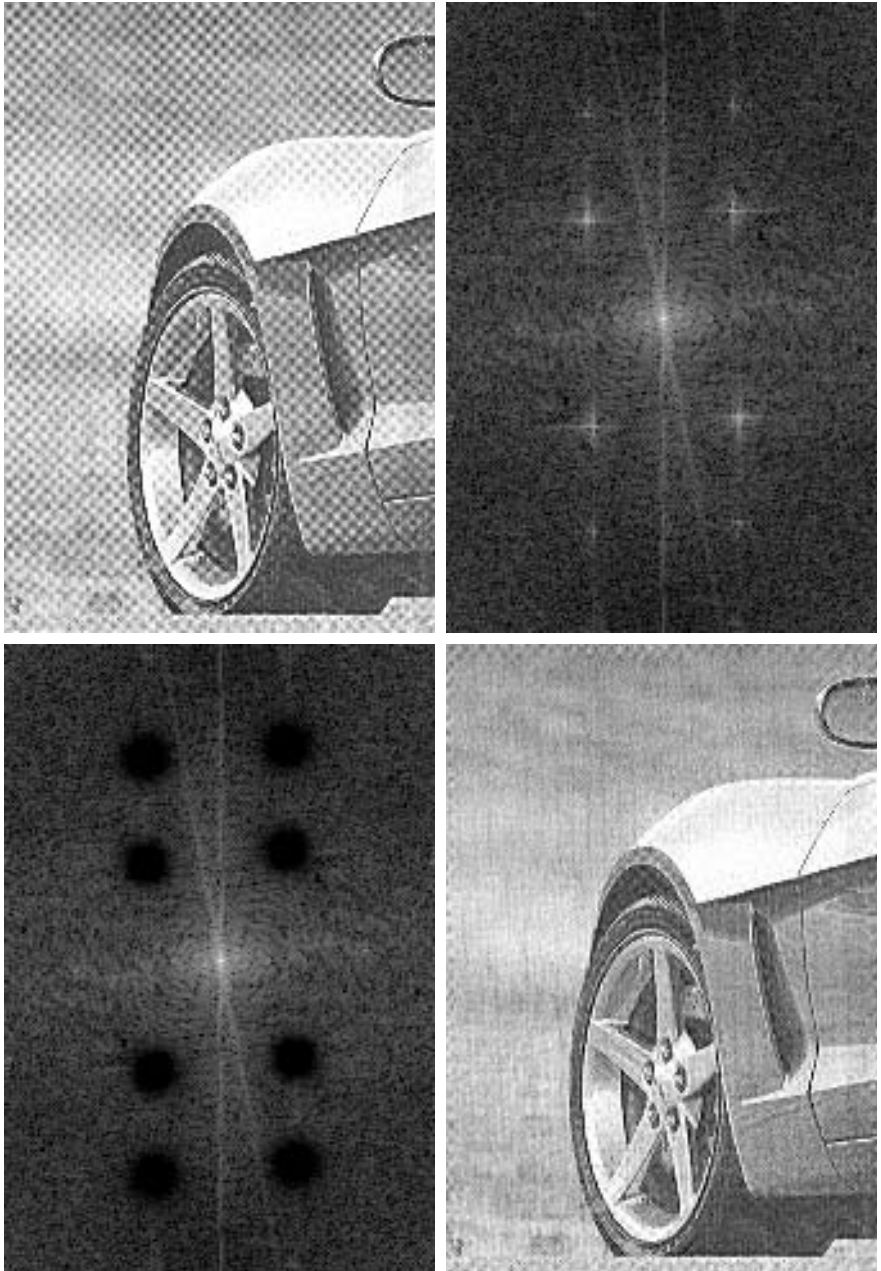


Figure 4.64(c) shows the result of multiplying the DFT of Fig. 4.64(a) by a Butterworth notch reject filter with $D_0 = 3$ and $n = 4$ for all notch pairs. The value of the radius was selected (by visual inspection of the spectrum) to encompass the energy bursts completely, and the value of n was selected to give notches with mildly sharp transitions. The locations of the center of the notches were determined interactively from the spectrum. Figure 4.64(d) shows the result obtained with this filter using the procedure outlined in Section 4.7.3. The improvement is significant, considering the low resolution and degradation of the original image.

■ Figure 4.65(a) shows an image of part of the rings surrounding the planet Saturn. This image was captured by *Cassini*, the first spacecraft to enter the planet's orbit. The vertical sinusoidal pattern was caused by an AC signal superimposed on the camera video signal just prior to digitizing the image. This was an unexpected problem that corrupted some images from the mission. Fortunately, this type of interference is fairly easy to correct by postprocessing. One approach is to use notch filtering.

Figure 4.65(b) shows the DFT spectrum. Careful analysis of the vertical axis reveals a series of small bursts of energy which correspond to the nearly sinusoidal

EXAMPLE 4.24: Enhancement of corrupted *Cassini* Saturn image by notch filtering.

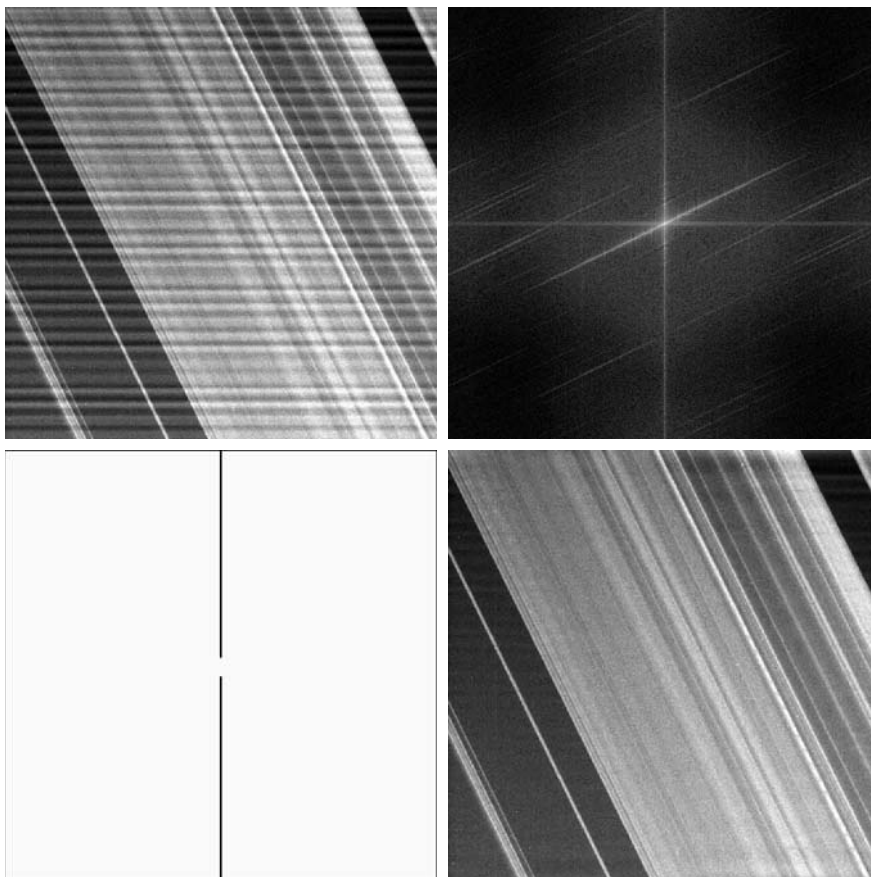
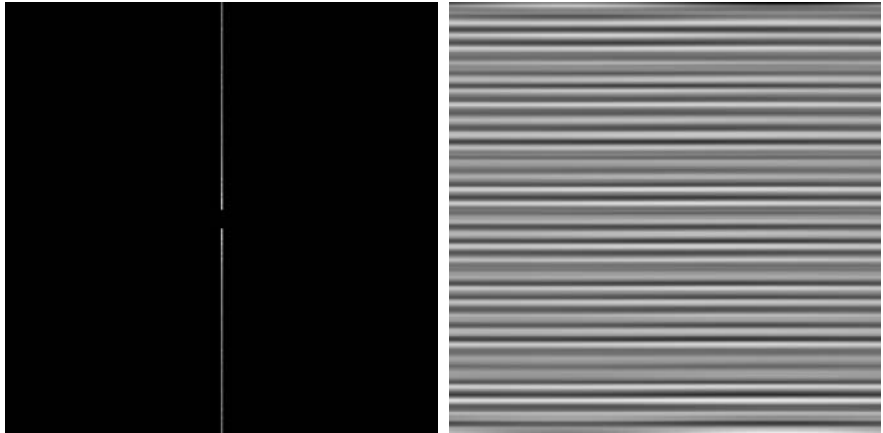


FIGURE 4.65
(a) 674×674 image of the Saturn rings showing nearly periodic interference. (b) Spectrum: The bursts of energy in the vertical axis near the origin correspond to the interference pattern. (c) A vertical notch reject filter. (d) Result of filtering. The thin black border in (c) was added for clarity; it is not part of the data. (Original image courtesy of Dr. Robert A. West, NASA/JPL.)

a b

FIGURE 4.66

(a) Result (spectrum) of applying a notch pass filter to the DFT of Fig. 4.65(a).
 (b) Spatial pattern obtained by computing the IDFT of (a).



interference. A simple approach is to use a narrow notch rectangle filter starting with the lowest frequency burst and extending for the remaining of the vertical axis. Figure 4.65(c) shows such a filter (white represents 1 and black 0). Figure 4.65(d) shows the result of filtering the corrupted image with this filter. This result is a significant improvement over the original image.

We isolated the frequencies in the vertical axis using a notch *pass* version of the same filter [Fig. 4.66(a)]. Then, as Fig. 4.66(b) shows, the IDFT of these frequencies yielded the spatial interference pattern itself. ■

4.11 Implementation

We have focused attention thus far on theoretical concepts and on examples of filtering in the frequency domain. One thing that should be clear by now is that computational requirements in this area of image processing are not trivial. Thus, it is important to develop a basic understanding of methods by which Fourier transform computations can be simplified and speeded up. This section deals with these issues.

4.11.1 Separability of the 2-D DFT

As mentioned in Table 4.2, the 2-D DFT is separable into 1-D transforms. We can write Eq. (4.5-15) as

$$\begin{aligned}
 F(u, v) &= \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N} \\
 &= \sum_{x=0}^{M-1} F(x, v) e^{-j2\pi ux/M}
 \end{aligned} \tag{4.11-1}$$

where

$$F(x, v) = \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi vy/N} \tag{4.11-2}$$

For each value of x and for $v = 0, 1, 2, \dots, N - 1$, we see that $F(x, v)$ is simply the 1-D DFT of a row of $f(x, y)$. By varying x from 0 to $M - 1$ in Eq. (4.11-2), we compute a set of 1-D DFTs for all rows of $f(x, y)$. The computations in Eq. (4.11-1) similarly are 1-D transforms of the columns of $F(x, v)$.

Thus, we conclude that the 2-D DFT of $f(x, y)$ can be obtained by computing the 1-D transform of each row of $f(x, y)$ and then computing the 1-D transform along each column of the result. This is an important simplification because we have to deal only with one variable at a time. A similar development applies to computing the 2-D IDFT using the 1-D IDFT. However, as we show in the following section, we can compute the IDFT using an algorithm designed to compute the DFT.

We could have expressed Eq. (4.11-1) and (4.11-2) in the form of 1-D column transforms followed by row transforms. The final result would have been the same.

4.11.2 Computing the IDFT Using a DFT Algorithm

Taking the complex conjugate of both sides of Eq. (4.5-16) and multiplying the results by MN yields

$$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)} \quad (4.11-3)$$

Multiplication by MN in this development assumes the forms in Eqs. (4.5-15) and (4.5-16). A different constant multiplication scheme is required if the constants are distributed differently between the forward and inverse transforms.

But, we recognize the form of the right side of this result as the DFT of $F^*(u, v)$. Therefore, Eq. (4.11-3) indicates that if we substitute $F^*(u, v)$ into an algorithm designed to compute the 2-D forward Fourier transform, the result will be $MNf^*(x, y)$. Taking the complex conjugate and multiplying this result by MN yields $f(x, y)$, which is the inverse of $F(u, v)$.

Computing the 2-D inverse from a 2-D forward DFT algorithm that is based on successive passes of 1-D transforms (as in the previous section) is a frequent source of confusion involving the complex conjugates and multiplication by a constant, neither of which is done in the 1-D algorithms. The key concept to keep in mind is that we simply input $F^*(u, v)$ into whatever forward algorithm we have. The result will be $MNf^*(x, y)$. All we have to do with this result to obtain $f(x, y)$ is to take its complex conjugate and multiply it by the constant MN . Of course, when $f(x, y)$ is real, as typically is the case, $f^*(x, y) = f(x, y)$.

4.11.3 The Fast Fourier Transform (FFT)

Work in the frequency domain would not be practical if we had to implement Eqs. (4.5-15) and (4.5-16) directly. Brute-force implementation of these equations requires on the order of $(MN)^2$ summations and additions. For images of moderate size (say, 1024×1024 pixels), this means on the order of a trillion multiplications and additions for just one DFT, excluding the exponentials, which could be computed once and stored in a look-up table. This would be a challenge even for super computers. Without the discovery of the *fast Fourier transform* (FFT), which reduces computations to the order of $MN \log_2 MN$ multiplications and additions, it is safe to say that the material presented in this chapter would be of little practical value. The computational reductions afforded by the FFT are impressive indeed. For example, computing the 2-D FFT of a 1024×1024 image would require on the order of 20 million multiplication and additions, which is a significant reduction from the one trillion computations mentioned above.

Although the FFT is a topic covered extensively in the literature on signal processing, this subject matter is of such significance in our work that this chapter would be incomplete if we did not provide at least an introduction explaining why the FFT works as it does. The algorithm we selected to accomplish this objective is the so-called *successive-doubling method*, which was the original algorithm that led to the birth of an entire industry. This particular algorithm assumes that the number of samples is an integer power of 2, but this is not a general requirement of other approaches (Brigham [1988]). We know from Section 4.11.1 that 2-D DFTs can be implemented by successive passes of the 1-D transform, so we need to focus only on the FFT of one variable.

When dealing with derivations of the FFT, it is customary to express Eq. (4.4-6) in the form

$$F(u) = \sum_{x=0}^{M-1} f(x) W_M^{ux} \quad (4.11-4)$$

$u = 0, 1, \dots, M - 1$, where

$$W_M = e^{-j2\pi/M} \quad (4.11-5)$$

and M is assumed to be of the form

$$M = 2^n \quad (4.11-6)$$

with n being a positive integer. Hence, M can be expressed as

$$M = 2K \quad (4.11-7)$$

with K being a positive integer also. Substituting Eq. (4.11-7) into Eq. (4.11-4) yields

$$\begin{aligned} F(u) &= \sum_{x=0}^{2K-1} f(x) W_{2K}^{ux} \\ &= \sum_{x=0}^{K-1} f(2x) W_{2K}^{u(2x)} + \sum_{x=0}^{K-1} f(2x+1) W_{2K}^{u(2x+1)} \end{aligned} \quad (4.11-8)$$

However, it can be shown using Eq. (4.11-5) that $W_{2K}^{2ux} = W_K^{ux}$, so Eq. (4.11-8) can be expressed as

$$F(u) = \sum_{x=0}^{K-1} f(2x) W_K^{ux} + \sum_{x=0}^{K-1} f(2x+1) W_K^{ux} W_{2K}^u \quad (4.11-9)$$

Defining

$$F_{\text{even}}(u) = \sum_{x=0}^{K-1} f(2x) W_K^{ux} \quad (4.11-10)$$

for $u = 0, 1, 2, \dots, K - 1$, and

$$F_{\text{odd}}(u) = \sum_{x=0}^{K-1} f(2x+1)W_K^{ux} \quad (4.11-11)$$

for $u = 0, 1, 2, \dots, K-1$, reduces Eq. (4.11-9) to

$$F(u) = F_{\text{even}}(u) + F_{\text{odd}}(u)W_{2K}^u \quad (4.11-12)$$

Also, because $W_M^{u+M} = W_M^u$ and $W_{2M}^{u+M} = -W_{2M}^u$, Eqs. (4.11-10) through (4.11-12) give

$$F(u+K) = F_{\text{even}}(u) - F_{\text{odd}}(u)W_{2K}^u \quad (4.11-13)$$

Analysis of Eqs. (4.11-10) through (4.11-13) reveals some interesting properties of these expressions. An M -point transform can be computed by dividing the original expression into two parts, as indicated in Eqs. (4.11-12) and (4.11-13). Computing the first half of $F(u)$ requires evaluation of the two $(M/2)$ -point transforms given in Eqs. (4.11-10) and (4.11-11). The resulting values of $F_{\text{even}}(u)$ and $F_{\text{odd}}(u)$ are then substituted into Eq. (4.11-12) to obtain $F(u)$ for $u = 0, 1, 2, \dots, (M/2-1)$. The other half then follows directly from Eq. (4.11-13) *without* additional transform evaluations.

In order to examine the computational implications of this procedure, let $m(n)$ and $a(n)$ represent the number of complex multiplications and additions, respectively, required to implement it. As before, the number of samples is 2^n with n a positive integer. Suppose first that $n = 1$. A two-point transform requires the evaluation of $F(0)$; then $F(1)$ follows from Eq. (4.11-13). To obtain $F(0)$ requires computing $F_{\text{even}}(0)$ and $F_{\text{odd}}(0)$. In this case $K = 1$ and Eqs. (4.11-10) and (4.11-11) are one-point transforms. However, because the DFT of a single sample point is the sample itself, no multiplications or additions are required to obtain $F_{\text{even}}(0)$ and $F_{\text{odd}}(0)$. One multiplication of $F_{\text{odd}}(0)$ by W_2^0 and one addition yield $F(0)$ from Eq. (4.11-12). Then $F(1)$ follows from (4.11-13) with one more addition (subtraction is considered to be the same as addition). Because $F_{\text{odd}}(0)W_2^0$ has already been computed, the total number of operations required for a two-point transform consists of $m(1) = 1$ multiplication and $a(1) = 2$ additions.

The next allowed value for n is 2. According to the above development, a four-point transform can be divided into two parts. The first half of $F(u)$ requires evaluation of two, two-point transforms, as given in Eqs. (4.11-10) and (4.11-11) for $K = 2$. As noted in the preceding paragraph, a two-point transform requires $m(1)$ multiplications and $a(1)$ additions, so evaluation of these two equations requires a total of $2m(1)$ multiplications and $2a(1)$ additions. Two further multiplications and additions are necessary to obtain $F(0)$ and $F(1)$ from Eq. (4.11-12). Because $F_{\text{odd}}(u)W_{2K}^u$ already has been computed for $u = \{0, 1\}$, two more additions give $F(2)$ and $F(3)$. The total is then $m(2) = 2m(1) + 2$ and $a(2) = 2a(1) + 4$.

When n is equal to 3, two four-point transforms are considered in the evaluation of $F_{\text{even}}(u)$ and $F_{\text{odd}}(u)$. They require $2m(2)$ multiplications and $2a(2)$ additions. Four more multiplications and eight more additions yield the complete transform. The total then is $m(3) = 2m(2) + 4$ and $a(3) = 2a(2) + 8$.

Continuing this argument for any positive integer value of n leads to recursive expressions for the number of multiplications and additions required to implement the FFT:

$$m(n) = 2m(n-1) + 2^{n-1} \quad n \geq 1 \quad (4.11-14)$$

and

$$a(n) = 2a(n-1) + 2^n \quad n \geq 1 \quad (4.11-15)$$

where $m(0) = 0$ and $a(0) = 0$ because the transform of a single point does not require any additions or multiplications.

Implementation of Eqs. (4.11-10) through (4.11-13) constitutes the successive doubling FFT algorithm. This name comes from the method of computing a two-point transform from two one-point transforms, a four-point transform from two two-point transforms, and so on, for any M equal to an integer power of 2. It is left as an exercise (Problem 4.41) to show that

$$m(n) = \frac{1}{2} M \log_2 M \quad (4.11-16)$$

and

$$a(n) = M \log_2 M \quad (4.11-17)$$

The computational advantage of the FFT over a direct implementation of the 1-D DFT is defined as

$$\begin{aligned} c(M) &= \frac{M^2}{M \log_2 M} \\ &= \frac{M}{\log_2 M} \end{aligned} \quad (4.11-18)$$

Because it is assumed that $M = 2^n$, we can write Eq. (4.11-18) in terms of n :

$$c(n) = \frac{2^n}{n} \quad (4.11-19)$$

Figure 4.67 shows a plot of this function. It is evident that the computational advantage increases rapidly as a function of n . For instance, when $n = 15$ (32,768 points), the FFT has nearly a 2,200 to 1 advantage over the DFT. Thus, we would expect that the FFT can be computed nearly 2,200 times faster than the DFT on the same machine.

There are so many excellent sources that cover details of the FFT that we will not dwell on this topic further (see, for example, Brigham [1988]). Virtually all comprehensive signal and image processing software packages have generalized implementations of the FFT that handle cases in which the number of points is not an integer power of 2 (at the expense of less efficient computation). Free FFT programs also are readily available, principally over the Internet.

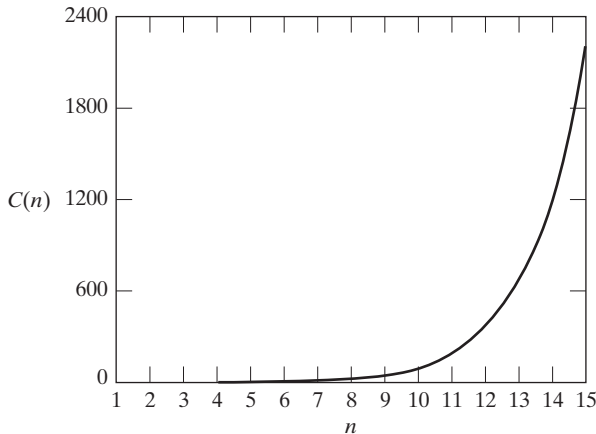


FIGURE 4.67
Computational advantage of the FFT over a direct implementation of the 1-D DFT. Note that the advantage increases rapidly as a function of n .

4.11.4 Some Comments on Filter Design

The approach to filtering discussed in this chapter is based strictly on fundamentals, the focus being specifically to explain the effects of filtering in the frequency domain as clearly as possible. We know of no better way to do that than to treat filtering the way we did here. One can view this development as the basis for “prototyping” a filter. In other words, given a problem for which we want to find a filter, the frequency domain approach is an ideal tool for experimenting, quickly and with full control over filter parameters.

Once a filter for a specific application has been found, it often is of interest to implement the filter directly in the spatial domain, using firmware and/or hardware. This topic is outside the scope of this book. Petrou and Bosdogianni [1999] present a nice tie between two-dimensional frequency domain filters and the corresponding digital filters. On the design of 2-D digital filters, see Lu and Antoniou [1992].

Summary

The material in this chapter is a progression from sampling to the Fourier transform, and then to filtering in the frequency domain. Some of the concepts, such as the sampling theorem, make very little sense if not explained in the context of the frequency domain. The same is true of effects such as aliasing. Thus, the material developed in the preceding sections is a solid foundation for understanding the fundamentals of digital signal processing. We took special care to develop the material starting with basic principles, so that any reader with a modest mathematical background would be in a position not only to absorb the material, but also to apply it.

A second major objective of this chapter was the development of the discrete Fourier transform and its use for filtering in the frequency domain. To get there, we had to introduce the convolution theorem. This result is the foundation of linear systems, and underlies many of the restoration techniques developed in Chapter 5. The types of filters we discussed are representative of what one finds in practice. The key point in presenting these filters, however, was to show how simple it is to formulate and implement filters in the frequency domain. While final implementation of a solution typically is based on spatial filters, the insight gained by working in the frequency domain as a guide in the selection of spatial filters cannot be overstated.

Although most filtering examples in this chapter are in the area of image enhancement, the procedures themselves are general and are utilized extensively in subsequent chapters.

References and Further Reading

For additional reading on the material in Section 4.1, see Hubbard [1998]. The books by Bracewell [2000, 1995] are good introductions to the continuous Fourier transform and its extensions to two dimensions for image processing. These two books, in addition to Lim [1990], Castleman [1996], Petrou and Bosdogianni [1999], Brigham [1988], and Smith [2003], provide comprehensive background for most of the discussion in Sections 4.2 through 4.6. For an overview of early work on the topic of moiré patterns, see Oster and Nishijima [1963]. Creath and Wyant [1992] discuss the state of the art in that field thirty years later. The sampling, aliasing, and image reconstruction issues discussed in Section 4.5 are also topics of significant interest in computer graphics, as exemplified by Shirley [2002].

For additional background on the material in Sections 4.7 through 4.11, see Castleman [1996], Pratt [2001], and Hall [1979]. To learn more about the imaging sensors in the Cassini spacecraft (Section 4.10.2), see Porco, West, et al. [2004]. Effective handling of issues on filter implementation (like ringing) still is a topic of interest, as exemplified by Bakir and Reeves [2000]. For unsharp masking and high-frequency-emphasis filtering, see Schowengerdt [1983]. The material on homomorphic filtering (Section 4.9.5) is based on a paper by Stockham [1972]; see also the books by Oppenheim and Schaffer [1975] and Pitas and Venetsanopoulos [1990]. Brinkman et al. [1998] combine unsharp masking and homomorphic filtering for the enhancement of magnetic resonance images.

As noted in Section 4.1.1, “discovery” of the Fast Fourier transform (Section 4.11.3) was a major milestone in the popularization of the DFT as a fundamental signal processing tool. Our presentation of the FFT in Section 4.11.3 is based on a paper by Cooley and Tukey [1965] and on the book by Brigham [1988], who also discusses several implementations of the FFT, including bases other than 2. Formulation of the fast Fourier transform is often credited to Cooley and Tukey [1965]. However, the FFT has an interesting history worth sketching here. In response to the Cooley–Tukey paper, Rudnick [1966] reported that he was using a similar technique, whose number of operations also was proportional to $N \log_2 N$ and which was based on a method published by Danielson and Lanczos [1942]. These authors, in turn, referenced Runge [1903, 1905] as the source of their technique. The latter two papers, together with the lecture notes of Runge and König [1924], contain the essential computational advantages of present FFT algorithms. Similar techniques also were published by Yates [1937], Stumpff [1939], Good [1958], and Thomas [1963]. A paper by Cooley, Lewis, and Welch [1967a] presents a historical summary and an interesting comparison of results prior to the 1965 Cooley–Tukey paper.

The FFT algorithm in Section 4.11.3 is from the original paper by Cooley and Tukey [1965]. See Brigham [1988] and Smith [2003] for complementary reading. For the design of digital filters (Section 4.11.4) based on the frequency domain formulations discussed in this chapter, see Lu and Antoniou [1992] and Petrou and Bosdogianni [1999]. For software implementation of many of the approaches discussed in Sections 4.7 through 4.11, see Gonzalez, Woods, and Eddins [2004].

Problems

- 4.1 Repeat Example 4.1, but using the function $f(t) = A$ for $0 \leq t \leq W$ and $f(t) = 0$ for all other values of t . Explain the reason for any differences between your results and the results in the example.
- ★4.2 Show that $\tilde{F}(\mu)$ in Eq. (4.4-2) is infinitely periodic in both directions, with period $1/\Delta T$.
- ★4.3 It can be shown (Bracewell [2000]) that $1 \Leftrightarrow \delta(\mu)$ and $\delta(t) \Leftrightarrow 1$. Use the first of these properties and the translation property from Table 4.3 to show that the Fourier transform of the continuous function $f(t) = \sin(2\pi nt)$, where n is a real number, is $F(\mu) = (j/2)[\delta(\mu + n) - \delta(\mu - n)]$.



Detailed solutions to the problems marked with a star can be found in the book Web site. The site also contains suggested projects based on the material in this chapter.

4.4 Consider the continuous function $f(t) = \sin(2\pi nt)$.

★(a) What is the period of $f(t)$?

★(b) What is the frequency of $f(t)$?

The Fourier transform, $F(\mu)$, of $f(t)$ is purely imaginary (Problem 4.3), and because the transform of the sampled data consists of periodic copies of $F(\mu)$, the transform of the sampled data, $\tilde{F}(\mu)$, will also be purely imaginary. Draw a diagram similar to Fig. 4.6, and answer the following questions based on your diagram (assume that sampling starts at $t = 0$).

★(c) What would the sampled function and its Fourier transform look like in general if $f(t)$ is sampled at a rate higher than the Nyquist rate?

(d) What would the sampled function look like in general if $f(t)$ is sampled at a rate lower than the Nyquist rate?

(e) What would the sampled function look like if $f(t)$ is sampled at the Nyquist rate with samples taken at $t = 0, \Delta T, 2\Delta T, \dots$?

★4.5 Prove the validity of the 1-D convolution theorem of a continuous variable, as given in Eqs. (4.2-21) and (4.2-22).

4.6 Complete the steps that led from Eq. (4.3-11) to Eq. (4.3-12).

4.7 As the figure below shows, the Fourier transform of a “tent” function (on the left) is a squared sinc function (on the right). Advance an argument that shows that the Fourier transform of a tent function can be obtained from the Fourier transform of a box function. (*Hint:* The tent itself can be generated by convolving two equal boxes.)



4.8 (a) Show that Eqs. (4.4-4) and (4.4-5) constitute a Fourier transform pair.

★(b) Repeat (a) for Eqs. (4.4-6) and (4.4-7). You will need the following orthogonality property of exponentials in both parts of this problem:

$$\sum_{x=0}^{M-1} e^{j2\pi r x/M} e^{-j2\pi u x/M} = \begin{cases} M & \text{if } r = u \\ 0 & \text{otherwise} \end{cases}$$

4.9 Prove the validity of Eqs. (4.4-8) and (4.4-9).

★4.10 Prove the validity of the discrete convolution theorem of one variable [see Eqs. (4.2-21), (4.2-22), and (4.4-10)]. You will need to use the translation properties $f(x)e^{j2\pi u_0 x/M} \Leftrightarrow F(u - u_0)$ and conversely, $f(x - x_0) \Leftrightarrow F(u)e^{-j2\pi x_0 u/M}$.

★4.11 Write an expression for 2-D continuous convolution.

4.12 Consider a checkerboard image in which each square is 1×1 mm. Assuming that the image extends infinitely in both coordinate directions, what is the minimum sampling rate (in samples/mm) required to avoid aliasing?

4.13 We know from the discussion in Section 4.5.4 that shrinking an image can cause aliasing. Is this true also of zooming? Explain.

★4.14 Prove that both the 2-D continuous and discrete Fourier transforms are linear operations (see Section 2.6.2 for a definition of linearity).

4.15 You are given a “canned” program that computes the 2-D, DFT pair. However, it is not known in which of the two equations the $1/MN$ term is included or if it

was split as two constants $1/\sqrt{MN}$ in front of both the forward and inverse transforms. How can you find where the term(s) is (are) included if this information is not available in the documentation?

4.16 ★(a) Prove the validity of the translation property in Eq. (4.6-3).

(b) Prove the validity of Eq. (4.6-4).

4.17 You can infer from Problem 4.3 that $1 \Leftrightarrow \delta(\mu, \nu)$ and $\delta(t, z) \Leftrightarrow 1$. Use the first of these properties and the translation property in Table 4.3 to show that the Fourier transform of the continuous function $f(t, z) = A \sin(2\pi\mu_0 t + 2\pi\nu_0 z)$ is

$$F(\mu, \nu) = \frac{j}{2} [\delta(\mu + \mu_0, \nu + \nu_0) - \delta(\mu - \mu_0, \nu - \nu_0)]$$

4.18 Show that the DFT of the discrete function $f(x, y) = 1$ is

$$\mathfrak{F}\{1\} = \delta(u, v) = \begin{cases} 1 & \text{if } u = v = 0 \\ 0 & \text{otherwise} \end{cases}$$

4.19 Show that the DFT of the discrete function $f(x, y) = \sin(2\pi u_0 x + 2\pi v_0 y)$ is

$$F(u, v) = \frac{j}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$$

4.20 The following problems are related to the properties in Table 4.1.

★(a) Prove the validity of property 2.

★(b) Prove the validity of property 4.

(c) Prove the validity of property 5.

★(d) Prove the validity of property 7.

(e) Prove the validity of property 9.

(f) Prove the validity of property 10.

★(g) Prove the validity of property 11.

(h) Prove the validity of property 12.

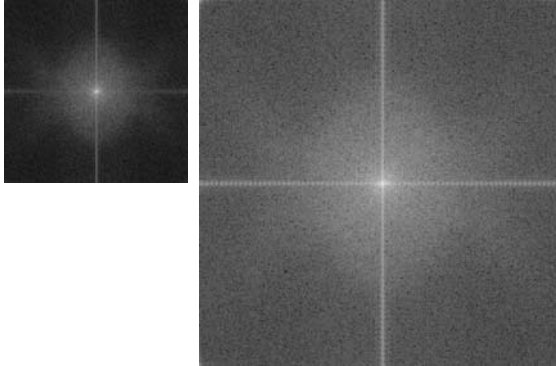
(i) Prove the validity of property 13.

★4.21 The need for image padding when filtering in the frequency domain was discussed in Section 4.6.6. We showed in that section that images needed to be padded by appending zeros to the ends of rows and columns in the image (see the following image on the left). Do you think it would make a difference if we



centered the image and surrounded it by a border of zeros instead (see image on the right), but without changing the total number of zeros used? Explain.

- ★4.22 The two Fourier spectra shown are of the same image. The spectrum on the left corresponds to the original image, and the spectrum on the right was obtained after the image was padded with zeros. Explain the significant increase in signal strength along the vertical and horizontal axes of the spectrum shown on the right.



- 4.23 You know from Table 4.2 that the dc term, $F(0, 0)$, of a DFT is proportional to the average value of its corresponding spatial image. Assume that the image is of size $M \times N$. Suppose that you pad the image with zeros to size $P \times Q$, where P and Q are given in Eqs. (4.6-31) and (4.6-32). Let $F_p(0, 0)$ denote the dc term of the DFT of the padded function.

- ★(a) What is the ratio of the average values of the original and padded images?
 (b) Is $F_p(0, 0) = F(0, 0)$? Support your answer mathematically.

- 4.24 Prove the periodicity properties (entry 8) in Table 4.2.

- 4.25 The following problems are related to the entries in Table 4.3.

- ★(a) Prove the validity of the discrete convolution theorem (entry 6) for the 1-D case.
 (b) Repeat (a) for 2-D.
 ★(c) Prove the validity of entry 7.
 ★(d) Prove the validity of entry 12.

(Note: Problems 4.18, 4.19, and 4.31 are related to Table 4.3 also.)

- 4.26 (a) Show that the Laplacian of a continuous function $f(t, z)$ of continuous variables t and z satisfies the following Fourier transform pair [see Eq. (3.6-3) for a definition of the Laplacian]:

$$\nabla^2 f(t, z) \Leftrightarrow -4\pi^2(\mu^2 + \nu^2)F(\mu, \nu)$$

[Hint: Study entry 12 in Table 4.3 and see Problem 4.25(d).]

- ★(b) The preceding closed form expression is valid only for continuous variables. However, it can be the basis for implementing the Laplacian in the discrete frequency domain using the $M \times N$ filter

$$H(u, v) = -4\pi^2(u^2 + v^2)$$

for $u = 0, 1, 2, \dots, M - 1$ and $v = 0, 1, 2, \dots, N - 1$. Explain how you would implement this filter.

- (c) As you saw in Example 4.20, the Laplacian result in the frequency domain was similar to the result of using a spatial mask with a center coefficient equal to -8 . Explain the reason why the frequency domain result was not similar instead to the result of using a spatial mask with a center coefficient of -4 . See Section 3.6.2 regarding the Laplacian in the spatial domain.
- ★4.27 Consider a 3×3 spatial mask that averages the four closest neighbors of a point (x, y) , but excludes the point itself from the average.
- (a) Find the equivalent filter, $H(u, v)$, in the frequency domain.
- (b) Show that your result is a lowpass filter.
- 4.28 Based on Eq. (3.6-1), one approach for approximating a discrete derivative in 2-D is based on computing differences of the form $f(x + 1, y) - f(x, y)$ and $f(x, y + 1) - f(x, y)$.
- (a) Find the equivalent filter, $H(u, v)$, in the frequency domain.
- (b) Show that your result is a highpass filter.
- 4.29 Find the equivalent filter, $H(u, v)$, that implements in the frequency domain the spatial operation performed by the Laplacian mask in Fig. 3.37(a).
- ★4.30 Can you think of a way to use the Fourier transform to compute (or partially compute) the magnitude of the gradient [Eq. (3.6-11)] for use in image differentiation? If your answer is yes, give a method to do it. If your answer is no, explain why.
- ★4.31 A continuous Gaussian lowpass filter in the continuous frequency domain has the transfer function

$$H(\mu, \nu) = Ae^{-(\mu^2 + \nu^2)/2\sigma^2}$$

Show that the corresponding filter in the spatial domain is

$$h(t, z) = A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2 + z^2)}$$

- 4.32 As explained in Eq. (4.9-1), it is possible to obtain the transfer function, H_{HP} , of a highpass filter from the transfer function of a lowpass filter as

$$H_{\text{HP}} = 1 - H_{\text{LP}}$$

Using the information given in Problem 4.31, what is the form of the *spatial domain* Gaussian highpass filter?

- 4.33 Consider the images shown. The image on the right was obtained by: (a) multiplying the image on the left by $(-1)^{x+y}$; (b) computing the DFT; (c) taking the complex conjugate of the transform; (d) computing the inverse DFT; and (e) multiplying the real part of the result by $(-1)^{x+y}$. Explain (mathematically) why the image on the right appears as it does.



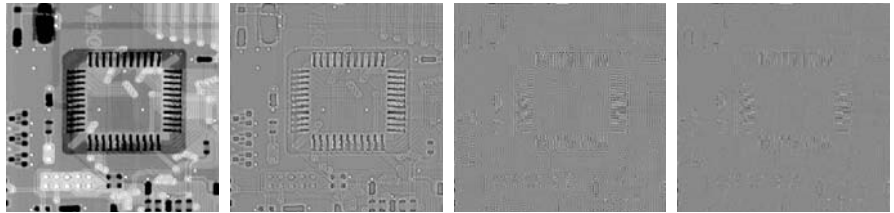
- 4.34** What is the source of the nearly periodic bright points in the horizontal axis of Fig. 4.41(b)?
- ★ **4.35** Each filter in Fig. 4.53 has a strong spike in its center. Explain the source of these spikes.
- 4.36** Consider the images shown. The image on the right was obtained by lowpass filtering the image on the left with a Gaussian lowpass filter and then highpass filtering the result with a Gaussian highpass filter. The dimension of the images is 420×344 , and $D_0 = 25$ was used for both filters.
- (a) Explain why the center part of the finger ring in the figure on the right appears so bright and solid, considering that the dominant characteristic of the filtered image consists of edges on the outer boundary of objects (e.g., fingers, wrist bones) with a darker area in between. In other words, would you not expect the highpass filter to render the constant area inside the ring dark, since a highpass filter eliminates the dc term?
- (b) Do you think the result would have been different if the order of the filtering process had been reversed?



(Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

- 4.37** Given an image of size $M \times N$, you are asked to perform an experiment that consists of repeatedly lowpass filtering the image using a Gaussian lowpass filter with a given cutoff frequency D_0 . You may ignore computational round-off errors. Let $c_{\min}$ denote the smallest positive number representable in the machine in which the proposed experiment will be conducted.
- ★ (a) Let K denote the number of applications of the filter. Can you predict (without doing the experiment) what the result (image) will be for a sufficiently large value of K ? If so, what is that result?
- (b) Derive an expression for the *minimum* value of K that will guarantee the result that you predicted.
- 4.38** Consider the sequence of images shown. The image on the left is a segment of an X-ray image of a commercial printed circuit board. The images following it are, respectively, the results of subjecting the image to 1, 10, and 100 passes of a Gaussian highpass filter with $D_0 = 30$. The images are of size 330×334 pixels, with each pixel being represented by 8 bits of gray. The images were scaled for display, but this has no effect on the problem statement.

- (a) It appears from the images that changes will cease to take place after some finite number of passes. Show whether or not this in fact is the case. You may ignore computational round-off errors. Let $c_{\min}$ denote the smallest positive number representable in the machine in which the proposed experiment will be conducted.
- (b) If you determined in (a) that changes would cease after a finite number of iterations, determine the minimum value of that number.



Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.

- 4.39 As illustrated in Fig. 4.59, combining high-frequency emphasis and histogram equalization is an effective method for achieving edge sharpening and contrast enhancement.
 - (a) Show whether or not it matters which process is applied first.
 - (b) If the order does matter, give a rationale for using one or the other method first.
- 4.40 Use a Butterworth highpass filter to construct a homomorphic filter that has the same general shape as the filter in Fig. 4.61.
- ★4.41 Show the validity of Eqs. (4.11-16) and (4.11-17). (*Hint:* Use proof by induction.)
- 4.42 Suppose that you are given a set of images generated by an experiment dealing with the analysis of stellar events. Each image contains a set of bright, widely scattered dots corresponding to stars in a sparsely occupied section of the universe. The problem is that the stars are barely visible, due to superimposed illumination resulting from atmospheric dispersion. If these images are modeled as the product of a constant illumination component with a set of impulses, give an enhancement procedure based on homomorphic filtering designed to bring out the image components due to the stars themselves.
- 4.43 A skilled medical technician is assigned the job of inspecting a certain class of images generated by an electron microscope. In order to simplify the inspection task, the technician decides to use digital image enhancement and, to this end, examines a set of representative images and finds the following problems: (1) bright, isolated dots that are of no interest; (2) lack of sharpness; (3) not enough contrast in some images; and (4) shifts in the average intensity, when this value should be V to perform correctly certain intensity measurements. The technician wants to correct these problems and then display in white all intensities in a band between I_1 and I_2 , while keeping normal tonality in the remaining intensities. Propose a sequence of processing steps that the technician can follow to achieve the desired goal. You may use techniques from both Chapters 3 and 4.